

GUIDELINES

AGA Living Clinical Practice Guideline on the Pharmacologic Management of Moderate-to-Severe Crohn's Disease



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BACKGROUND & AIMS: This American Gastroenterological Association (AGA) living guideline is intended to support practitioners in the pharmacologic management of moderate-to-severely active Crohn's disease (CD). **METHODS:** A multidisciplinary panel of clinical experts and methodologists used the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework to prioritize clinical questions, identify patient-centered outcomes, conduct an evidence synthesis, and develop recommendations. **RESULTS:** The guideline panel agreed on 16 recommendations, of which 1 is a strong recommendation, 9 are conditional recommendations, and 6 were identified as knowledge gaps. In adult patients with moderate-to-severely active CD, the AGA recommends the use of infliximab, adalimumab, ustekinumab, risankizumab, mirikizumab, guselkumab, or upadacitinib over no treatment, and suggests the use of certolizumab pegol or vedolizumab over no treatment. In individuals who are naïve to advanced therapies, the AGA suggests using a higher efficacy medication (infliximab, adalimumab, vedolizumab, ustekinumab, risankizumab, mirikizumab, or guselkumab) rather than a lower-efficacy medication (certolizumab pegol or upadacitinib). In individuals who have previously been exposed to 1 or more advanced therapies, the AGA suggests using a higher efficacy medication (adalimumab, risankizumab, guselkumab, or upadacitinib) or intermediate efficacy medication (ustekinumab or mirikizumab) rather than a lower efficacy medication (vedolizumab or certolizumab pegol). In adult outpatients with moderate-to-severely active CD, the AGA suggests against using thiopurine monotherapy for induction of remission, but suggests using thiopurine monotherapy over no treatment for maintenance of (typically corticosteroid-induced) remission. The AGA suggests using subcutaneous methotrexate for induction and maintenance of remission, but suggests against using oral methotrexate. The AGA suggests using combination therapy with infliximab and thiopurines over infliximab monotherapy, particularly in those naïve to thiopurines. The AGA also suggests using initial advanced therapy over step therapy

involving corticosteroids and/or immunomodulators in patients with moderate-to-severely active CD. The panel also proposed key implementation considerations for optimal use of these medications, identified several knowledge gaps, and areas for future research. **CONCLUSIONS:** This guideline provides a comprehensive, patient-centered, evidence-based approach to the pharmacologic management of patients with moderate-to-severely active CD.

Keywords: Inflammatory Bowel Disease; Network Meta-Analysis; Evidence Synthesis; Positioning.

Executive Summary

Crohn's disease (CD) affects more than 1 million individuals in the United States and more than 3 million individuals worldwide.^{1–3} CD-related inflammation can result in penetrating and stricturing complications, often resulting in hospitalization and surgery.^{4,5} Advanced therapies, including monoclonal antibodies (also referred to as biologics) targeting tumor necrosis factor (TNF)- α (infliximab, adalimumab, and certolizumab pegol), leukocyte

Abbreviations used in this paper: AGA, American Gastroenterological Association; CD, Crohn's disease; CDAI, Crohn's Disease Activity Index; CGC, Clinical Guidelines Committee; CMD, clinically meaningful difference; COI, conflict of interest; FDA, US Food and Drug Administration; GRADE, Grading of Recommendations Assessment, Development and Evaluation; HR, hazard ratio; IBD, inflammatory bowel disease; IL, interleukin; IR, incidence ratio; JAK, Janus kinase; NMA, network meta-analysis; OR, odds ratio; PICO, population, intervention, comparator, outcome; RCT, randomized controlled trial; RR, relative risk; SAE, serious adverse event; TNF, tumor necrosis factor.



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trafficking (vedolizumab), and interleukins (ILs) 12 and 23 (ustekinumab, risankizumab, mirikizumab, and guselkumab), as well as small molecules targeting Janus kinase (JAK; upadacitinib), have improved clinical outcomes in patients with CD.

The goal of these evidence-based guidelines from the American Gastroenterological Association (AGA) is to provide comprehensive recommendations for the pharmacologic management of moderate-to-severely active CD in adults, as well as recommendations in those without and those with prior exposure to an advanced therapy, the timing of using these therapies, their withdrawal, and appropriate goals of medical management.

How to Use These Guidelines

Table 1 provides an overview of each guideline recommendation, the strength of the recommendation, and the certainty of evidence used to inform it. Recommendations are accompanied by implementation considerations (based on the collective experience of the panel members) that are meant to help provide guidance regarding the recommendations. Broad overarching considerations for implementing these recommendations in clinical practice are also provided (**Table 2**). Two clinical decision support tools, which may assist clinicians in making pharmacologic management decisions for patients with moderate-to-severely active CD, are presented in **Figures 1** and **2**.

Additional information about the background, methods, evidence reviews, and detailed justifications for each recommendation are provided in these full guidelines. Corresponding forest plots for each pharmacologic intervention and the studies included to inform them are provided both in the main document and the **Supplementary Materials**. The methods, evidence reviews, and analyses related to network meta-analyses conducted for this guideline are published alongside this guideline. The term “recommend” was used to indicate strong recommendations, and the term “suggest” was used to indicate conditional recommendations. The interpretation of certainty of evidence and implications of strong and conditional recommendations for health care providers, patients, and policy makers are presented in **Table 3**.

Objectives and Scope of These Guidelines

The approval of advanced therapies in CD has significantly impacted the course of disease. Since the publication of the previous AGA guidelines for moderate-to-severely active CD in 2021, two additional classes of advanced therapies with 5 novel agents have been approved for treating moderate-to-severely active CD.⁶ These therapies were followed by the introduction of biosimilars for several agents over the past decade, including for infliximab, adalimumab, and ustekinumab, as well as subcutaneous delivery options for infliximab and vedolizumab.

The purpose of these guidelines is to provide evidence-based recommendations for the pharmacologic management

of adults with moderate-to-severely active CD, defined as moderate to severe daily abdominal pain and diarrhea⁷ coupled with confirmed active intestinal inflammation. These guidelines may also apply to individuals with CD with mild symptoms but with predictors of future disease-related complications (such as high burden of inflammation, extensive disease, or ileal/ileocolonic or upper gut involvement),⁸ patients who are corticosteroid-dependent, or when the disease has a significant impact on quality of life. These guidelines are intended to be used in the ambulatory setting. This guideline is not intended to inform decision making related to the management of postoperative, perianal, or internally penetrating or stricturing CD.

Target Audience

This guideline, like all AGA guidelines, is primarily targeted toward health care providers in the fields of gastroenterology and primary care who depend on our expert, evidence-based recommendations to inform their clinical practice and shared decision making with patients. Rather than represent a specific standard to adhere to, we intend these recommendations to be used by clinicians to guide their patient management decisions and to inform considerations of benefits and harms of treatments in each individual case.

Guideline Development Procedures

Overview

This document represents the official recommendations of the AGA. It was developed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework, and adheres to best practices in guideline development as outlined by the National Academy of Medicine (formerly Institute of Medicine), using a process outlined previously.⁹

Organization and Panel Composition

The guideline panel was composed of 10 members, each selected based on their specific expertise. The panel included 8 clinical content experts with both clinical and research expertise, and 2 methodologists with specialized GRADE guideline development skills. An information specialist assisted the panel members with designing and executing the required literature searches. All panel members participated in evidence review and the senior methodologist oversaw data synthesis and analysis. All members of the panel reviewed analyses, contributed to consensus development, constructed the final recommendations, and contributed to manuscript development.

Document Review

The guideline underwent several levels of review, including an open, 30-day public comment period, external peer review by 2 topic experts, and patient review. A network meta-analysis (NMA) that informed focused questions on positioning of advanced therapies underwent separate editorial and peer review per *Gastroenterology's*

Table 1. Executive Summary of Recommendations for the Management of Adult Outpatients With Moderate-to-Severe Crohn's Disease

Recommendations
Use and positioning of advanced therapies
1. In adult outpatients with moderate-to-severely active CD, the AGA <i>recommends</i> the use of infliximab, adalimumab, ustekinumab, risankizumab, mirikizumab, guselkumab, or upadacitinib,^a over no treatment. (<i>Strong recommendation, moderate to high certainty of evidence</i>) 2. In adult outpatients with moderate-to-severely active CD, the AGA <i>suggests</i> the use of certolizumab pegol or vedolizumab, over no treatment. (<i>Conditional recommendation, moderate certainty of evidence</i>)
Implementation Considerations
• Biosimilars of infliximab, adalimumab, and ustekinumab can be considered equivalent to their originator drug in their efficacy in terms of therapy selection. • Subcutaneous formulations of infliximab and vedolizumab have shown comparable efficacy to their respective FDA-approved intravenous maintenance doses. • In some patients with partial response to FDA-approved dosing, particularly those with more severe disease, extended induction regimens or dose escalation may be beneficial for most advanced therapies. • There are 2 dosing options available for maintenance therapy for risankizumab, guselkumab, and upadacitinib. Higher maintenance doses may be preferred in patients with a high burden of inflammation and/or more severe disease, and those who have previously failed TNF antagonists. 3. In adult outpatients with moderate-to-severely active CD who are naïve to advanced therapies, the AGA <i>suggests</i> using a HIGHER efficacy medication (infliximab, adalimumab, vedolizumab, ustekinumab, risankizumab, mirikizumab, guselkumab), rather than a LOWER efficacy medication (certolizumab pegol, upadacitinib). (<i>Conditional recommendation, low to high certainty of evidence</i>)
Implementation Considerations
• Individual patient factors (eg, age, comorbidities, frailty, pregnancy, adherence) and preferences (eg, route of administration, ease of access) should be incorporated within a shared decision framework in selection of advanced therapies. • There are limited data on the safety of JAK inhibitors in pregnancy. These drugs should generally be avoided in women contemplating pregnancy in the near future. 4. In adult outpatients with moderate-to-severely active CD who have previously been exposed to 1 or more advanced therapies, particularly TNF antagonists, the AGA <i>suggests</i> using a HIGHER efficacy medication (adalimumab, risankizumab, guselkumab, upadacitinib) OR an INTERMEDIATE efficacy medication (ustekinumab, mirikizumab), rather than a LOWER efficacy medication (vedolizumab, certolizumab pegol). (<i>Conditional recommendation, low to moderate certainty of evidence</i>)
Implementation Considerations
• Second-line TNF antagonists (especially infliximab or adalimumab) may be effective in patients who discontinued their first TNF antagonist after initial response due to immunogenicity or intolerance. They may not be effective in patients with primary non-response to TNF antagonists, or those who lose response despite adequate drug trough levels. In such patients alternative mechanisms of action should be considered. • Individual patient factors (eg, age, comorbidities, frailty, pregnancy, adherence) and preferences (eg, route of administration, ease of access) should be incorporated within a shared decision framework in selection of advanced therapies. • Some patients, such as those with multiple prior biologic failures, may require longer duration of induction therapy for response or higher dosing for maintenance therapy for response. • JAK inhibitors may be associated with higher risk of major adverse cardiovascular events and cancer than TNF antagonists in older adults with cardiovascular risk factors (smoking, hypertension, high-density lipoprotein cholesterol level of <40 mg/dl, diabetes mellitus, family history of premature coronary heart disease, extra-articular rheumatoid arthritis, or prior cardiovascular disease). • There are limited data on the safety of JAK inhibitors in pregnancy. These drugs should be avoided in women of childbearing age contemplating pregnancy in the near future.
Use of immunomodulators
5. In adult outpatients with moderate-to-severely active CD, the AGA <i>suggests AGAINST</i> using thiopurines monotherapy over no treatment for inducing remission. (<i>Conditional recommendation, very low certainty of evidence</i>) 6. In adult outpatients with moderate-to-severely active CD who have achieved remission, the AGA <i>SUGGESTS</i> using thiopurine monotherapy over no treatment for maintaining remission. (<i>Conditional recommendation, low certainty of evidence</i>) 7. In adult outpatients with moderate-to-severely active CD, the AGA <i>SUGGESTS</i> using subcutaneous or intramuscular methotrexate monotherapy over no treatment. (<i>Conditional recommendation, moderate certainty of evidence</i>) 8. In adult outpatients with moderate-to-severely active CD, the AGA <i>suggests AGAINST</i> using oral methotrexate monotherapy over no treatment. (<i>Conditional recommendation, very low certainty of evidence</i>)

Table 1. Continued

Recommendations
Implementation Considerations
• The typical dose of thiopurines is 2–2.5 mg/kg/d for azathioprine and 1–1.5 mg/kg/d for mercaptopurine when used as monotherapy for those with normal drug metabolism. • In clinical trials of thiopurines for preventing relapse in patients who achieved remission, remission was typically induced with corticosteroids. • The typical dose of methotrexate is 25 mg weekly during induction for 16–24 weeks, and 15 mg weekly for maintenance. Methotrexate should be accompanied by daily folic acid supplementation. • Routine monitoring of complete blood counts and liver function tests is recommended when using thiopurines and methotrexate. • Use of methotrexate is contraindicated in women actively considering pregnancy or who are pregnant.
Combination therapy of biologics and immunomodulators
9. In adult outpatients with moderate-to-severely active CD who are NAÏVE to thiopurines and starting infliximab, the AGA SUGGESTS using infliximab in combination with thiopurines rather than infliximab monotherapy. (<i>Conditional recommendation, low certainty of evidence</i>) 10. In adult outpatients with moderate-to-severely active CD, the AGA makes NO RECOMMENDATION on using infliximab in combination with methotrexate over infliximab monotherapy. (<i>No recommendation, knowledge gap</i>) 11. In adult outpatients with moderate-to-severely active CD, the AGA makes NO RECOMMENDATION on using adalimumab in combination with thiopurines or methotrexate over adalimumab monotherapy. (<i>No recommendation, knowledge gap</i>) 12. In adult outpatients with moderate-to-severely active CD, the AGA makes NO RECOMMENDATION in favor of, or against, using non-TNF-targeting biologics (vedolizumab, ustekinumab, risankizumab, mirikizumab, guselkumab) in combination with thiopurines or methotrexate or corresponding biologic monotherapy. (<i>No recommendation, knowledge gap</i>)
Implementation Considerations
• The benefit of routinely combining immunomodulators with TNF antagonists in patients who have previously failed immunomodulator monotherapy is uncertain. • In patients who previously failed immunomodulator monotherapy, there may be benefits of adding immunomodulators when starting TNF antagonists in specific situations where patients may be at a higher risk for immunogenicity. These include patients with history of immunogenicity with a TNF antagonist, patients being re-exposed to TNF antagonists after a prolonged lapse in therapy, patients carrying HLA-DQ-A1*05 variants and patients with high drug clearance, such as those with more severe disease, high burden of inflammation, or low albumin.
De-escalation of therapy
13. In adult outpatients with moderate-to-severely active CD who are in corticosteroid-free clinical remission for at least 6 months on combination therapy with TNF antagonists and an immunomodulator, the AGA SUGGESTS withdrawing the IMMUNOMODULATOR. (<i>Conditional recommendation, low certainty of evidence</i>)
Comment: Patients, particularly those with difficult-to-treat disease, who place a lower value on the trivial-to-small increase in risk of long-term adverse effects of continuing immunomodulators (such as risk of malignancy or infection), and a higher value on avoiding a trivial-to-small increase in risk of relapse with withdrawal of immunomodulators, may reasonably choose to continue combination therapy
Implementation Considerations
• There may be benefit in continuing combination therapy with TNF antagonists and immunomodulators in those who are felt to be at higher risk of immunogenicity, such as those with prior immunogenic failure to a biologic therapy (ie, anti-drug antibody formation), those with lower trough TNF antagonist concentrations despite dose escalation, or HLA-DQA1*05 carriers. 14. In adult outpatients with moderate-to-severely active Crohn's disease who are in corticosteroid-free clinical remission for at least 6 months on combination therapy of TNF antagonists and an immunomodulator, the AGA suggests AGAINST withdrawal of the TNF ANTAGONIST. (<i>Conditional recommendation, low certainty of evidence</i>)
Early advanced therapy
15. In adult outpatients with moderate-to-severely active CD, the AGA SUGGESTS upfront use of advanced therapy compared with step-up therapy with initial use of corticosteroids and/or immunomodulator monotherapy. (<i>Conditional recommendation, very low certainty of evidence</i>)
Implementation Considerations
• All trials that informed the evidence used combination therapy with TNF antagonists with immunomodulators. • Some individuals may still derive benefit from the use of corticosteroids for symptomatic management while initiating advanced therapy.
Treating to endoscopic remission vs clinical remission
16. In adult outpatients with moderate-to-severely active CD, the AGA makes NO RECOMMENDATION in favor, or against, treating to a target of endoscopic remission, compared with treating to a target of symptomatic remission. (<i>No recommendation, knowledge gap</i>)

^aIn the United States, the FDA recommends reserving use of JAK inhibitors in patients with failure or intolerance to TNF-antagonist therapy.

Table 2. Key Overarching Considerations in the Management of Adult Outpatients With Moderate-to-Severe Crohn’s Disease

Considerations
Patients should have confirmation of active inflammation based on CD-related symptoms, biomarkers, and/or endoscopic evaluation before starting advanced therapies.
Patients should have both general and therapy-specific pretreatment workup before initiation of such treatments. These include screening for hepatitis B and tuberculosis exposure before any biologic or advanced small molecule treatments, and thiopurine methyl transferase testing before initiation of thiopurines. There may be other treatment- and patient-specific tests that should be performed in accordance with the labels from regulatory agencies.
It is important to evaluate for factors influencing risk of treatment-related complications, including assessment of comorbidities, frailty, and functional status and concomitant medications, and assessment of thromboembolic and cardiovascular risk factors.
Vaccination against influenza, pneumococcal pneumonia, and herpes zoster should be considered before initiation of immunosuppressive therapies to decrease risk of serious infections.
Initiation of advanced therapy should be followed by monitoring for both symptomatic response and/or remission and improvement in objective markers of inflammation during follow-up.
Periodic laboratory monitoring for potential therapy-related toxicity should be performed while receiving immunosuppressive therapies, according to drug label.

policy.¹⁰ Organization-level review was carried out by the CGC and AGA’s Governing Board. At each stage, the guideline panel considered all reviewer comments and feedback and revised the guideline manuscript in response, as needed.

Guideline Updates

The AGA developed these guidelines as living guidelines, given the rapid pace of innovation and new therapies in CD.¹¹ A living guideline is defined as one that allows for optimization of guidelines during the development process with updating of individual recommendations based on the availability of new evidence. The evidence for focused questions in the living mode will be reviewed every 6 months; these questions are denoted as living guidelines in Table 4.

Methods

Formulation of Clinical Questions

The clinical research questions that underpin these guidelines were developed by the guideline panel, with methodologists and clinical content experts working together to develop specific questions that address current knowledge gaps in the pharmacologic management of moderate-to-severely active CD. The PICO format was used to outline the specific patient population (P), intervention (I), comparator (C), and outcome(s) for each clinical question (Table 4).

Outcomes of Interest

The panel selected desirable and undesirable patient-important outcomes (benefits and harms). Critical outcomes for decision making for the interventions included in this review are summarized in the evidence profiles. In individuals with moderate-to-severely active CD in the ambulatory setting, induction and maintenance of clinical remission were considered critical, patient-centric outcomes for decision making.

Achieving endoscopic remission, corticosteroid-free remission and serious adverse events (SAEs) were considered important outcomes. Although the discordance between clinical symptoms and endoscopic disease activity has been well described in patients with CD,¹² clinical remission was deemed to be a more patient-centered outcome that was more uniformly described across clinical trials as the primary study outcome and has been used to inform regulatory approval of all advanced therapies. Research suggests that patients perceive improving quality of life and symptom resolution as their primary treatment objectives; only 12.8% prioritize normalization of colonoscopy as a treatment objective.¹³ In clinical trials, clinical remission was most commonly measured using the Crohn’s Disease Activity Index (CDAI), based on abdominal pain, bowel movements, general well-being, complications of disease, abdominal mass, anemia, and weight change. CDAI scores <150 suggest clinical remission, and scores 150–220, 221–450, and >450 denote mild, moderate, and severe disease, respectively.¹⁴ For questions on efficacy of a strategy of initial advanced therapy vs gradual step-up therapy and treat-to-target of clinical vs endoscopic remission, maintaining clinical remission and preventing disease-related complications at 1 year were deemed to be the critical outcomes. The prioritization of the outcome of clinical remission is also consistent with prior guidelines in both CD (2021) and ulcerative colitis (2024).^{6,15}

Search Strategy

The systematic review process was guided by a search protocol developed *a priori* by the guideline panel members in collaboration with the medical librarian. The librarian conducted a comprehensive search of the following databases Ovid MEDLINE, EMBASE, and Wiley Cochrane Library from inception to August 14, 2024, using a combination of controlled vocabulary terms supplemented with keywords. The search was limited to English language and humans. References from previous guidelines and consensus statements, as well as from conference proceedings and press releases

Adult outpatients with moderate-to-severely active Crohn's disease

Moderate-to-severely active Crohn's disease defined as:

- Moderate-to-severe abdominal pain and/or diarrhea due to inflammation
- Mild symptoms, with high burden of inflammation
- Patients with corticosteroid-dependence, or refractory to oral corticosteroids
- Significant extent of disease or upper GI involvement

SUGGEST upfront use of advanced therapies, rather than step-up therapy after corticosteroids and/or immunomodulator monotherapy

(Conditional recommendation, very low certainty of evidence)

RECOMMEND using any of the following, over no treatment:

Infliximab, adalimumab, ustekinumab, risankizumab, guselkumab, mirikizumab, or upadacitinib*

(Strong recommendation, moderate to high certainty of evidence)

SUGGEST using any of the following, over no treatment:

Certolizumab pegol, vedolizumab

(Conditional recommendation, moderate certainty of evidence)

Implementation considerations:

- Biosimilars of infliximab, adalimumab, and ustekinumab can be considered equivalent to their originator drug in their efficacy in terms of therapy selection
- Subcutaneous formulations of infliximab and vedolizumab have shown comparable efficacy to the respective intravenous maintenance doses
- In some patients with suboptimal response to standard dosing, particularly those with more severe disease, extended induction regimens or dose escalation may be beneficial for most advanced therapies
- There are two dosing options available for maintenance therapy for risankizumab, guselkumab, and upadacitinib. Higher maintenance doses may be preferred in patients with high burden of inflammation and/or more severe disease, and those who have previously failed TNF antagonists

Advanced therapy-naïve patients (first-line therapy)

SUGGEST using a HIGHER efficacy rather than a lower efficacy medication

(Conditional recommendation, low to high certainty of evidence)

HIGHER EFFICACY MEDICATIONS: Infliximab, adalimumab, vedolizumab, ustekinumab, risankizumab, mirikizumab, guselkumab

LOWER EFFICACY MEDICATIONS: Certolizumab pegol, upadacitinib*

Prior exposure to one or more advanced therapies, particularly TNF antagonists

SUGGEST using a HIGHER efficacy, or INTERMEDIATE efficacy medication, rather than a lower efficacy medication

(Conditional recommendation, low to moderate certainty of evidence)

HIGHER EFFICACY MEDICATIONS: Adalimumab, risankizumab, guselkumab, upadacitinib

INTERMEDIATE EFFICACY MEDICATIONS: Ustekinumab, mirikizumab

LOWER EFFICACY MEDICATIONS: Certolizumab pegol, vedolizumab

*The FDA label recommends the use of JAK inhibitors only in patients with prior failure or intolerance to TNF antagonists

Figure 1. Clinical decision support tool for use and positioning of advanced therapies in the management of adult outpatients with moderate-to-severely active CD.

through April 1, 2025, were also reviewed. The final strategy is available in [Supplementary Table 2](#). The included references of prior guidelines on this topic were searched to identify relevant studies that may have been missed. In addition, content experts helped identify any ongoing studies with results expected soon.

Study Selection, Data Collection, and Analysis

These recommendations are based on evidence derived from randomized controlled trials (RCTs) that were synthesized in an updated systematic review, meta-analysis, and NMA conducted specifically for this guideline. Details regarding the NMA are published in an accompanying article. Data from previously

published systematic reviews were also used if no new data were identified to inform a specific PICO.

The inclusion and exclusion criteria were based on the clinical research questions developed by the guideline panel. All searches were combined using *Covidence* bibliographic software. The minimum trial duration for induction and maintenance therapy required was 4 weeks and 24 weeks, respectively. Adult and pediatric studies were considered when appropriate. Efficacy trials exclusively in ulcerative colitis were excluded. Phase 2 and 3 efficacy trials were eligible for inclusion. All medications considered for inclusion in this guideline required US Food and Drug Administration (FDA) approval. Phase 2 trial data were only considered when representative outcomes were included, dosing was relatively

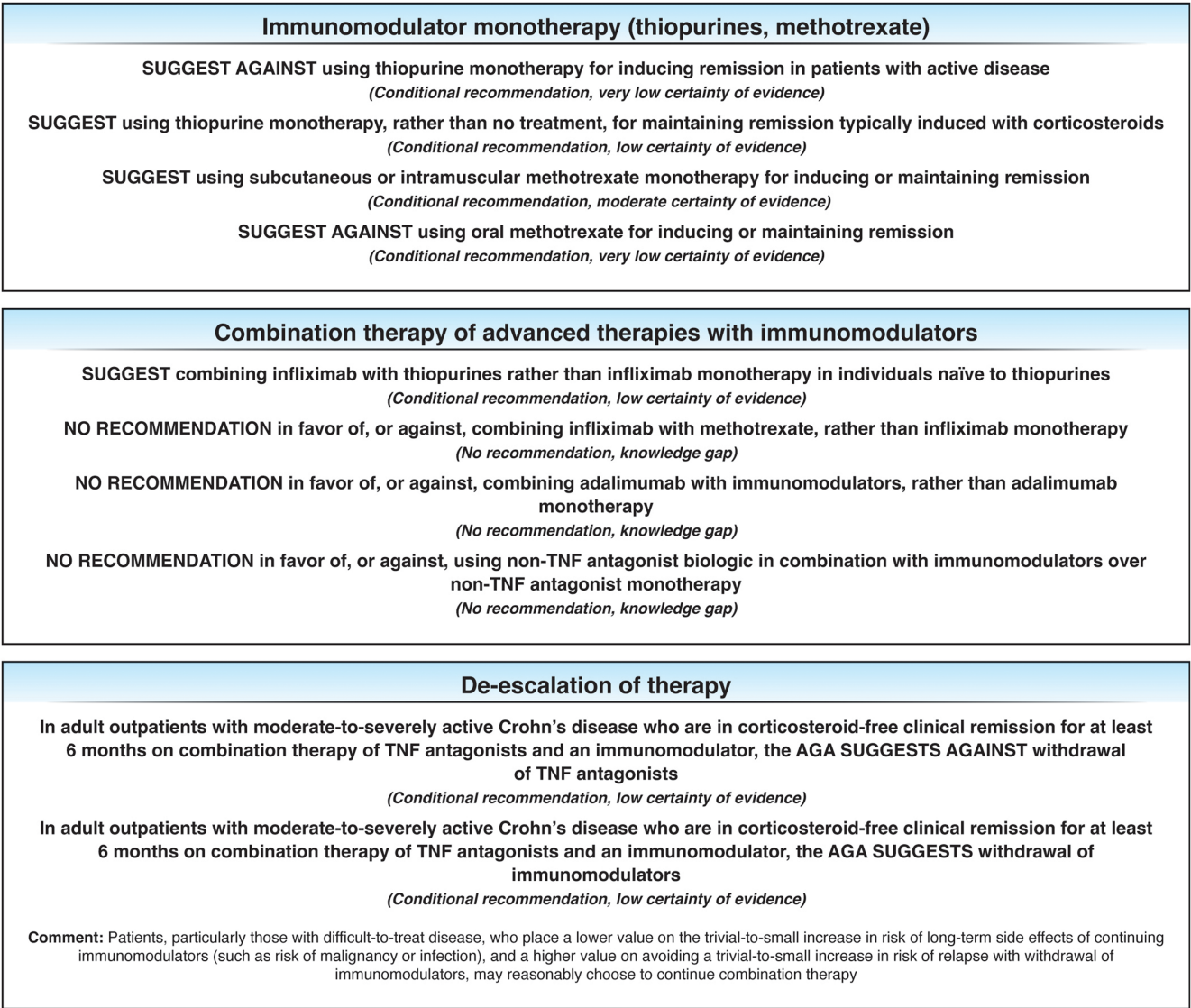


Figure 2. Clinical decision support tool for the use of combination therapy and immunomodulator therapy, and de-escalation of therapy in the management of adult outpatients with moderate-to-severely active CD.

Table 3. Interpretation of Strong and Conditional Recommendations Using the Grading of Recommendations Assessment, Development and Evaluation Framework

Implications	Strong recommendation	Conditional recommendation
For patients	Most individuals in this situation would want the recommended course of action and only a small proportion would not.	The majority of individuals in this situation would want the suggested course of action, but many would not.
For clinicians	Most individuals should receive the intervention. Formal decision aids are not likely to be needed to help individuals make decisions consistent with their values and preferences.	Different choices will be appropriate for individual patients consistent with their values and preferences. Use shared decision making. Decision aids may be useful in helping patients make decisions consistent with their individual risks, values, and preferences.
For policy makers	The recommendation can be adapted as policy or performance measure in most situations	Policy making will involve various stakeholders. Performance measures should assess whether decision making is appropriate.

Table 4. Focused Questions and Corresponding Patient Population, Intervention, Comparator, and Outcomes Addressed in the Guidelines

Question no. ^a	Focused question	Patients	Intervention	Comparator	Critical outcomes
1 (Living)	In adult outpatients with moderate-to-severely active CD, what is the efficacy of TNF antagonists (infliximab, adalimumab, certolizumab pegol), vedolizumab, ustekinumab, upadacitinib, and IL23 antagonists (risankizumab, mirikizumab, guselkumab), for induction and maintenance of remission?	Adult outpatients with moderate-to-severely active CD	TNF antagonists (infliximab, adalimumab, certolizumab pegol) Vedolizumab Ustekinumab Upadacitinib IL23 antagonists (risankizumab, mirikizumab, guselkumab)	Placebo	Induction of clinical remission Maintenance of clinical remission
2 (Living)	In adult outpatients with moderate-to-severely active CD who are naïve to advanced therapies, what is the comparative efficacy of infliximab, adalimumab, certolizumab pegol, vedolizumab, ustekinumab, upadacitinib, risankizumab, mirikizumab and guselkumab, for induction and maintenance of remission?	Adult outpatients with moderate-to-severely active CD who are naïve to advanced therapies	Infliximab Adalimumab Certolizumab pegol Vedolizumab Ustekinumab Upadacitinib Risankizumab Mirikizumab Guselkumab	Placebo or another active comparator	Induction of clinical remission Maintenance of clinical remission
3 (Living)	In adult outpatients with moderate-to-severely active CD who have been exposed to advanced therapies, what is the comparative efficacy of infliximab, adalimumab, certolizumab pegol, vedolizumab, ustekinumab, upadacitinib, risankizumab, mirikizumab and guselkumab for induction and maintenance of remission?	Adult outpatients with moderate-to-severely active CD who have been exposed to advanced therapies	Infliximab Adalimumab Certolizumab pegol Vedolizumab Ustekinumab Upadacitinib Risankizumab Mirikizumab Guselkumab	Placebo or another active comparator	Induction of clinical remission Maintenance of clinical remission
4	In adult outpatients with moderate-to-severely active CD, what is the efficacy of immunomodulator monotherapy (thiopurines, methotrexate) for induction and maintenance of remission?	Adult outpatients with moderate-to-severely active CD	Thiopurines (azathioprine, mercaptopurine) Methotrexate (oral or subcutaneous)	Placebo (or 5-aminosalicylates [5-ASA])	Achieving remission Prevention of relapse (≈ maintenance of remission)
5 (Living)	In adult outpatients with moderate-to-severely active CD, is combination therapy of TNF antagonists with an immunomodulator (thiopurines or methotrexate) superior to TNF monotherapy or immunomodulator monotherapy for induction and maintenance of remission?	Adult outpatients with moderate-to-severely active CD	Combination therapy with a TNF antagonist and an immunomodulator (thiopurines or methotrexate)	TNF-antagonist monotherapy Immunomodulator monotherapy (thiopurines or methotrexate)	Induction of clinical remission Maintenance of clinical remission

Table 4. Continued

Question no. ^a	Focused question	Patients	Intervention	Comparator	Critical outcomes
6 (Living)	In adult outpatients with moderate-to-severely active CD, is combination therapy of a non-TNF biologic with an immunomodulator (thiopurines or methotrexate) superior to TNF monotherapy or immunomodulator monotherapy for induction and maintenance of remission?	Adult outpatients with moderate-to-severely active CD	Combination therapy with a non-TNF-antagonist biologic and an immunomodulator (thiopurines or methotrexate)	Non-TNF-antagonist biologic monotherapy Immunomodulator monotherapy (thiopurines or methotrexate)	Induction of clinical remission Maintenance of clinical remission
7 (Living)	In adult outpatients with moderate-to-severely active CD in steroid-free remission on combination therapy of TNF antagonist plus immunomodulator, is discontinuation of (a) an immunomodulator or (b) discontinuation of a TNF antagonist, inferior to continuation of combination therapy?	Adult outpatients with moderate-to-severely active CD in steroid-free remission on combination therapy	Discontinuation of an immunomodulator Discontinuation of a biologic	Continuation of combination therapy	Prevention of relapse
8	In adult outpatients with moderate-to-severely active CD, is upfront use of advanced therapies superior to step-up therapy for decreasing the risk of disease-related complications?	Adult outpatients with moderate-to-severely active CD	Top-down therapy Upfront use of advanced therapies with or without immunomodulator therapy	Step therapy Acceleration to advanced therapy only after failure of immunomodulators	Disease-related complications Maintenance of remission
9	In adult outpatients with moderate-to-severely active CD, is treat-to-target of endoscopic remission (resolution of inflammation on endoscopy) superior to treat-to-target of symptomatic remission, for maintenance of remission and decreasing risk of disease-related complications?	Adult outpatients with moderate-to-severely active CD	Treat-to-target of endoscopic remission Systematic assessment for symptoms and endoscopic inflammation, followed by treatment escalation in those with evidence of inflammation, regardless of presence or absence of symptoms	Treat-to-target of symptomatic remission Systematic assessment for symptoms, followed by treatment escalation in those with ongoing symptoms	Disease-related complications Maintenance of remission

^aQuestions in the “living” mode will be reviewed every 6 months for new evidence. Evidence synthesis will be updated when new phase 3 or phase 4 data of a relevant intervention or new agents become publicly available, there is change in regulatory guidance, or there are large studies suggesting meaningful change in safety of existing therapies or treatment strategies.

consistent with the FDA-approved dosing regimen, and when phase 3 data were also available.

Given the typically inadequate sample size and follow-up time in RCTs in relation to serious but infrequent adverse events, such as serious infection or neoplasm, observational studies, systematic reviews, and meta-analyses were considered when evaluating risks of therapies. Observational studies were not included in assessments of efficacy. A consensus was reached on study inclusion. Any disagreements were resolved with adjudication by the guideline panel chair and senior methodologist.

Study characteristics, including year of publication, study site, study population, intervention, comparison group, outcomes, and methods for risk-of-bias assessment were extracted from each study. Meta-analyses were conducted when more than 1 study contributed data for the same intervention and outcome. Dichotomous outcomes were combined to obtain a relative risk (RR) or odds ratio (OR) and 95% confidence interval (CI). These effect estimates were calculated using the Mantel-Haenszel fixed-effects model (in the absence of conceptual heterogeneity and if fewer than 5 studies) or the DerSimonian-Laird random-effects model.¹⁶

Pooled RR estimates were derived from clinical trials for each individual therapy. These risk estimates are influenced by both the efficacy of individual therapies as well as the placebo rates in each of these trials. RR estimates were then applied to a standardized, pooled placebo rate derived from placebo rates across all phase 2 and 3 RCTs in moderate-to-severely active CD. The absolute risk difference derived from RR estimates standardized by the pooled placebo rate was then used to inform the strength of evidence for efficacy for each treatment. Use of a pooled standardized placebo response rate allows for a common reference against which to estimate the predicted benefit under the assumption that RRs and ORs are translatable across studies. This choice of pooled placebo rates is consistent with prior AGA guidelines for management of ulcerative colitis and CD and recommendations from GRADE.^{6,15} For RCTs assessing induction, the pooled placebo rate of induction of clinical remission was set at 15%, and for trials assessing maintenance of remission, a rate of 22% was used.¹⁷

Statistical heterogeneity was assessed using the I^2 statistic.¹⁸ Small study effects were examined using funnel plot symmetry and Egger's regression test, although it is important to recognize that these tests are unreliable when the number of studies is fewer than 10.¹⁹ Direct comparisons were performed using Comprehensive Meta-Analysis software, version 2.0 (Biostat, Inc). Due to a paucity of head-to-head trials of active agents, to inform comparative efficacy of different pharmacologic interventions we performed an NMA using the frequentist approach, with the statistical package "netmeta" (version 9.0, <https://cran.r-project.org/web/packages/netmeta/index.html>) in R (version 4.0.2). Details of the NMA are reported in the accompanying 2025 AGA Evidence Synthesis document on comparative efficacy of different advanced therapies for management of moderate-to-severely active CD published in *Gastroenterology*.¹⁰

Certainty of the Evidence

The guideline panel used the GRADE approach to assess the certainty of evidence for the effect of the intervention on each outcome using the GradePro Guideline Development Tool software (<https://grade.pro.org>). The GRADE approach considers factors such as study design, population studied, risk of

bias, inconsistency, indirectness, imprecision, and risk of publication bias to rate the certainty of evidence as high, moderate, low, or very low (Supplementary Table 3).

For questions of comparative efficacy of different pharmacologic interventions, we used a standardized GRADE approach for NMA.^{20,21} The guideline panel set a clinically meaningful difference (CMD) threshold of 10% for comparisons with no treatment, based on consensus and in line with thresholds used in prior guidelines for moderate-to-severe ulcerative colitis.¹⁵ If the effect size was below this CMD threshold, then the benefit was deemed to be trivial to small. These thresholds are similar to thresholds for small effect size in a Delphi consensus for guideline development and clinical decision making by Gordon et al.²² Similarly, for comparisons between 2 active therapies in meta-analyses, the guideline panel set a CMD threshold of 5%, based on consensus, that is, we considered the difference between an active agent vs comparator as "important" if the absolute risk difference of achieving remission crossed the CMD threshold of >50 per 1000 patients treated (5%). In using NMA for evidence synthesis, we relied on direct evidence when it was available from head-to-head comparisons and provided at least moderate certainty evidence. If there were no direct comparisons between 2 interventions or if the evidence from direct comparisons was very low or low certainty evidence, then effect estimates from the NMA were used.

Development of Recommendations

Evidence was translated into guideline recommendations following the GRADE Evidence-to-Decision framework via discussion among guideline panel members over several virtual meetings.²³ The Evidence-to-Decision framework considers the certainty of evidence, balance of benefits and harms, patient values and preferences, feasibility, acceptability, equity, and resource use. Evidence profiles and evidence-to-decision tables are presented in Tables 5–19. Group consensus was reached for all of the recommendations.

The interpretation of strength of recommendations is summarized in Table 3. The certainty of evidence and the strength of recommendation are provided for each clinical question. As per GRADE methodology, recommendations are labeled as "strong" or "conditional." The words "we recommend" indicate strong recommendations and "we suggest" indicate conditional recommendations.

To aide clinicians in clinical decision making regarding appropriate therapy selection when considering treatment positioning stratified by prior exposures, and in the context of potential intra-class differences among specific therapies in inflammatory bowel disease (IBD), the panel rated each therapy individually as opposed to grouping by class. Separate groupings were then created when considering treatment selection in those who were advanced therapy-naïve and those who had previously received 1 or more advanced therapies. For each group, *a priori* criteria were defined for group inclusion.

When considering advanced therapy-naïve individuals, 2 groups were created: (1) higher efficacy and (2) lower efficacy. Classification criteria for higher efficacy were defined *a priori* and included:

1. An absolute difference in efficacy in comparison with lower efficacy therapies of 5% (CMD $\geq 5\%$)

2. A *P* score in the NMA of 0.49–1.0 in comparison with placebo
3. An absolute difference vs placebo in phase 3 clinical trials for that agent of $\geq 15\%$

Inclusion in the higher efficacy group required meeting these criteria.

When considering individuals who are advanced therapy-exposed, 3 therapeutic groups were constructed, including higher efficacy, intermediate efficacy, and lower efficacy. Inclusion in the higher efficacy group required meeting all 3 criteria below and intermediate efficacy therapies met some but not all 3 criteria. These criteria included:

1. An absolute difference in efficacy in comparison with lower efficacy therapies of at least 5%
2. A *P* score in the NMA of 0.49–1.0 in comparison with placebo
3. An absolute difference in phase 3 clinical trials for that agent of $\geq 10\%$ against placebo

A smaller absolute difference was felt to be more appropriate for advanced therapy-exposed individuals than for advanced therapy-naïve individuals.

Recommendations

General Implementation Considerations

There are several overarching considerations for implementing these recommendations in clinical practice (Table 2). First, as noted above, it is important to consider appropriate categorization of disease activity when considering advanced therapy initiation and that treatment decisions are made based on confirmed active inflammation. Concomitant functional gastrointestinal symptoms may be seen in up to one-third of patients with CD.²⁴ Therefore, before considering an advanced therapy, the presence of inflammation should be confirmed via objective measures of biochemical inflammation (C-reactive protein or fecal calprotectin) or with structural assessment via ileocolonoscopy or imaging studies (computed tomography, magnetic resonance enterography, or intestinal ultrasound, dependent on center availability and expertise).¹²

It is advised to assess for comorbidities that may increase the risk of advanced therapies. This includes screening for hepatitis B and tuberculosis, given the risk of reactivation with exposure to advanced therapies. In addition, it is appropriate to review vaccination status for pneumococcus, influenza, COVID-19, herpes zoster, respiratory syncytial virus, and human papillomavirus, and ensure these are up to date, consistent with general health care maintenance guidelines in IBD.²⁵ Baseline laboratory assessments, concordant with medication label, should also be performed. This would include assessing complete blood counts and liver function tests before using an immunomodulator or documenting lipid levels before therapy initiation with a JAK inhibitor, and repeating these levels after starting therapy. Lastly, comorbidities that may be exacerbated by specific classes of therapies should be considered when selecting an advanced therapy. For example, TNF antagonists

may be contraindicated in the setting of advanced congestive heart failure or multiple sclerosis, while they and other advanced therapies may be associated with an increased risk of lymphoma and nonmelanoma and melanoma skin cancer.^{26–28} When prescribing JAK inhibitors, additional risks of venous thromboembolic events, cardiovascular events, and stroke should be considered.²⁹

Once an advanced therapy is initiated, disease activity monitoring to ensure treatment outcomes are achieved should continue. International societal guidance documents, such as STRIDE and STRIDE-II, recommended periodic assessment of clinical response, as well as repeat biochemical assessment (C-reactive protein or fecal calprotectin) after initiating an advanced therapy.^{12,30,31} Repeat structural assessment (ie, ileocolonoscopy or enterography, dependent on disease location) is generally considered appropriate, although there are limited real-world data to date confirming that this improves clinical outcomes over time.

Use of Advanced Therapies

Question 1: What is the efficacy of advanced therapies for induction and maintenance of remission in moderate-to-severely active luminal CD?

Recommendation 1: In adult outpatients with moderate-to-severely active CD, the AGA *recommends* the use of infliximab, adalimumab, ustekinumab, risankizumab, mirikizumab, guselkumab, or upadacitinib,^a over no treatment. (*Strong recommendation, moderate to high certainty of evidence*)

Recommendation 2: In adult outpatients with moderate-to-severely active CD, the AGA *suggests* the use of certolizumab pegol or vedolizumab, over no treatment. (*Conditional recommendation, moderate certainty of evidence*)

Implementation Considerations

- Biosimilars of infliximab, adalimumab, and ustekinumab can be considered equivalent to their originator drug in their efficacy in terms of therapy selection.
- Subcutaneous formulations of infliximab and vedolizumab have shown comparable efficacy to their respective FDA-approved intravenous maintenance doses.
- In some patients with partial response to FDA-approved dosing, particularly those with more severe disease, extended induction regimens or dose escalation may be beneficial for most advanced therapies.
- There are 2 dosing options available for maintenance therapy for risankizumab, guselkumab, and upadacitinib. Higher maintenance doses may be preferred in patients with a high burden of inflammation and/or more severe disease, and those who have previously failed TNF antagonists.

^aIn the United States, the FDA recommends reserving use of JAK inhibitors in patients with failure or intolerance to TNF-antagonist therapy.

Source of Evidence

The data examining the efficacy of advanced therapies in comparison with no treatment was derived from 45 placebo-controlled RCTs for moderate-to-severely active CD. This included data from 15 trials of TNF antagonists (5 infliximab,^{32–36} 7 adalimumab,^{37–43} and 3 certolizumab pegol^{44–46}), 4 trials of vedolizumab,^{47–50} 12 trials from ustekinumab,^{51–56} 10 trials from IL23p19 antagonists (2 mirikizumab,^{56,57} 4 guselkumab,^{54,55,58} and 4 risankizumab^{59–61}), and 4 trials for upadacitinib.^{62,63} All trials were conducted in patients with moderate-to-severely active CD with placebo-based controls. Patient populations were largely similar across trials in terms of patient and disease characteristics, although it is important to note that later trials had higher percentages of individuals with prior advanced therapy exposures, in comparison with early trials, such as those for TNF antagonists. The majority of trials provided information regarding stratification of prior advanced therapy exposure. Biosimilar data and data related to reformulations for alternative modes of delivery were included when appropriate. Efficacy data for the primary outcomes of induction and maintenance of clinical remission are presented by study and with summary estimates for each therapy in [Supplementary Figures 1–9](#). GRADE evidence profiles for each agent are presented in [Tables 5–8](#).

Benefits

On meta-analysis, infliximab, adalimumab, ustekinumab, risankizumab, guselkumab, and upadacitinib met the predefined threshold of a clinically meaningful benefit (CMD) >10% for at least moderate benefit, with the magnitude of benefit in inducing remission over placebo ranging from 119 to 353 per 1000 patients treated. The magnitude of benefit with certolizumab pegol and vedolizumab did not meet the *a priori* defined threshold of a CMD >10% for inducing remission, at 48 and 90 per 1000 patients treated, respectively, suggesting a trivial-to-small benefit. The pivotal phase 3 RCT of mirikizumab, VIVID-1, was designed as a treat-straight-through trial in which patients were randomized to mirikizumab, ustekinumab, or placebo, and treated through for 52 weeks. With mirikizumab, the magnitude of benefit for inducing remission at 12 weeks, compared with placebo, was 92 per 1000 patients; however, with continued follow-up in a treat-straight-through design, the magnitude of benefit over placebo in achieving remission at week 52 was 387 per 1000 patients. Hence, the guideline panel deemed that the overall certainty of evidence for benefit with mirikizumab is moderate.

Harms

Adverse events considered associated with advanced therapies include serious infection and malignancies. Although the incidence of these events in the included clinical trials was low and often not significantly increased relative to placebo, it is important to note that RCTs are often underpowered for detection of adverse events and

have relatively short follow-up times. Observational data can be used to better inform these risks. In a systematic review and meta-analysis of 20 head-to-head studies comparing risk of infections between different advanced therapies for treatment of IBD, in patients with CD, ustekinumab was associated with 51% lower risk of serious infections compared with TNF antagonists, and 60% lower risk compared with vedolizumab.⁶⁴ There was no difference in the risk of serious infections between TNF antagonists and vedolizumab. There have been limited comparative safety studies of JAK inhibitors and IL23 p19 antagonists. TNF antagonists have also been associated with increased risk of lymphoma and melanoma. In a French population-based study, the estimated annual incidence of lymphoma in patients treated with TNF-antagonist monotherapy vs unexposed patients was 0.41 per 1000 person-years vs 0.26 per 1000 person-years; after adjusting for potential confounders, the risk of lymphoma was 2.4-times higher in patients treated with TNF-antagonist monotherapy.²⁸ This risk was comparable to the risk observed in patients treated with thiopurine monotherapy. The FDA has issued a black box warning on the increased risk of malignancy with TNF antagonists.⁶⁵ Currently, there is a paucity of population-representative data to inform risk estimates related to malignancy for other classes of advanced therapies.

When considering these adverse events, it is also important to balance them with the potential deleterious consequence of inadequately treating patients with moderate-to-severely active CD. Effective therapies to control disease may lower the risk of serious infections.^{66–68} In addition, active CD is associated with reduced quality of life, decreased functional status, increased risk of corticosteroid exposure, increased risk of hospitalization, or surgery.^{5,69,70} These risks of poorly controlled CD were felt to exceed risks that may be attributable to advanced therapies in CD.

Certainty in Evidence of Effects

Overall, ustekinumab, guselkumab, and mirikizumab had high certainty of evidence for induction, and infliximab, adalimumab, risankizumab, and upadacitinib had moderate certainty of evidence, with evidence being rated down for imprecision due to low number of events (optimal information size). Infliximab and guselkumab had high certainty of evidence for maintenance of remission, whereas adalimumab, certolizumab, vedolizumab, risankizumab, mirikizumab, and upadacitinib had moderate certainty of evidence. Full evidence profiles for each therapy in comparison with placebo, as well as justifications for rating of the evidence, are presented in [Tables 5–8](#). The GRADE Evidence-to-Decision judgments for all advanced therapies are presented in [Table 9](#).

Discussion

When interpreting these recommendations, there are several important factors to consider. Although the evidence synthesis used to inform these recommendations

Table 5. Grading of Recommendations Assessment, Development and Evaluation Evidence Profile Comparing Tumor Necrosis Factor Antagonists (Infliximab, Adalimumab, and Certolizumab Pegol) With Placebo for Induction and Maintenance of Remission in Patients With Moderate-to-Severely Active Crohn's Disease

Outcomes	Study event rates, n/N (%)		Relative effect, RR (95% CI)	Absolute effect ^a	No. of participants (studies)	Quality of the evidence (GRADE) ^b
	Risk with placebo	Risk with infliximab				
Infliximab compared with placebo for moderate-to-severely active CD						
Induction of clinical remission (CRITICAL)	11/54 (20.4)	29/52 (55.8)	2.16 (1.24–3.77)	174 more per 1000 (from 36 more to 416 more)	106 (2 RCTs)	⊕⊕⊕○ ^c MODERATE
Maintenance of clinical remission (CRITICAL)	66/258 (25.6)	208/381 (54.6)	1.99 (1.58–2.49)	218 more per 1000 (from 128 more to 328 more)	639 (3 RCTs)	⊕⊕⊕⊕ HIGH
Outcomes	Study event rates, n/N (%)		Relative effect, RR (95% CI)	Absolute effect ^a	No. of participants (studies)	Quality of the evidence (GRADE) ^b
	Risk with placebo	Risk with adalimumab				
Adalimumab compared with placebo for moderate-to-severely active CD						
Induction of clinical remission (CRITICAL)	31/366 (8.5)	110/370 (29.7)	3.35 (2.31–4.86)	353 more per 1000 (from 197 more to 579 more)	736 (5 RCTs)	⊕⊕⊕○ ^d MODERATE
Maintenance of clinical remission (CRITICAL)	36/275 (14.3)	106/276 (40.1)	2.67 (1.94–3.68)	367 more per 1000 (from 207 more to 590 more)	551 (4 RCTs)	⊕⊕⊕○ ^d MODERATE
Outcomes	Study event rates, n/N (%)		Relative effect, RR (95% CI)	Absolute effect ^a	No. of participants (studies)	Quality of the evidence (GRADE) ^b
	Risk with placebo	Risk with infliximab				
Certolizumab pegol compared with placebo for moderate-to-severely active CD						
Induction of clinical remission (CRITICAL)	119/608 (19.6)	161/616 (26.1)	1.32 (1.07–1.62)	48 more per 1000 (from 11 more to 93 more)	1224 (3 RCTs)	⊕⊕⊕○ ^e MODERATE

Table 5. Continued

Outcomes	Study event rates, n/N (%)		Relative effect, RR (95% CI)	Absolute effect ^a	No. of participants (studies)	Quality of the evidence (GRADE) ^b
	Risk with placebo	Risk with infliximab				
Maintenance of clinical remission (CRITICAL)	61/210 (29.0)	103/215 (47.9)	1.65 (1.28–2.12)	143 more per 1000 (from 62 more to 246 more)	425 (1 RCT)	⊕⊕⊕⊙ ^f MODERATE

^aAbsolute effects estimated based on the following placebo rates across trials: induction of clinical remission 15%; maintenance of clinical remission 22%.

^bGRADE Working Group grades of evidence. High quality: We are very confident that the true effect lies close to that of the estimate of the effect. Moderate quality: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low quality: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect. Very low quality: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect.

^cLower limit of 95% of absolute effect crosses the CMD of drug over placebo (10%).

^dRated down for imprecision because optimal information size not met (<200 events).

^eRated down for serious imprecision because magnitude of benefit is below the 100 per 1000 absolute benefit rate of CMD threshold over placebo, identified by the guideline panel.

^fLower limit of 95% of absolute effect crosses the CMD of drug over placebo.

Table 6. Grading of Recommendations Assessment, Development and Evaluation Evidence Profile Comparing Vedolizumab With Placebo for Induction and Maintenance of Remission in Patients With Moderate-to-Severely Active Crohn's Disease

Outcomes	Study event rates, n/N (%)		Relative effect, RR (95% CI)	Absolute effect ^a	No. of participants (studies)	Quality of the evidence (GRADE) ^b
	Risk with placebo	Risk with vedolizumab				
Vedolizumab compared with placebo for moderate-to-severely active CD						
Induction of clinical remission (CRITICAL)	45/433 (10.4)	83/508 (16.3)	1.60 (1.13–2.25)	90 more per 1000 (from 19 more to 188 more)	941 (3 RCTs)	⊕⊕⊕○ ^c MODERATE
Maintenance of clinical remission (CRITICAL)	81/299 (18.3)	253/595 (45.2)	1.54 (1.25–1.89)	119 more per 1000 (from 55 more to 196 more)	894 (3 RCTs)	⊕⊕⊕○ ^d MODERATE

^aAbsolute effects estimated based on the following placebo rates across trials: induction of clinical remission 15%; maintenance of clinical remission 22%.

^bGRADE Working Group grades of evidence. High quality: We are very confident that the true effect lies close to that of the estimate of the effect. Moderate quality: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low quality: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect. Very low quality: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect.

^cRated down for serious imprecision because magnitude of benefit is below the 100 per 1000 absolute benefit rate of CMD threshold over placebo, identified by the guideline panel.

^dLower limit of 95% of absolute effect crosses the CMD of drug over placebo for clinical remission outcome.

Table 7. Grading of Recommendations Assessment, Development and Evaluation Evidence Profile Comparing Interleukin 12/23 Antagonist (Ustekinumab) and Interleukin 23 Antagonists (Mirikizumab, Risankizumab, Guselkumab) With Placebo for Induction and Maintenance of Remission in Patients With Moderate-to-Severely Active Crohn's Disease

	Study event rates, n/N (%)					
Outcomes	Risk with placebo	Risk with ustekinumab	Relative effect, RR (95% CI)	Absolute effect ^a	No. of participants (studies)	Quality of the evidence (GRADE) ^b
Ustekinumab compared with placebo for moderate-to-severely active CD						
Induction of clinical remission (CRITICAL)	69/589 (11.7)	148/589 (25.1)	2.12 (1.63–2.74)	224 more per 1000 (from 126 more to 348 more)	1178 (2 RCTs)	⊕⊕⊕⊕ HIGH
Maintenance of clinical remission (CRITICAL)	67/204 (18.3)	99/200 (45.2)	1.51 (1.19–1.91)	128 more per 1000 (from 47 more to 227 more)	404 (2 RCTs)	⊕⊕⊕○ ^c MODERATE
	Study event rates, n/N (%)					
Outcomes	Risk with placebo	Risk with mirikizumab	Relative effect, RR (95% CI)	Absolute effect ^a	No. of participants (studies)	Quality of the evidence (GRADE) ^b
Risankizumab compared with placebo for moderate-to-severely active CD						
Induction of clinical remission (CRITICAL)	86/401 (21.4)	247/568 (43.5)	1.98 (1.60–2.44)	147 more per 1000 (from 90 more to 216 more)	969 (3 RCTs)	⊕⊕⊕○ ^d MODERATE
Maintenance of clinical remission (CRITICAL)	67/164 (40.8)	180 mg: 87/157 (55.4) 360 mg: 74/141 (52.5)	180 mg:1.36 (1.08–1.71) 360 mg:1.28 (1.01–1.64)	180 mg: 79 more per 1000 (from 18 more to 156 more) 360 mg: 62 more per 1000 (from 2 more to 141 more)	462 (1 RCT)	180 mg: ⊕⊕⊕○ ^e MODERATE 360 mg: ⊕⊕⊕○ ^e MODERATE
	Study event rates, n/N (%)					
Outcomes	Risk with placebo	Risk with risankizumab	Relative effect, RR (95% CI)	Absolute effect ^a	No. of participants (studies)	Quality of the evidence (GRADE) ^b
Mirikizumab compared with placebo for moderate-to-severely active CD						
Induction of clinical remission (CRITICAL)	56/263 (21.3)	248/675 (36.7)	1.61 (1.26–2.07)	92 more per 1000 (from 39 more to 160 more)	938 (2 RCTs)	⊕⊕⊕○ ^f MODERATE

Table 7. Continued

Outcomes	Study event rates, n/N (%)		Relative effect, RR (95% CI)	Absolute effect ^a	No. of participants (studies)	Quality of the evidence (GRADE) ^b
	Risk with placebo	Risk with risankizumab				
Maintenance of clinical remission (CRITICAL)	39/199 (19.6)	313/579 (54.1)	2.76 (2.06–3.69)	387 more per 1000 (from 233 more to 592 more)	778 (1 RCT)	⊕⊕⊕⊕ HIGH

Outcomes	Study event rates, n/N (%)		Relative effect, RR (95% CI)	Absolute effect ^a	No. of participants (studies)	Quality of the evidence (GRADE) ^b
	Risk with placebo	Risk with guselkumab				
Guselkumab compared with placebo for moderate-to-severely active CD						
Induction of clinical remission (CRITICAL)	38/209 (18.2)	309/643 (48.1)	2.66 (1.97–3.58)	249 more per 1000 (from 146 more to 387 more)	852 (3 RCTs)	⊕⊕⊕⊕ HIGH
Maintenance of clinical remission (CRITICAL)	18/148 (12.2)	289/582 (49.7)	4.08 (2.62–6.34)	678 more per 1000 (from 356 more to 1000 more)	730 (2 RCTs)	⊕⊕⊕⊕ HIGH

^aAbsolute effects estimated based on the following placebo rates across trials: induction of clinical remission 15%; maintenance of clinical remission 22%.

^bGRADE Working Group grades of evidence. High quality: We are very confident that the true effect lies close to that of the estimate of the effect. Moderate quality: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low quality: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect. Very low quality: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect.

^cLower limit of 95% of absolute effect crosses the CMD of drug over placebo for clinical remission outcome.

^dLower limit of 95% of absolute effect crosses the CMD of drug over placebo for induction of clinical remission outcome.

^eRated down for serious imprecision because magnitude of benefit is below the 100 per 1000 absolute benefit rate of minimal clinically important difference threshold over placebo, identified by the guideline panel.

^fRated down for serious imprecision because magnitude of benefit is below the 100 per 1000 absolute benefit rate of minimal clinically important difference threshold over placebo, identified by the guideline panel.

Table 8. Grading of Recommendations Assessment, Development and Evaluation Evidence Profile Comparing Upadacitinib With Placebo for Induction and Maintenance of Remission in Patients With Moderate-to-Severely Active Crohn's Disease

	Study event rates, n/N (%)					
Outcomes	Risk with placebo	Risk with upadacitinib	Relative effect, RR (95% CI)	Absolute effect ^a	No. of participants (studies)	Quality of the evidence (GRADE) ^b
Upadacitinib compared with placebo for moderate-to-severely active CD						
Induction of clinical remission (CRITICAL)	93/384 (24.2)	313/710 (44.1)	1.79 (1.47–2.17)	119 more per 1000 (from 71 more to 176 more)	1094 (3 RCTs)	⊕⊕⊕○ ^c MODERATE
Maintenance of clinical remission (CRITICAL)	25/165 (15.2)	15 mg: 63/169 (37.3) 30 mg: 80/168 (47.6)	15 mg: 2.46 (1.63–3.71) 30 mg: 3.14 (2.12–4.66)	15 mg: 321 more per 1000 (from 139 more to 596 more) 30 mg: 471 more per 1000 (from 246 more to 805 more)	502 (1 RCT)	15 mg: ⊕⊕⊕○ ^d MODERATE 30 mg: ⊕⊕⊕○ ^d MODERATE

^aAbsolute effects estimated based on the following placebo rates across trials: induction of clinical remission 15%; maintenance of clinical remission 22%.

^bGRADE Working Group grades of evidence. High quality: We are very confident that the true effect lies close to that of the estimate of the effect. Moderate quality: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low quality: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect. Very low quality: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect.

^cLower limit of 95% of absolute effect crosses the CMD of drug over placebo for induction of clinical remission outcome.

^dRated down for imprecision because optimal information size not met (<200 events).

Table 9. Grading of Recommendations Assessment, Development and Evaluation Evidence-to-Decision Framework for Use of Advanced Therapies Over No Intervention for the Management of Patients With Moderate-to-Severely Active Crohn's Disease

Domain	Criteria	Judgment					
Problem	Is the problem a priority?	No	Probably No	Probably yes	Yes	Varies	Do not know
Desirable effects	How substantial are the desirable anticipated effects?	Trivial to small (<10%) CZP, VDZ		Moderate (10%–30%) IFX, UST, RIS, MIRI, GUS, UPA	Large (>30%) ADA	Varies	Do not know
Undesirable effects	How substantial are the undesirable anticipated effects?	Trivial IFX, ADA, CZP, VDZ, UST, RIS, GUS, MIRI, UPA	Small	Moderate	Large	Varies	Do not know
Certainty of evidence	What is the overall certainty of the evidence of effects	Very low		Low	Moderate IFX, ADA, CZP, VDZ, RIS, MIRI, UPA	High UST, GUS	No included studies
Values	Is there important uncertainty about or variability in how much people value the main outcomes?	Important uncertainty or variability		Possibly important uncertainty or variability	Probably no important uncertainty or variability		No important uncertainty or variability
Balance of effects	Does the balance between desirable and undesirable effects favor the intervention or the comparison?	Favors comparison/control	Probably favors comparison	Does not favor either comparison or intervention	Probably favors intervention CZP, VDZ	Favors intervention IFX, ADA, UST, RIS, MIRI, GUS, UPA	Varies/do not know
Resource use	Is the incremental cost small relative to the net benefits?	No	Probably No	Uncertain	Probably yes	Yes	Uncertain
Equity	What would be the impact on health inequities?	Increased	Probably increased	Probably no impact	Probably reduced	Reduced	Varies/ Do not know
Acceptability	Is the option acceptable to key stakeholders?	No	Probably no	Probably yes	Yes	Varies	Do not know
Feasibility	Is the option feasible to implement?	No	Probably no	Probably yes	Yes	Varies	Do not know

NOTE. CMD threshold of drug over placebo was set at 10%. Judgments made by guideline panel are in bold type.

ADA, adalimumab; CZP, certolizumab pegol; GUS, guselkumab; IFX, infliximab; MIRI, mirikizumab; RIS, risankizumab; UPA, upadacitinib; UST, ustekinumab; VDZ, vedolizumab.

used the available data from phase 2 and 3 clinical trials, heterogeneity across trials, including variability in patient populations, prior and concurrent medication use, and timing of clinical remission end points, persists. Phase 2 clinical trial data were only incorporated when co-existing phase 3 data were present for an FDA-approved therapy with similar dosing and clinical remission endpoints. As noted in [Supplementary Figures 1–9](#), effect estimates were similar for phase 2 and phase 3 data across studies, consistent with prior meta-analyses.⁷¹ Prior exposures, in relation to advanced therapies and concurrent immunomodulator use, are addressed in subsequent recommendations in this guideline.

When considering implementing these recommendations, there are several important factors to consider. As noted above, therapy selection and initiation should be conducted with confirming active disease and performing appropriate baseline laboratory evaluation. Medication selection should also incorporate a patient's baseline preferences regarding risk to benefit balance, as well as mode of delivery (ie, intravenous vs subcutaneous vs oral therapy). Such decisions should also integrate potential interactions with competing comorbidities or extraintestinal manifestations. In addition, choice of therapy may be influenced by the prior therapies the patient has tried, the initial response, and reasons for discontinuation. These issues are addressed further in Questions 2 and 3. Lastly, impending life events, such as pregnancy planning in the near future, should be considered.

Infliximab, adalimumab, and ustekinumab biosimilars have been approved by the FDA for CD. Several studies have demonstrated the efficacy and safety of these biosimilars in CD, and switching from an originator to a biosimilar has not been found to be associated with an increased risk of subsequent clinical, biochemical, or endoscopic relapse.⁷² Furthermore, originator biologics and biosimilars can be used interchangeably, as they possess similar efficacy and safety profiles. The preference of the panel was that individuals initiate the first available therapy when considering originator compounds vs biosimilars as opposed to delaying care.

Subcutaneous formulations of infliximab and vedolizumab are FDA-approved in CD. They have comparable efficacy to their respective intravenous formulations and can be used interchangeably with intravenous formulations after induction for most patients. In the LIBERTY trial, individuals with moderate-to-severely active CD received open-label 5 mg/kg induction with CT-P13, an infliximab biosimilar, at weeks 0, 2, and 6, and responders were then randomized to subcutaneous CT-P13 every 2 weeks or placebo, with 62.3% achieving clinical remission in the treatment arm compared with 32.1% in the control group at 54 weeks.³⁵ In the VISIBLE 2 trial, participants with moderate-to-severely active CD received open-label induction with intravenous vedolizumab at weeks 0 and 2, and responders were then randomized to every other week subcutaneous vedolizumab vs placebo. Those receiving vedolizumab were significantly more likely to be in remission at 1 year in comparison with placebo (43.2% vs

20.8%).⁵⁰ However, there are limited data regarding transitioning from intravenous to subcutaneous therapies in those who are already receiving dose-escalated regimens of infliximab or vedolizumab.

Emerging data suggest that in some patients with suboptimal response to standard induction therapy, particularly those with severe disease, extended induction therapy may be beneficial. Observational cohorts have demonstrated the potential benefit of extended induction periods, particularly when considering JAK inhibitors or IL23 p19 antagonists, in those with more severe disease or who do not tolerate transition to maintenance dosing.^{73–75} Beyond induction, some patients may also derive benefit from increased dosing during maintenance therapy as well. In the U-EXCEED trial of maintenance upadacitinib, rates of maintenance of remission were higher in those receiving 30 mg daily (47.6%) than those receiving 15 mg daily (37.3%) at 52 weeks.⁶² However, one cannot extrapolate these findings to all therapies. Data on the potential clinical benefit of higher maintenance dosing for guselkumab may also demonstrate benefit.⁷⁶ Further real-world evidence assessing dose-escalation strategies is needed for more recently approved agents. It is important to recognize that higher dosing of these agents may also modify their safety profile. Although increased trough levels of TNF antagonists have not been associated with increased SAEs over time, clinical trials of JAK inhibitors have suggested that there may be higher rates of infectious complications at the higher dose with this class.⁶²

Evidence Gaps and Future Research

Several knowledge gaps and opportunities for future research currently exist. Clinical trials emphasizing potential effect modifiers on clinical and endoscopic response and remission rates will assist clinicians in appropriate medication selection. Examples of such factors include stratification by disease location, severity, or behavior (inflammatory vs penetrating vs stricturing), race and ethnicity, and the efficacy and safety in older individuals with CD. In addition, long-term comparative safety data for recently FDA-approved therapy classes, such as JAK inhibitors and IL23 p19 antagonists, against older agents will better inform clinical decision making.

In addition, there is a significant knowledge gap regarding how these therapies may be most effectively combined with each other in severe CD. Several small observational studies of various combinations of biologics and/or small molecules have demonstrated potential therapeutic benefit and limited adverse events in comparison with advanced monotherapy. For example, in a recent meta-analysis of 13 observational studies comprising 266 total patients with IBD, of whom 7% were diagnosed with CD, clinical remission rates to combination advanced therapy ranged from 40.4% to 76.5%, and SAE rates ranged from 0% to 12.3%.⁷⁷ Dedicated RCTs are ongoing to better understand how to best maximize additive or synergistic effects of therapies across classes.⁷⁸

Positioning of Advanced Therapies in Individuals Who Are Naïve to Advanced Therapies

Question 2: In adult outpatients with moderate-to-severely active CD who are naïve to advanced therapies, what is the comparative efficacy of infliximab, adalimumab, certolizumab pegol, vedolizumab, ustekinumab, upadacitinib, risankizumab, mirikizumab and guselkumab, for induction and maintenance of remission?

Recommendation 3: In adult outpatients with moderate-to-severely active CD who are naïve to advanced therapies, the AGA suggests using a **HIGHER** efficacy medication (infliximab, adalimumab, vedolizumab, ustekinumab, risankizumab, mirikizumab, guselkumab), rather than a **LOWER** efficacy medication (certolizumab pegol, upadacitinib). (*Conditional recommendation, low to high certainty of evidence*)

Implementation Considerations

- Individual patient factors (eg, age, comorbidities, frailty, pregnancy, adherence) and preferences (eg, route of administration, ease of access) should be incorporated within a shared decision framework in selection of advanced therapies.
- There are limited data on the safety of JAK inhibitors in pregnancy. These drugs should generally be avoided in women contemplating pregnancy in the near future.

Source of Evidence

With the expanding armamentarium of available therapies for the treatment of moderate-to-severely active CD, understanding how to position these agents among each other becomes paramount. To date, head-to-head trials have been limited in CD. To inform this guideline, the panel relied on phase 2 and 3 RCTs and conducted an NMA, stratified by prior exposure to advanced therapies. RCTs were used to inform these analyses, with an emphasis on induction of clinical remission as a patient-centric, uniformly reported outcome; for trials with treat-straight-through design, stratified analyses based on prior exposure to advanced therapies was also performed and informed recommendations. Separate NMAs for maintenance of remission were also conducted based on trial design (treat-straight-through vs re-randomization of responders). Pairwise comparisons were performed between active therapies, with an *a priori* threshold superiority of a CMD of >5%. The full results are of this NMA are available separately and summarized below.

The NMA included 8 placebo-controlled trials of TNF antagonists (2 infliximab, 4 adalimumab, and 2 certolizumab pegol),^{32,33,37,39–41,44,79} 3 trials of anti-integrins (3 vedolizumab),^{47–49} 3 trials of IL12/23 antagonists (3 ustekinumab),^{51,52} 9 trials of IL23p19 antagonists (3 risankizumab, 2 mirikizumab, and 4 guselkumab),^{54–56,58–60} and 3

trials of JAK inhibitors (3 upadacitinib).^{62,63} One head-to-head trial without placebo comparator (SEAVUE) that compared ustekinumab with adalimumab among 386 individuals was also included in the analysis.⁸⁰ In that trial, similar rates of clinical remission were observed with ustekinumab and adalimumab at 52 weeks (65% vs 61%).

Benefits

In advanced therapy-naïve individuals, most therapies demonstrated clinically important benefit compared with certolizumab pegol or upadacitinib. Adalimumab demonstrated important benefit when compared with certolizumab pegol with a high certainty of evidence, and probably important benefit when compared with upadacitinib with moderate certainty of evidence. With regards to anti-IL therapies, ustekinumab demonstrated probably important benefit compared with certolizumab pegol and upadacitinib with moderate certainty of evidence. Guselkumab demonstrated probably important benefit compared with certolizumab pegol with moderate certainty of evidence, as well as possibly important benefit compared with upadacitinib with low certainty of evidence. Risankizumab demonstrated possible important benefit compared with certolizumab with low certainty of evidence, and mirikizumab demonstrated possible important benefit compared with both certolizumab pegol and upadacitinib with low certainty of evidence. Lastly, vedolizumab demonstrated possibly important benefit compared with certolizumab pegol, with low certainty of evidence. Differences between other advanced therapies were either of very low certainty and/or were deemed to have trivial-to-small benefit (<5% risk difference).

When analyzing 4 trials with treat-straight-through designs,^{54,56,80,81} guselkumab every 8 weeks demonstrated possibly trivial benefit in comparison with adalimumab (low certainty of evidence). Guselkumab dosed at 200 mg every 4 weeks demonstrated possibly trivial benefit in comparison to 100 mg every 8 week dosing (low certainty of evidence).

Harms

In all maintenance studies assessed in the NMA, risks of serious infection between advanced therapies and placebo were comparable, and there were no significant differences when comparing advanced therapies. When examining per-trial defined SAEs, the risk of SAEs was lower in those treated with guselkumab 200 mg every 4 weeks in comparison with infliximab, vedolizumab, ustekinumab, and guselkumab 100 mg over 48 weeks. It is important to reiterate that RCTs are often underpowered and have inadequate follow-up time to fully capture important but uncommon adverse events, particularly serious infections, neoplastic complications, thromboembolic events, or drug-induced liver injury.

Certainty in Evidence of Effects

The certainty of evidence varied based on the comparative agents in the NMA.¹⁰ In general, this ranged from low to moderate certainty as described above with each

pairwise comparison. The only exception to this was the high level of certainty that adalimumab had an important level of benefit over certolizumab pegol.

Discussion

To aide clinicians in clinical decision making regarding appropriate therapy selection, and in the context of potential intra-class differences among specific therapies in IBD, the panel rated each therapy individually as opposed to grouping by class. Two groups were created: (1) higher efficacy and (2) lower efficacy. Classification criteria for higher efficacy were defined *a priori* as described in the Methods section, Development of Recommendation subsection. Based on these predefined criteria, infliximab, adalimumab, vedolizumab, ustekinumab, risankizumab, mirikizumab, and guselkumab were classified as higher efficacy medications for the induction of remission in advanced therapy-naïve individuals, as each of these therapies met these predefined criteria using the available RCT data. Conversely, upadacitinib and certolizumab were classified as lower efficacy medications. The panel considered both the relative efficacy of therapies in induction and maintenance of clinical remission, though greater emphasis was placed on effect estimates at the end of induction for categorization, consistent with prior guidelines.¹⁵ Treat-straight-through designs were also considered, which incorporate both induction and maintenance of remission.

The lower efficacy of upadacitinib in biologic-naïve patients was noteworthy, especially given a significantly higher efficacy in patients with prior exposure to TNF antagonists. Several factors may influence the results of the NMA with regards to upadacitinib positioning when stratified by prior advanced therapy exposure. For example, in phase 3 RCTs for upadacitinib, striking differences were observed in the efficacy of upadacitinib vs placebo based on disease location; while upadacitinib was highly effective in patients with isolated colonic disease, there was no significant efficacy signal in patients with ileal CD. Combined stratification by both disease location and prior therapy exposure was not possible with reported data, but recently published meta-analyses suggest similar differential efficacy comparing colonic and ileal disease for JAK inhibitors.⁸² Other recent international societal guidelines have also provided upadacitinib a conditional recommendation in CD.⁸³ Importantly, FDA labeling in the United States also recommends the use of this therapy after TNF-antagonist failure.

Although RCT data were used to inform the NMA that was used in creating these guidelines, it remains possible that changes in trial participants toward those with more advanced disease or multiple therapy failures over time could differentially influence positioning of therapies assessed in older trials, such as the TNF antagonists, vs those more recently. It is also possible, as with any *in silico* analysis, that methodologic biases could influence the results of this model. Although multiple sensitivity analyses were conducted assessing assumptions used in these analyses, the panel uniformly felt that NMA was the ideal approach for providing guidance for clinicians with the available evidence at this

time. These analyses will be updated as new data, particularly head-to-head trials, become available.

There are several factors that should be considered when implementing the recommendations of the panel with regards to positioning therapies in advanced therapy-naïve individuals. Individual patient factors, such as age, active and prior comorbidities, frailty, interest in future pregnancy, preferred route of administration, cost considerations, and ease of access should be incorporated into a shared decision framework. Comparative efficacy of different medications based on specific disease phenotypes, such as disease location (small bowel dominant vs colon-dominant), behavior (inflammatory vs fibrostenotic and penetrating), and burden of inflammation (moderate vs severe), is unclear, and may drive choice of therapies once such data are available. Such data are still sparsely reported from clinical trials, limiting systematic evidence synthesis.

Although clinical trial data demonstrated minimal differences in serious infections and SAEs, it is important to recognize such studies are underpowered for measuring effect estimates for such events. When considering individual patient decision making, there may be specific patients at higher risk for treatment related complications. These patients and their providers may place higher value on safety-related considerations when choosing a therapy.

The relative positioning of different therapies was informed primarily by comparative efficacy in inducing clinical remission which was defined by the panel *a priori* as a critical outcome of interest. The panel recognized that other patient-important end points, including achieving corticosteroid-free remission, maintenance of clinical remission, avoiding surgery and hospitalization, and achieving endoscopic remission are important treatment goals for moderate-to-severely active CD. The heterogeneity in trial designs (responder re-randomization or treat-straight-through) prevented robust comparisons for treatments for longer-term end points. Where data were available, relative efficacy for maintenance end points was broadly consistent with induction data although in treat-through design trials, greater efficacy was evident with longer follow-up for mirikizumab. Similarly, data on achieving endoscopic improvement were also largely consistent with clinical remission end points where available, although the magnitude of benefit in achieving endoscopic outcomes was higher in treat-through trials comparing IL23 p19 antagonists with ustekinumab. There was a lack of systematic reporting of other end points, particularly for older clinical trials, that precluded using such data to inform relative positioning. Recent NMAs of RCTs focusing on endoscopic remission and response outcome have also been conducted. In an analysis by Vuyyuru et al,⁸⁴ JAK inhibitors and IL23 p19 antagonists were more efficacious than etrolizumab, a lymphocyte trafficking inhibitor, and JAK inhibitors were more efficacious than IL12/23p40 antagonists for inducing endoscopic response. JAK inhibitors and IL23 p19 antagonists ranked highest for induction of endoscopic response. On NMA of 6 RCTs, all agents except vedolizumab were effective in maintaining endoscopic remission compared with placebo.

TNF antagonists, IL12/23p40 antagonists, and JAK inhibitors were ranked highest for maintenance of endoscopic remission. Findings from NMAs of endoscopic outcomes should be interpreted with caution, however, as there is a paucity of RCTs of TNF antagonists reporting endoscopic outcomes with induction therapy. Although such RCT data are not available, these agents have been well-established to induce endoscopic remission in observational studies. Moreover, these results were not consistently stratified by prior exposure to advanced therapies, which can influence the magnitude of benefit.

Evidence Gaps and Future Research

The guideline panel employed NMA methods in order to inform advanced therapy positioning in these recommendations, comparing multiple placebo-controlled trials and four active comparator trials. It is important to reiterate that there are several evidence gaps to consider in this approach. There may be baseline heterogeneity across trials in terms of patients, such as differences in disease location, duration, behavior, and prior failure of IBD-directed therapies, that is not fully adjusted for in this approach. In addition, inherent differences in trial design may influence the NMA's findings. Although we attempted to control for 1 such factor through stratification (re-randomization vs treat-straight-through designs), it remains possible that other trial features may differ across studies that could impact our findings. Similarly, we used a pooled placebo rate for NMA-related analyses, and it is possible that the selected rate may have influenced our results; however, sensitivity analyses related to this value, exploring varying placebo response rates, yielded predominantly similar findings. Lastly, although the best available clinical trial data were used to inform this model, it will be important to re-evaluate the positioning of these therapies when any of the following become available: additional active comparator data, longer-term observational studies characterizing the adverse event profiles of each therapy, and when new therapies are FDA-approved.

Future research priorities should target these gaps. For example, large scale, pragmatic studies comparing different therapies both in advanced therapy-naïve and exposed individuals would serve to inform therapy positioning, while also incorporating more heterogeneous patient populations than those routinely included in current studies, accounting for real-world dose-escalation strategies across multiple potential outcomes of interest (eg, remission, response, endoscopic end points, and adverse events). Such studies would ideally identify ideal first-line therapies when considering specific disease characteristics, as well as address knowledge gaps such as TNF antagonist use after failure of another advanced therapy or differential efficacy of upadacitinib after failure of a first-line IL12/23 antagonist or IL23 p19 antagonist in relation to TNF antagonists.

Positioning of Advanced Therapies in Individuals Who Have Been Exposed to Advanced Therapies

Question 3: In adult outpatients with moderate-to-severely active CD who have been exposed to advanced therapies, what is the comparative efficacy of infliximab, adalimumab, certolizumab pegol, vedolizumab, ustekinumab, upadacitinib, risankizumab, mirikizumab, and guselkumab for induction and maintenance of remission?

Recommendation 4: In adult outpatients with moderate-to-severely active CD who have previously been exposed to 1 or more advanced therapies, particularly TNF antagonists, the AGA suggests using a **HIGHER efficacy medication** (adalimumab, risankizumab, guselkumab, upadacitinib) **OR an INTERMEDIATE efficacy medication** (ustekinumab, mirikizumab), rather than a **LOWER efficacy medication** (vedolizumab, certolizumab pegol). (*Conditional recommendation, low to moderate certainty of evidence*)

Implementation Considerations

- Second-line TNF antagonists (especially infliximab or adalimumab) may be effective in patients who discontinued their first TNF antagonist after initial response due to immunogenicity or intolerance. They may not be effective in patients with primary non-response to TNF antagonists, or those who lose response despite adequate drug trough levels. In such patients alternative mechanisms of action should be considered.
- Individual patient factors (eg, age, comorbidities, frailty, pregnancy, adherence) and preferences (eg, route of administration, ease of access) should be incorporated within a shared decision framework in selection of advanced therapies.
- Some patients, such as those with multiple prior biologic failures, may require longer duration of induction therapy for response or higher dosing for maintenance therapy for response.
- JAK inhibitors may be associated with higher risk of major adverse cardiovascular events and cancer than TNF antagonists in older adults with cardiovascular risk factors (smoking, hypertension, high-density lipoprotein cholesterol level of <40 mg/dL, diabetes mellitus, family history of premature coronary heart disease, extra-articular rheumatoid arthritis, or prior cardiovascular disease).
- There are limited data on the safety of JAK inhibitors in pregnancy. These drugs should be avoided in women of childbearing age contemplating pregnancy in the near future.

Source of Evidence

The evidence used to inform therapy positioning for those who have been exposed to a prior advanced therapy with moderate-to-severely active CD has been summarized in our accompanying NMA. These analyses included 8 placebo-controlled trials of TNF antagonists,^{32,33,37,39–41,44,79} 3 trials assessing vedolizumab,^{47–49} 3 trials of ustekinumab,^{51,52} 9 trials assessing anti-IL23p19s,^{54–56,58–60} and 3 trials of upadacitinib,^{62,63} as well as 2 head-to-head-trials^{80,81} and 5 trials of IL23p19 antagonists that included both placebo and an active comparator (ustekinumab). As with the analyses presented for advanced therapy-naïve individuals, analyses were conducted stratified by re-randomization vs treat-straight-through design.

Benefits

Overall, the panel agreed that a $\geq 5\%$ risk difference over an active comparator was clinically meaningful. When considering the results of the NMA, adalimumab demonstrated probably important benefit compared with vedolizumab (moderate certainty of evidence) and possibly important benefit compared with mirikizumab (low certainty of benefit) in advanced therapy exposed individuals. Risankizumab demonstrated probable important benefit compared with both vedolizumab and ustekinumab (moderate certainty of evidence), and possibly important benefit compared with mirikizumab (low certainty of evidence). Guselkumab demonstrated probably important benefit compared with vedolizumab, ustekinumab, and mirikizumab (moderate certainty of evidence). Ustekinumab demonstrated possibly important benefit compared with vedolizumab (low certainty of evidence). Upadacitinib demonstrated possibly important benefit compared with vedolizumab and mirikizumab (low certainty of evidence). Importantly, there were no RCT data to inform positioning infliximab among advanced therapy exposed individuals.

When examining studies with a treat-straight-through design and follow-up to weeks 48–52, risankizumab demonstrated important benefit compared with ustekinumab (high certainty of evidence), and possibly trivial benefit compared with guselkumab 100 mg every 8 weeks (low certainty of evidence). Both guselkumab 200 mg every 4 weeks and 100 mg every 8 weeks demonstrated probable important benefit compared with ustekinumab every 8 weeks (moderate certainty of evidence). Lastly, mirikizumab demonstrated possibly important benefit compared with ustekinumab (low certainty of evidence).

Harms

Potential adverse events associated with advanced therapies included in this guideline were discussed in the Harms subsection of the Use of Advanced Therapies section. The panel agreed that although there may be small differences among adverse events and infectious rates among therapies included in the NMA, that these risks were

likely trivial across drug classes, and unlikely to inform choice of therapy for most patients. As noted when considering therapy selection in advanced therapy-naïve individuals, any therapeutic decision making that considers adverse events of therapy must also balance these factors against the potential for immediate and longer-term consequences of incomplete disease control. This is particularly relevant in those who have already had an inadequate response to prior therapies or treatment-related complications and thus are more likely to experience harm with delays in appropriate therapy.

Certainty in Evidence of Effects

The certainty of evidence for all pairwise comparisons of advanced therapies ranged from low to moderate, as described above. The only exception to this was the high certainty of evidence noted for important benefit of risankizumab over ustekinumab when considering treat-straight-through trials.

Discussion

To aid clinicians in clinical decision making, and in the context of potential intra-class differences among specific therapies in IBD, the panel rated each therapy individually as opposed to grouping by therapeutic class. As discussed previously in relation to advanced therapy-naïve individuals (see Discussion subsection above), the guideline panel felt it appropriate to consider grouping available therapies into therapeutic categories using criteria that were defined *a priori*. These criteria are presented above in the Methods (Development of Recommendation subsection). Three therapeutic groups were constructed, including “higher efficacy,” “intermediate efficacy,” and “lower efficacy.” Inclusion in the higher-efficacy group required meeting all 3 criteria noted above, while intermediate efficacy therapies met some but not all 3 criteria. Using these criteria, adalimumab, risankizumab, guselkumab, and upadacitinib were considered higher efficacy, ustekinumab and mirikizumab were considered intermediate efficacy, and vedolizumab and certolizumab pegol were considered lower efficacy therapies. The panel recommended that, when selecting an advanced therapy, it would be appropriate to consider either a higher efficacy or intermediate efficacy medication over a lower efficacy therapy.

There are several factors to consider in the implementation of these recommendations. First, almost all enrolled individuals in the clinical trials included in this systematic review and NMA had been exposed to a TNF antagonist. Therefore, there are limited data to inform therapy selection in those whose prior failure consisted of a non-TNF antagonist. In addition, the results of the NMA are stratified by any prior advanced therapy. It is possible that the efficacy of given therapies decreases with the number of prior failures to advanced therapies; unfortunately, the published data regarding this potential effect are limited. There are no RCT data regarding the positioning of infliximab in individuals who are advanced therapy-exposed. The panel felt it would be

reasonable to extrapolate recommendations from adalimumab when considering infliximab, however; and this is supported by some observational data demonstrating preserved efficacy after failure of 1 advanced therapy.^{85,86} On the contrary, evidence for adalimumab in bio-exposed patients is largely derived from the GAIN trial, which specifically included individuals who had responded to infliximab and subsequently developed intolerance or lost response.⁴¹ It is possible that this population may be biased toward responsiveness to this class of medications, and may not be representative of individuals enrolled in more recent studies, which included those with mechanistic failures to TNF antagonists. Thus, use of adalimumab as second-line therapy after prior infliximab failure should primarily be considered in those whose reason for discontinuation of infliximab was logistical (such as insurance coverage), due to intolerance, or loss of response due to drug-specific immunogenicity rather than mechanistic failure.

As discussed previously with advanced therapy-naïve individuals, treatment selection for those who have failed advanced therapies should incorporate a careful consideration of individual patient factors, including but not limited to age, active or prior comorbidities, frailty, and medication access, and interest in future pregnancy. Patients' preferences with regard to preferred route of administration should be considered within this shared decision-making framework. In addition, there may be variability with regard to the risks of adverse events or serious or opportunistic infections with specific medication classes. As noted above, medication selection when incorporating these factors into decision making should also be coupled with consideration of potential risks in delays in clinical response and the possibility for short- and long-term disease-related complications. With decreasing efficacy, there may be the potential for longer time periods for onset of action, which may expose individuals to prolonged periods of reduced quality of life or extended courses of corticosteroids, particularly in those with more severe disease or a higher number of prior advanced therapy failures.

It is also important to incorporate family planning into conversations regarding advanced therapies. All agents categorized as higher efficacy are currently felt to be safe in pregnancy and lactation, with robust data supporting this for TNF antagonists, vedolizumab, and ustekinumab.⁸⁷ Although not the primary focus of the data review, the committee members felt that it was appropriate to extrapolate these recommendations to newer IL23 p19 antagonists, while formal data supporting this are forthcoming. There are concerns related to the use of JAK inhibitors in pregnancy; there are limited human data demonstrating safety and some animal models suggesting increased risk.⁸⁸ Providers and their patients should carefully consider available therapeutic agents and their risks when individualizing a therapeutic plan; panel members agreed that it would be reasonable at this time to avoid JAK

inhibitors in women contemplating pregnancy in the near future.

Lastly, it is important to recognize that JAK inhibitors, such as upadacitinib, may possess a unique risk profile in comparison with other therapies. Specifically, this class may be associated with an increased risk of major adverse cardiovascular events. Prior long-term follow-up data from another JAK inhibitor, tofacitinib, demonstrated a potential increased risk of major adverse cardiovascular events and venous thromboembolic events, as well as malignancy, in those using tofacitinib 10 mg twice daily.²⁹ It is important to consider the patient population in which this signal was detected to place these findings in context: the cohort consisted of individuals with rheumatoid arthritis who were aged 50 years or older and with at least 1 cardiovascular risk factor (ie, prior cardiovascular or coronary artery disease, tobacco use, high-density lipoprotein <40 mg/dL, a history of diabetes mellitus, family history of coronary heart disease, or extra-articular rheumatoid arthritis). Although it is unclear whether these risks extend to other JAK inhibitors and other patient populations, it would be reasonable to use these drugs with caution in patients at high risk for atherosclerotic cardiovascular disease.

Evidence Gaps and Future Research

There are several knowledge gaps and opportunities for future research. There remains a lack of robust head-to-head clinical trial data comparing therapies in those with prior advanced therapy exposure. Fortunately, this is changing over time, with 2 recent head-to-head studies, as well as several new IL23p19 antagonists including active comparators in their RCTs. These studies should continue to incorporate clinical remission-related outcomes, as well as outcomes related to endoscopic remission, rates of health care utilization and/or surgery, and impacts on quality of life. Second, trials assessing treatment sequencing for first-line and subsequent therapies are needed to guide clinicians. Such studies may help confirm the efficacy of JAKs such as upadacitinib as first- or second-line therapies and explore the biologic underpinnings of differential efficacy that appear to be dependent on such positioning. Third, although many clinical trials enroll patient populations with significant prior advanced therapy exposure, the majority of these prior exposures were to TNF antagonists. The generalizability of these results to patients who have been treated with only non-TNF-antagonist therapies is unknown. Lastly, the long-term safety profile of newer classes of therapies in CD, in particular JAK inhibitors and IL23p19 antagonists, is incompletely understood at the time of this guideline development. Although clinical trial safety data and data from other disease states are reassuring, these studies are also almost certainly underpowered for safety signals related to rare events, and further research is indicated.⁸⁹⁻⁹²

Use of Immunomodulators

Question 4: In adult outpatients with moderate-to-severely active CD, what is the efficacy of immunomodulator monotherapy (thiopurines, methotrexate) for induction and maintenance of remission?

Recommendation 5: In adult outpatients with moderate-to-severely active CD, the AGA suggests **AGAINST** using thiopurines monotherapy over no treatment for inducing remission. (*Conditional recommendation, very low certainty of evidence*)

Recommendation 6: In adult outpatients with moderate-to-severely active CD who have achieved remission, the AGA **SUGGESTS** using thiopurine monotherapy over no treatment for maintaining remission. (*Conditional recommendation, low certainty of evidence*)

Recommendation 7: In adult outpatients with moderate-to-severely active CD, the AGA **SUGGESTS** using subcutaneous or intramuscular methotrexate monotherapy over no treatment. (*Conditional recommendation, moderate certainty of evidence*)

Recommendation 8: In adult outpatients with moderate-to-severely active CD, the AGA suggests **AGAINST** using oral methotrexate monotherapy over no treatment. (*Conditional recommendation, very low certainty of evidence*)

Implementation Considerations

- The typical dose of thiopurines is 2.0–2.5 mg/kg/d for azathioprine and 1.0–1.5 mg/kg/d for mercaptopurine when used as monotherapy for those with normal drug metabolism.
- In clinical trials of thiopurines for preventing relapse in patients who achieved remission, remission was typically induced with corticosteroids.
- The typical dose of methotrexate is 25 mg weekly during induction for 16–24 weeks, and 15 mg weekly for maintenance. Methotrexate should be accompanied by daily folic acid supplementation.
- Routine monitoring of complete blood counts and liver function tests is recommended when using thiopurines and methotrexate
- Use of methotrexate is contraindicated in women actively considering pregnancy or who are pregnant.

Source of Evidence

To inform the recommendations related to induction of remission of moderate-to-severely active CD with thiopurines, data from 6 clinical trials were included.^{93–98} One trial of thiopurines for induction and maintenance of remission in pediatric CD was added since the last guidelines were published in 2021.⁹⁸ For maintenance of remission with thiopurines, data from 6 trials were included.^{33,93,97–100}

For induction of remission with methotrexate, we identified 1 RCT of subcutaneous methotrexate¹⁰¹ (unchanged from previous guideline) and 2 RCTs of oral methotrexate^{95,102}

(1 additional trial published after the previous guideline). In the 2 trials of oral methotrexate, the dose of methotrexate ranged from a fixed dose 12.5 mg weekly to 15 mg weekly escalated to 22.5 mg weekly in case of suboptimal response. For maintenance of remission with methotrexate, 1 RCT of subcutaneous methotrexate and 2 RCT of oral methotrexate were included.^{95,103}

Benefits

Thiopurines. Across the 6 trials assessing induction of remission with thiopurines in comparison with placebo, 380 individuals were enrolled. On meta-analysis of these studies, thiopurines were not more likely to induce clinical remission (RR, 1.11; 95% CI, 0.96–1.30) ([Supplementary Figure 10A](#)). Across 5 trials of maintenance of remission with thiopurines, 347 individuals were enrolled. On meta-analysis, thiopurines were associated with maintenance of remission (RR, 1.23; 95% CI, 1.00–1.50) ([Supplementary Figure 10B](#)).

Subcutaneous methotrexate. In a clinical trial of 141 individuals with CD, patients treated with subcutaneous methotrexate (25 mg weekly) were more likely to achieve remission compared with placebo (RR, 2.06; 95% CI, 1.09–3.89).¹⁰¹ In another trial of 76 patients, subcutaneous methotrexate (15 mg weekly) was more effective than placebo in preventing relapse in patients who achieved clinical remission after 16–24 weeks of open-label subcutaneous methotrexate (25 mg weekly) (RR, 1.67; 95% CI, 1.05–2.67).¹⁰³

Oral methotrexate. In 2 clinical trials, oral methotrexate was not more effective than placebo in inducing remission (RR, 0.77; 95% CI, 0.55–1.08).^{95,102} In 1 clinical trial, oral methotrexate 12.5 mg weekly was not more effective than placebo in preventing relapse in patients with CD in remission (RR, 1.35; 95% CI, 0.86–2.12).

Harms

The risks associated with thiopurines and methotrexate are well characterized. For thiopurines, these risks include hepatotoxicity, pancreatitis, bone marrow suppression, opportunistic infections, nonmelanoma skin cancers, and lymphoproliferative disorders. Methotrexate is also associated with nonmelanoma skin cancer, bone marrow suppression, and drug-induced liver injury, and is also associated with both pulmonary and hepatic fibrosis with long-term use.^{26–28,104–107} Methotrexate should also be avoided in women contemplating pregnancy or who are pregnant, due to its teratogenicity. There are also risks of selecting ineffective therapies for moderate-to-severely active CD, with the potential for increased risks of disease-related complications.

Certainty in Evidence of Effects

The overall benefit of thiopurines for inducing remission was uncertain with very low certainty of evidence ([Table 10](#)). Evidence was rated down for risk of bias due to inadequate blinding and allocation concealment, indirectness due to concomitant corticosteroid use during induction, heterogeneity in outcome definitions, and imprecision. For maintenance of remission with thiopurines, the certainty of evidence was rated as low, and was rated down for risk of bias as well as imprecision. The certainty of

Table 10. Grading of Recommendations Assessment, Development and Evaluation Evidence Profile Comparing Thiopurines With No Thiopurines for Achieving Steroid-Free Remission, and Preventing Relapse in Patients With Moderate-to-Severely Active Crohn's Disease

Outcomes	Study event rates, n/N (%)		Relative effect, RR (95% CI)	Absolute effect	No. of participants (studies)	Quality of the evidence (GRADE) ^a
	Risk with placebo	Risk with thiopurines				
Thiopurines compared with placebo for moderate-to-severely active CD						
Achieving clinical remission (CRITICAL)	68/183 (37.1)	95/197 (48.2)	1.23 (0.97–1.54)	85 more per 1000 (from 11 fewer to 201 more)	380 (5 RCTs)	⊕○○○ ^{b,c,d} VERY LOW
Relapse after achieving clinical remission (CRITICAL)	97/172 (56.4)	127/175 (72.6)	1.23 (1.00–1.50)	130 more per 1000 (from 0 fewer to 282 more)	347 (5 RCT)	⊕⊕○○ ^{b,e} LOW

^aGRADE Working Group grades of evidence. High quality: We are very confident that the true effect lies close to that of the estimate of the effect. Moderate quality: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low quality: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect. Very low quality: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect.

^bRated down for risk of bias (due to inadequate blinding and allocation concealment).

^cRated down for indirectness (because these trials did not truly assess induction of remission, but rather the ability to achieve corticosteroid-free clinical remission, over a wide range of time, using a variety of disease activity indices with definitions inconsistent with modern definitions of remission).

^dRated down for imprecision because 95% CI crosses unity.

^eRated down for imprecision because 95% CI crosses minimally clinically important difference of 10%.

evidence supporting the use of subcutaneous methotrexate for inducing and maintaining remission was moderate, rated down for imprecision because the 95% CI crossed the clinically meaningful benefit of 10% compared with placebo (Table 11). The overall certainty of evidence supporting the use of oral methotrexate was very low, rated down for indirectness (low dose of oral methotrexate used in 1 trial, variability in outcome definition) and very serious imprecision (Table 11). The GRADE Evidence-to-Decision framework is presented in Table 12.

Discussion

In weighing benefits and harms associated with thiopurines, the panel suggested against the use of thiopurines for induction of remission but suggested its use for maintenance of remission in patients with clinically quiescent CD. Thiopurines have a slow onset of action, potentially requiring concomitant corticosteroids for induction, while also delaying the use of appropriate advanced therapies. Real-world studies and previously published systematic reviews and meta-analyses have confirmed the benefit of azathioprine when used to maintain remission in CD,¹⁰⁸ and the studies included in our analyses demonstrated a benefit as well.

With regard to methotrexate, there was moderate certainty evidence supporting the use of subcutaneous methotrexate for

both induction and maintenance of remission. For induction, this was informed by 1 study using 25 mg/wk, whereas for maintenance of remission, the single trial evaluated used 15 mg/wk. In contrast, the data supporting oral dosing of methotrexate was less robust, with the available studies showing no benefit for either inducing or maintaining remission, but with limited sample sizes for adequate assessment.

There are several implementation considerations with regard to these recommendations. Before initiating thiopurines, AGA guidelines recommend thiopurine methyltransferase testing,¹⁰⁹ as well as baseline and close subsequent monitoring of complete blood counts and liver function tests. When using azathioprine as monotherapy, the typical dose is 2.0–2.5 mg/kg/d, and for 6-mercaptopurine the typical dose is 1.0–1.5 mg/kg/d.

When considering the use of methotrexate for inducing and maintaining remission, baseline assessments of complete blood counts and liver function tests are recommended. Methotrexate use should also be coupled with daily oral folic acid supplementation, starting at 1 mg/d, to minimize the potential for folate deficiency and symptoms secondary to this. Routine subsequent laboratory monitoring is advised while using methotrexate. Given its known teratogenicity, counseling regarding avoiding pregnancy and contraception is also advised. Risks of hepatic fibrosis, which are relatively low with weekly dosing as used in CD,

Table 11. Grading of Recommendations Assessment, Development and Evaluation Evidence Profile Assessing Subcutaneous and Oral Methotrexate for Achieving Steroid-Free Remission, and Preventing Relapse in Patients With Moderate-to-Severely Active Crohn's Disease

	Study event rates, n/N (%)					
Outcomes	Risk with placebo	Risk with subcutaneous methotrexate	Relative effect, RR (95% CI)	Absolute effect	No. of participants (studies)	Quality of the evidence (GRADE) ^a
Subcutaneous methotrexate compared with placebo for moderate-to-severe luminal CD						
Achieving clinical remission (CRITICAL)	9/47 (19.1)	37/94 (39.4)	2.06 (1.09–3.89)	203 more per 1000 (from 17 more to 553 more)	141 (1 RCT)	⊕⊕⊕○ ^b MODERATE
Relapse after achieving clinical remission (CRITICAL)	14/36 (38.9)	26/40 (65.0)	1.67 (1.05–2.67)	261 more per 1000 (from 19 more to 649 more)	76 (1 RCT)	⊕⊕⊕○ ^b MODERATE
	Study event rates, n/N (%)					
Outcomes	Risk with placebo	Risk with oral methotrexate	Relative effect, RR (95% CI)	Absolute effect	No. of participants (studies)	Quality of the evidence (GRADE) ^a
Oral methotrexate compared with placebo for moderate-to-severe luminal CD						
Achieving clinical remission (CRITICAL)	12/26 (46.2)	10/26 (38.5)	0.83 (0.44–1.58)	78 fewer per 1000 (from 258 fewer to 268 more)	52 (1 RCT)	⊕○○○ ^{c,d} VERY LOW
Relapse after achieving clinical remission (CRITICAL)	8/12 (66.7)	9/10 (90.0)	1.35 (0.86–2.12)	233 more per 1000 (from 93 fewer to 747 more)	22 (1 RCT)	⊕○○○ ^{c,d} VERY LOW

^aGRADE Working Group grades of evidence. High quality: We are very confident that the true effect lies close to that of the estimate of the effect. Moderate quality: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low quality: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect. Very low quality: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect.

^bRated down for imprecision because 95% CI crosses minimally clinically important difference of 10% and optimal information size not met (<200 events).

^cRated down for indirectness (used low dose oral methotrexate <15 mg/wk).

^dRated down for very serious imprecision due to very wide 95% CI (unable to rule out significant risk of harm with intervention).

Table 12. Grading of Recommendations Assessment, Development and Evaluation Evidence-to-Decision Framework for Use of Immunomodulator Monotherapy for the Management of Patients With Moderate-to-Severely Active Crohn's Disease

Domain	Criteria			Judgment			
Problem	Is the problem a priority?	No	Probably No	Probably Yes	Yes	Varies	Do not know
Desirable effects	How substantial are the desirable anticipated effects?	Trivial to small (<10%) TP (Ind) PO MTX (Ind)		Moderate (10%–30%) TP (M) SC MTX (Ind + M) PO MTX (M)	Large (>30%)	Varies	Do not know
Undesirable effects	How substantial are the undesirable anticipated effects?	Trivial to small TP, PO, or SC MTX		Moderate	Large	Varies	Do not know
Certainty of evidence	What is the overall certainty of the evidence of effects	Very low TP (Ind) PO MTX (Ind + M)	Low TP (M)	Moderate SC MTX (Ind + M)	High	No included studies	
Values	Is there important uncertainty about or variability in how much people value the main outcomes?	Important uncertainty or variability		Possibly important uncertainty or variability	Probably no important uncertainty or variability		No important uncertainty or variability
Balance of effects	Does the balance between desirable and undesirable effects favor the intervention or the comparison?	Favors comparison/control	Probably favors comparison	Does not favor either comparison or intervention TP (Ind) PO MTX (Ind)	Probably favors intervention TP (M) SC MTX (Ind + M) PO MTX (M)	Favors intervention	Varies/do not know
Resource use	Is the incremental cost small relative to the net benefits?	No	Probably No	Uncertain	Probably yes	Yes	Uncertain
Equity	What would be the impact on health inequities?	Increased	Probably increased	Probably no impact	Probably reduced	Reduced	Varies/do not know
Acceptability	Is the option acceptable to key stakeholders?	No	Probably No	Probably yes	Yes	Varies	Do not know
Feasibility	Is the option feasible to implement?	No	Probably No	Probably yes	Yes	Varies	Do not know

NOTE. CMD threshold of active interventions was set at 10%. Judgments made by guideline panel are in bold type. Ind, induction; M, maintenance, MTX, methotrexate; PO, per os; SC, subcutaneous; TP, thiopurine.

can also be mitigated through the avoidance of alcohol. Lastly, it is reasonable to counsel patients regarding the potential for gastrointestinal adverse effects and fatigue. Although these symptoms are common and may impact quality of life, patients may favor avoidance of some adverse events over others, including those related to inadequate control of disease.

Evidence Gaps and Future Research

Although subcutaneous methotrexate was found to be beneficial in these analyses, the trial population in which these effects were studied is likely different than the cohort of patients with moderate-to-severely active CD entering modern clinical trials. Individuals enrolled in prior methotrexate trials were also treatment-naïve. It is unclear whether similar benefits would be appreciated in those with prior corticosteroid, thiopurine, or advanced therapy treatment. There is also likely a role for comparative effectiveness and long-term safety trials comparing subcutaneous methotrexate with biologic or small molecule therapies, as methotrexate may still have a substantive role to play in the management of moderate-to-severely active CD in resource limited settings.

Combination Therapy With Tumor Necrosis Factor Antagonists and Immunomodulators

Question 5: In adult outpatients with moderate-to-severely active CD, is combination therapy of TNF antagonists with an immunomodulator (thiopurines or methotrexate) superior to TNF monotherapy or immunomodulator monotherapy for induction and maintenance of remission?

Recommendation 9: In adult outpatients with moderate-to-severely active CD who are NAÏVE to thiopurines and starting infliximab, the AGA SUGGESTS using infliximab in combination with thiopurines rather than infliximab monotherapy. (Conditional recommendation, low certainty of evidence)

Recommendation 10: In adult outpatients with moderate-to-severely active CD, the AGA makes NO RECOMMENDATION on using infliximab in combination with methotrexate over infliximab monotherapy. (No recommendation, knowledge gap)

Recommendation 11: In adult outpatients with moderate-to-severely active CD, the AGA makes NO RECOMMENDATION on using adalimumab in combination with thiopurines or methotrexate over adalimumab monotherapy. (No recommendation, knowledge gap)

Implementation Considerations

- The benefit of routinely combining immunomodulators with TNF antagonists in patients who have previously failed immunomodulator monotherapy is uncertain.
- In patients who previously failed immunomodulator monotherapy, there may be benefits of adding immunomodulators when starting TNF antagonists in

specific situations where patients may be at a higher risk for immunogenicity. These include patients with history of immunogenicity with a TNF antagonist, patients being re-exposed to TNF antagonists after a prolonged lapse in therapy, patients carrying HLA-DQ-A1*05 variants and patients with high drug clearance, such as those with more severe disease, high burden of inflammation, or low albumin.

Source of Evidence

Four clinical trials assessed the benefit of combination TNF antagonists and immunomodulators, 1 of which had been published since the prior guidelines. Of these studies, 2 focused on infliximab (1 in combination with azathioprine, 1 in combination with methotrexate), 1 focused on adalimumab in combination with azathioprine, and the last included both infliximab or adalimumab in combination with methotrexate.^{110–113} Three studies focused on adult populations and 1 included pediatric patients. Across these trials, follow-up times for induction of remission (ranging from 10–16 weeks after treatment initiation) and maintenance of remission (ranging from 26–52 weeks) were variable.

Benefits

Two trials focused on the combination of infliximab and an immunomodulator. The SONIC trial randomized 508 biologic- and immunomodulator-naïve adults with moderate-to-severely active CD to combination infliximab plus azathioprine ($n = 169$) vs infliximab monotherapy ($n = 169$) vs azathioprine monotherapy ($n = 170$).¹¹¹ Combination therapy with infliximab and azathioprine was associated with higher rates of remission during induction (RR, 1.29; 95% CI, 1.03–1.54), and sustained remission at 26 weeks (RR, 1.28; 95% CI, 1.03–1.59) compared with infliximab monotherapy. The COMMIT trial compared 63 individuals randomized to receive combination therapy with infliximab plus methotrexate (dosing ranging from 10 mg to 25 mg weekly) to 63 individuals receiving infliximab monotherapy, employing a treat-straight-through design.¹¹² Although individuals were advanced therapy-naïve, up to 25% in each arm had received prior thiopurine therapy. At week 14, there was no significant difference between treatment groups (RR, 0.98; 95% CI, 0.81–1.19), and this persisted through week 50 (RR, 0.97; 95% CI, 0.71–1.32).

One RCT assessed combination therapy with adalimumab and azathioprine compared with adalimumab monotherapy. The DIAMOND trial randomized 91 individuals to combination therapy and 85 to adalimumab monotherapy in an open label trial with a treat-straight-through design.¹¹³ Enrolled individuals were naïve to TNF antagonists and immunomodulators. There was no difference between the treatment arms for induction of clinical remission (RR, 0.85; 95% CI, 0.70–1.03) or in maintenance of clinical remission at 52 weeks (RR, 0.95; 95% CI, 0.75–1.21). Combination therapy was associated with higher

rates of endoscopic remission at week 26 (84.2% vs 63.2%; $P = .02$), although when analyzing the subgroup with endoscopic evaluation at both randomization and at week 52, no differences were appreciated in this outcome (79.6% vs 69.8%; $P = .36$). This was an open-label study and had high dropout in the combination therapy arm: by week 26, 15 individuals (16.5%) receiving combination therapy and 1 (1.2%) receiving monotherapy had dropped out. Such high rates of treatment-related drug withdrawals have not been observed with prior trials of thiopurines or combination therapy with infliximab.

One study in pediatrics was reviewed that was not included in the prior guidelines. The COMBINE trial randomized 297 children with CD starting infliximab or adalimumab to adding either low-dose oral methotrexate or placebo and followed them over 12–36 months.¹¹⁰ Enrolled individuals were TNF antagonist-naïve but could have previously been exposed to thiopurines or methotrexate, with prior exposure rates ranging from 12% to 17%. The outcome was failure to achieve or maintain steroid-free remission, defined as a composite of failure to achieve remission (Short Pediatric Crohn's Disease Activity Index score <15) by week 26; failure to complete a steroid taper by week 16; Short Pediatric Crohn's Disease Activity Index score ≥ 15 attributed to active CD at 2 or more consecutive visits beyond week 26; hospitalization or surgery for CD beyond week 26; use of corticosteroids for CD for ≥ 10 weeks cumulatively, beyond week 16; and/or discontinuation of TNF antagonists and/or study drug for lack of effectiveness or toxicity. A total of 40 of 156 participants (26%) in the combination therapy group and 48 of 141 participants (34%) in the monotherapy group experienced treatment failure (hazard ratio [HR], 0.69; 95% CI, 0.45–1.05). The most common component of the composite end point experienced by study participants was hospitalization for active IBD after week 25. On prespecified subgroup analysis, the risk of treatment failure with combination therapy was lower in adalimumab-treated patients (combination therapy vs adalimumab monotherapy: 11 of 46 vs 20 of 39; HR, 0.40; 95% CI, 0.19–0.82), but not in infliximab-treated patients (combination therapy vs infliximab monotherapy: 29 of 110 vs 28 of 102; HR, 0.69; 95% CI, 0.45–1.05). It should be noted that this study was not powered to evaluate differences in these subgroups.

Harms

The potential risks of both TNF antagonists and immunomodulators when used as monotherapy have been described previously. Although underpowered to assess rare outcomes, the data from the 4 clinical trials assessed for this guideline were reassuring with regard to SAE risk. In SONIC and COMMIT, there were no statistically significant differences in SAEs for combination therapy compared with TNF-antagonist monotherapy.^{111,112} In COMMIT, there was a numerical increase in overall adverse events with combination therapy, but the rate of SAEs remained similar between groups (11% vs 15%). Discontinuation rates in DIAMOND were high (22%–24%), and adverse events associated with combination

therapy were higher than with TNF-antagonist monotherapy (16.5% vs 1.2%).¹¹³

Large observational studies with longer follow-up suggest a higher risk of serious infections, lymphoma, and nonmelanoma skin cancers with combination therapy in contrast to TNF-antagonist monotherapy.^{28,114,115} For example, in a retrospective cohort study of 182,289 individuals with IBD and a median follow-up time of 6.7 years, the incidence rate (IR) per 1000 person-years of lymphoma with TNF-antagonist monotherapy was 0.41 (95% CI, 0.27–0.55) in comparison with combination therapy with an TNF antagonist and immunomodulator (IR, 0.95; 95% CI, 0.45–1.45), with a 2.5-fold increased risk in multivariable modeling (adjusted HR, 2.53; 95% CI, 1.35–4.77).²⁸

Certainty in Evidence of Effects

The certainty of evidence for induction of remission with combined TNF-antagonist and immunomodulator therapy in comparison with monotherapy was felt to be very low, and was rated down due to heterogeneity in treatment effects, patient populations, and study design, as well as for very serious imprecision (Table 13). The quality of the evidence for maintenance of remission with combination TNF-antagonist and immunomodulator therapy was rated as low quality of evidence, due to heterogeneity and serious imprecision. The GRADE evidence-to-decision judgments are presented in Table 14.

Discussion

When examining the totality of evidence, the magnitude of benefit with the combination of TNF antagonists with immunomodulators over TNF-antagonist monotherapy was uncertain (Supplementary Figure 11). However, there were differences in trial design, patient population (prior exposure to immunomodulators), and interventions (concomitant treatment with corticosteroids at trial entry). Hence, the guideline panel opted to examine each trial in isolation and issue separate recommendations for specific combinations of therapy. Based on the SONIC trial, combination therapy of infliximab plus azathioprine is probably beneficial in achieving remission compared with infliximab monotherapy in infliximab- and immunomodulator-naïve patients. Based on the DIAMOND trial, the benefit of combining adalimumab with thiopurines vs adalimumab monotherapy was uncertain for achieving remission, although rates of achieving endoscopic remission were higher with combination therapy. Based on the COMMIT and COMBINE trials, the benefit of combination of infliximab and adalimumab with methotrexate over corresponding monotherapy was uncertain.

There are several mechanisms by which combining TNF antagonists with an immunomodulator may increase the effectiveness of the TNF antagonist. First, immunomodulators have their own efficacy in treating CD, as discussed previously. This may result in an additive effect and may also explain, in part, why those who have previously failed immunomodulators before beginning combination therapy may not experience as robust a benefit.^{110,112} This may

Table 13. Grading of Recommendations Assessment, Development and Evaluation Evidence Profile Comparing Tumor Necrosis Factor Antagonists Plus Immunomodulators vs Tumor Necrosis Factor Antagonist Monotherapy for Achieving Remission in Patients With Moderate-to-Severely Active Crohn’s Disease

Outcomes	Study event rates, n/N (%)		Relative effect, RR (95% CI)	Absolute effect	No. of participants (studies)	Quality of the evidence (GRADE) ^a
	Risk with placebo	Risk with combination therapy				
Combination therapy with TNF antagonists + immunomodulators compared with TNF-antagonist monotherapy for moderate-to-severe CD						
Induction of clinical remission, wk 10–16 (CRITICAL)	287/458 (62.7)	324/479 (67.6)	1.04 (0.89–1.23)	25 more per 1000 (from 69 fewer to 144 more)	937 (4 RCTs)	⊕⊕○○ ^{b,c} LOW
Maintenance of clinical remission, wk 26–52 (CRITICAL)	257/458 (56.1)	301/479 (62.8)	1.10 (0.97–1.25)	56 more per 1000 (from 17 fewer to 140 more)	937 (4 RCTs)	⊕⊕○○ ^{b,c} LOW

^aGRADE Working Group grades of evidence. High quality: We are very confident that the true effect lies close to that of the estimate of the effect. Moderate quality: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low quality: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect. Very low quality: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect.

^bRated down for inconsistency (difference in interventions, patients).

^cRated down for serious imprecision with wide 95% CI.

explain differential findings appreciated across studies reviewed here: combination therapy demonstrated clinical benefit in SONIC, where individuals were treatment-naïve, but no clear benefit was appreciated in COMMIT or COMBINE, where there was prior immunomodulator exposure. This has also been observed in regulatory clinical trials of TNF antagonists where no differences in treatment efficacy have been observed in subsets of patients who were vs were not on concomitant immunomodulators at time of trial enrollment; of note, all patients on concomitant immunomodulators in these trials were required to have moderate-to-severe activity despite immunomodulator monotherapy.¹¹⁶

Second, immunomodulators decrease the risk of immunogenicity to TNF antagonists and may favorably alter pharmacokinetic properties, leading to higher trough concentration.^{117–119} Recent studies have identified HLA-DQ-A1*05 as a genetic risk factor associated with increased risk of immunogenicity, where the addition of an immunomodulator may be associated with lower risk of clinical failure and immunogenicity.^{120,121} In a clinical trial of 98 patients with IBD with immune-mediated loss of response to prior TNF antagonists, the addition of azathioprine at the time of starting a second TNF antagonist was associated with markedly lower rates of clinical failure and unfavorable pharmacokinetic properties, compared with initiation of TNF antagonist

monotherapy.¹²² In these instances, combination therapy with immunomodulators may decrease the risk of immunogenicity with TNF antagonists and may be preferred.

Evidence Gaps and Future Research

Several evidence gaps should be addressed through future research. Although the guideline committee recommends the use of combination thiopurines with infliximab, particularly in those without prior thiopurine failure, there are limited data to guide clinicians regarding the ideal duration of combination therapy and optimal thiopurine dosing. There is significant practice heterogeneity regarding the use of traditional weight-based dosing for thiopurines. Although 6-thioguanine metabolite concentrations are correlated with TNF-antagonist trough concentrations, 6-thioguanine concentrations may be lower than those achieved with the traditional weight-based dose of 2.0–2.5 mg/kg azathioprine (1.0–1.5 mg/kg 6-mercaptopurine).¹¹⁸ This is further complicated by the specific TNF antagonist used, as the effects of concomitant immunomodulator therapy may differ between adalimumab and infliximab, and we did not find data assessing combination therapy with certolizumab pegol.¹²³ Future prospective RCTs could better determine the clinical implications of these

Table 14. Grading of Recommendations Assessment, Development and Evaluation Evidence-to-Decision Framework for use of Combination Therapy Over Tumor Necrosis Factor Monotherapy for the Management of Patients With Moderate-to-Severe Crohn's Disease

Domain	Criteria	Judgment					
Problem	Is the problem a priority?	No	Probably no	Probably yes	Yes	Varies	Do not know
Desirable effects	How substantial are the desirable anticipated effects?	Trivial to small (<10%)		Moderate (10%–30%)	Large (>30%)	Varies	Do not know
Undesirable effects	How substantial are the undesirable anticipated effects?	Trivial to small		Moderate	Large	Varies	Do not know
Certainty of evidence	What is the overall certainty of the evidence of effects	Very low		Low	Moderate	High	No included studies
Values	Is there important uncertainty about or variability in how much people value the main outcomes?	Important uncertainty or variability		Possibly important uncertainty or variability	Probably no important uncertainty or variability		No important uncertainty or variability
Balance of effects	Does the balance between desirable and undesirable effects favor the intervention or the comparison?	Favors comparison/control	Probably favors comparison	Does not favor either comparison or intervention	Probably favors intervention	Favors intervention	Varies/do not know
Resource use	Is the incremental cost small relative to the net benefits?	No	Probably No	Uncertain	Probably yes	Yes	Uncertain
Equity	What would be the impact on health inequities?	Increased	Probably increased	Probably no impact	Probably reduced	Reduced	Varies/do not know
Acceptability	Is the option acceptable to key stakeholders?	No	Probably no	Probably yes	Yes	Varies	Do not know
Feasibility	Is the option feasible to implement?	No	Probably no	Probably yes	Yes	Varies	Do not know

NOTE. CMD threshold of active interventions was set at 5%. Judgments made by guideline panel are in bold type.

differences, as well as identify the differences in duration of combination therapy, with significant interest among both patients and providers in therapy de-escalation. Studies specifically in patient populations who have had unexpected gaps in TNF antagonist or other biologic therapy would also inform clinical decision making regarding using combination therapy in these individuals as well. In addition, there are limited data to inform decision making regarding combination therapy for the subcutaneous formulation of infliximab.

Although these guidelines provide specific recommendations related to the use of immunomodulators with TNF antagonists, they do not specifically address the utilization of biologic and small molecule therapies in combination with each other. Observational studies assessing combining these agents have had reassuring safety and efficacy estimates.^{124,125} There are also future ongoing clinical trials assessing this approach in IBD. Future studies will serve to better describe those in whom this approach is most advantageous, which combinations may be most effective based on mechanism of action, and in whom the risks may outweigh the benefit.

Combination Therapy With Non-Tumor Necrosis Factor Targeting Biologics and Immunomodulators

Question 6: In adult outpatients with moderate-to-severely active CD, is combination therapy of a non-TNF-targeting biologic with an immunomodulator (thiopurines or methotrexate) superior to non-TNF monotherapy or immunomodulator monotherapy for induction and maintenance of remission?

Recommendation 12: In adult outpatients with moderate-to-severely active CD, the AGA makes **NO RECOMMENDATION** in favor of, or against using non-TNF-targeting biologics (vedolizumab, ustekinumab, risankizumab, mirikizumab, guselkumab) in combination with thiopurines or methotrexate or corresponding biologic monotherapy. (No recommendation, knowledge gap)

Summary of the Evidence

There are currently no RCTs comparing combination therapy with a non-TNF-targeting biologic and an immunomodulator to monotherapy with a non-TNF-targeting biologic. However, several post-hoc analyses from clinical trials and observational studies have assessed the impact of concomitant immunomodulators in combination with vedolizumab or ustekinumab. These subgroup analyses were assessed for the purposes of this guideline. In GEMINI, approximately 33% (n = 370) of individuals randomized to receive vedolizumab or placebo were also concurrently receiving an immunomodulator or an immunomodulator in combination with corticosteroids.⁴⁷ Similar rates of immunomodulator use were seen in clinical trials of ustekinumab (30%–39%, n = 304).⁵² Multiple observational studies have assessed if the use of combination

therapy with non-TNF-targeting biologics results in improved rates of remission as well. For example, a multicenter cohort study followed 546 individuals with IBD (CD = 286) who initiated vedolizumab or ustekinumab for 1 year, with 251 receiving combination therapy with either a thiopurine or methotrexate.¹²⁶ Similarly, a recently published meta-analysis included 2053 individuals with CD across 33 studies, combining both 6 RCTs with 28 observational studies.¹²⁷

Benefits

There have been heterogeneous results regarding the benefits of combination therapy of non-TNF-targeting biologics and immunomodulators in clinical trials. In subgroup analyses examining individuals receiving immunomodulators in combination with vedolizumab in GEMINI, there were similar rates of remission in those receiving immunomodulators compared with those who did not at 10 weeks (32.4% vs 26.8%).⁴⁸ Similarly, there were variable differences in response rates in those receiving an immunomodulator vs not, when stratified by concomitant steroids at week 6 for ustekinumab in UNITI.⁵² Meta-analyses and observational data to date suggest that there may be no benefit of combining an immunomodulator with a non-TNF-targeting biologic. In a meta-analysis conducted by Yzet et al,¹²⁷ incorporating data from 6 clinical trials as well as 28 observational studies, there was no significant difference between combination therapy and monotherapy for vedolizumab (OR, 0.84; 95% CI, 0.68–1.05) or ustekinumab (OR, 1.10; 95% CI, 0.87–1.38). Given their recent FDA approval, there remains a paucity of data regarding IL23 p19 antagonists.

Harms

There are limited available data regarding the rates of adverse events when receiving combination therapy with non-TNF antagonists and immunomodulators. Subgroup analyses from clinical trials as well as observational studies to date demonstrate no significant increase in serious infection or other SAEs, though long-term follow-up data are lacking and it would be appropriate to consider the previously described risks of adverse events associated with immunomodulators.

Certainty in Evidence of Effects

The evidence reviewed was derived from post-hoc subgroup analyses of RCTs in patients who have previously failed immunomodulators, as well as systematic reviews and meta-analyses of observational studies, and was rated as very low.

Discussion

Due to uncertainty in the magnitude of benefit and harm, the panel felt this was a knowledge gap and did not issue a recommendation in favor of, or against, the use of combining non-TNF-targeting biologics and immunomodulators. One of the putative mechanisms through which immunomodulators may enhance the effectiveness of biologic agents, by improving the pharmacokinetic properties of the biologic agent, may not apply to non-TNF-targeting biologics. As discussed previously, vedolizumab and ustekinumab appear to be less immunogenic

than TNF antagonists.^{128,129} In a multicenter retrospective cohort study of 369 patients with IBD, Yarur et al¹¹⁹ examined the association between thiopurine exposure and the pharmacokinetic properties of infliximab, vedolizumab, and ustekinumab. Although concomitant thiopurines were associated with higher trough concentration of infliximab, there was no association between the use of immunomodulators and vedolizumab or ustekinumab trough concentrations, nor was there an association with the risk of loss of response. The prevalence of anti-drug antibodies was <1% with vedolizumab and ustekinumab. However, there may be other mechanisms by which immunomodulators may enhance the effectiveness of biologics. As seen in the SONIC trial, in immunomodulator- and biologic-naïve patients with CD, the combination of thiopurines and infliximab was more effective than infliximab monotherapy, with plausibly some benefit attributed to the independent therapeutic effect of thiopurines on CD. Although such a trial does not exist for non-TNF-targeting biologics, real-world trial emulation studies comparing outcomes with combination therapy of vedolizumab and immunomodulators vs vedolizumab monotherapy suggest lower risk of treatment failure with combination therapy in patients with CD.¹³⁰ Although the benefit of combination therapy of immunomodulators with non-TNF-targeting biologics is not well-defined, there may be specific patient populations, such as those with medically refractory disease, prior immunogenic treatment failures to biologic therapies, or with concomitant medical diagnosis, such as autoimmune hepatitis or rheumatoid arthritis, that may benefit from this approach. There is a limited role for combining immunomodulators with JAK inhibitors. Because there is no risk of immunogenicity with small molecule drugs, and there is a clearer dose-response relationship (rather than an exposure-response relationship), any purported benefit with adding immunomodulators is unlikely to be due to improving the pharmacokinetic properties. In trials of upadacitinib in patients with CD, immunomodulators were discontinued before starting upadacitinib due to concern for adverse effects.

Evidence Gaps and Future Research

Clinical trials examining the efficacy and safety of combining non-TNF-targeting biologics with thiopurines and/or methotrexate vs monotherapy with non-TNF-targeting biologics, in both immunomodulator-naïve and immunomodulators-exposed patients are warranted to inform this question.

De-Escalation of Combination Therapy

Question 7: In adult outpatients with moderate-to-severely active CD in steroid-free remission on combination therapy with a TNF antagonist and an immunomodulator, is discontinuation of (a) an immunomodulator or (b) discontinuation of the TNF antagonist, inferior to continuation of combination therapy?

Recommendation 13: In adult outpatients with moderate-to-severely active CD who are in corticosteroid-free clinical remission for at least 6 months on combination therapy with TNF antagonists

and an immunomodulator, the AGA SUGGESTS withdrawing the IMMUNOMODULATOR. (*Conditional recommendation, low certainty of evidence*)

Comment: Patients, particularly those with difficult-to-treat disease, who place a lower value on the trivial-to-small increase in risk of long-term adverse effects of continuing immunomodulators (such as risk of malignancy or infection), and a higher value on avoiding a trivial-to-small increase in risk of relapse with withdrawal of immunomodulators, may reasonably choose to continue combination therapy.

Implementation Considerations

- There may be benefit in continuing combination therapy with TNF antagonists and immunomodulators in those who are felt to be at higher risk of immunogenicity, such as those with prior immunogenic failure to a biologic therapy (ie, anti-drug antibody formation), those with lower trough TNF-antagonist concentrations despite dose escalation, or HLA-DQA1*05 carriers.

Recommendation 14: In adult outpatients with moderate-to-severely active CD who are in corticosteroid-free clinical remission for at least 6 months on combination therapy with TNF antagonists and an immunomodulator, the AGA suggests AGAINST withdrawal of the TNF ANTAGONIST. (*Conditional recommendation, low certainty of evidence*)

Source of Evidence

Seven RCTs were identified comparing withdrawal of either immunomodulators or TNF antagonists to continued combination therapy.^{131–137} Five of these studies enrolled only individuals with CD.^{131,133,134,136,137} In 4 of these trials, the primary intervention compared withdrawing an immunomodulator to continued combination therapy, and in 2 trials the primary intervention was TNF-antagonist withdrawal. One trial included 3 arms, comparing immunomodulator withdrawal, TNF-antagonist withdrawal, and continued combination therapy.¹³⁶ One trial included immunomodulator dose reduction.¹³²

There was some heterogeneity regarding patients across these studies. Most patients enrolled in these studies were in corticosteroid-free remission for at least 6 months, with the average duration of remission exceeding 2 years; however, 1 included study of pediatric patients allowed clinical response.¹³³ Four of the reviewed studies required not only clinical remission but also biochemical and/or structural remission at randomization. Prior immunomodulator use/failure was also variable, with 1 study including individuals who had all failed an immunomodulator, another including only immunomodulator naïve individuals, and two studies not reporting results stratified by this prior exposure^{131–134}; 1 study reported no significant difference when stratifying results among those who had or had not been previously immunomodulator exposed.¹³⁶ Follow-up time ranged from 12 to 24 months.

Benefits

Immunomodulator withdrawal. In total, 404 individuals from 5 studies were included in a meta-analysis of immunomodulator withdrawal (202 with withdrawal, 202 with continued combination therapy). There was no significant difference in the overall risk of relapse with immunomodulator withdrawal compared with continued combination therapy (16.8% vs 14.9%; RR, 1.15; 95% CI, 0.75–1.76) ([Supplementary Figure 12](#)).

TNF-antagonist withdrawal. When examining outcomes related to TNF-antagonist withdrawal, 3 trials were evaluated, with 339 individuals. The rate of clinical relapse was higher in those with TNF-antagonist withdrawal than with continued combination therapy (30.6% vs 11.2%; RR, 2.23; 95% CI, 1.08–4.61) ([Supplementary Figure 12](#)).

Harms

Immunomodulator withdrawal. The risks associated with immunomodulators or TNF antagonists, as well as their combination are discussed previously, and include increased risks of infection as well as malignancies, including, but not limited to, nonmelanoma skin cancer and lymphoproliferative disorders. In the included RCTs, 6 studies examined rates of SAEs during follow-up and found no significant difference (immunomodulator withdrawal vs continued combination therapy: RR, 1.21; 95% CI, 0.68–2.17). However, it is important to emphasize that event rates were low, sample sizes were small, and duration of follow-up was limited to 12–24 months. One could hypothesize that withdrawing 1 of the 2 therapies may potentially mitigate the long-term risks of combination therapy. Several observational cohorts have demonstrated an increased risk of adverse events with combination therapy in relation to monotherapy. For example, several population-based studies have highlighted the increased risk of serious infections, such as pneumonia over time with combination therapy in comparison to monotherapy.^{138,139} Neoplastic risks may increase with cumulative exposure to combination therapy, such as the risk of lymphoproliferative disorders with combination TNF antagonist and thiopurine therapy.^{28,107} This effect may be cumulative over time, with at least 1 study demonstrating that the incidence rate of acute myeloid leukemia increased at least 2 years after thiopurine exposure in a cohort of patients with IBD.¹⁴⁰ Lymphoproliferative risks may decrease over time after thiopurine cessation, however.¹⁰⁷

Tumor necrosis factor-antagonist withdrawal. In the included RCTs, the risk of SAEs was similar between groups, although the number of events was relatively low (RR, 0.82; 95% CI, 0.36–1.88). As with any change in therapy in CD, the increase or decrease of adverse event risk associated with treatment modifications must be balanced against the long-term risk of disease progression. This may be of particular importance when considering TNF-antagonist withdrawal during combination therapy, as the studies that assessed biologic withdrawal here demonstrated a significant increased risk of flare.

Certainty in Evidence of Effects

Immunomodulator withdrawal. The certainty of evidence for withdrawal of immunomodulators was judged to be low ([Table 15](#)). This was primarily secondary to risks of bias due to the inclusion of open-label study designs with subjective, symptom-based outcomes and serious imprecision. When considering the evidence-to-decision framework, the desirable effects of immunomodulator withdrawal (lowering risk of infectious and neoplastic complications) and undesirable anticipated effects (higher risk of CD relapse) was felt to be trivial-to-small. Based on the potential for reducing long-term risks of immunomodulator therapy, and the evidence demonstrating no significant increase in relapse rates in CD in available clinical trials, the panel suggested in favor of withdrawal of immunomodulators in patients in clinical remission on combination therapy.

TNF-antagonist withdrawal. Regarding the withdrawal of TNF antagonists, the certainty of evidence was judged to be low ([Table 15](#)). Besides risk of bias, evidence was rated down for imprecision as the optimal information size was not met. When considering the evidence-to-decision framework, the desirable effects were felt to be trivial, but the undesirable effects, as demonstrated by the increased risk of flare with TNF-antagonist withdrawal, was felt to be moderate. Therefore, the balance of effects favor continuation of combination therapy over withdrawal of TNF antagonists. The GRADE evidence-to-decision judgments are presented in [Table 16](#).

Discussion

Immunomodulator withdrawal. Clinical trial evidence suggests that clinical outcomes are similar when withdrawing immunomodulators in those in corticosteroid-free remission on combination therapy. This serves as a potential opportunity to mitigate longer-term risks of combination TNF-antagonist and immunomodulator therapy. However, there are several important aspects to consider when implementing these recommendations. First, although most patients included in these trials had CD, analyses for several studies reviewed also included patients with ulcerative colitis. There was heterogeneity across assessed trials with regards to enrolled patient characteristics. This includes prior immunomodulator failure and follow-up time. However, point estimates across studies were relatively consistent ([Supplementary Figure 12](#)). The impacts of immunomodulator withdrawal on both TNF-antagonist trough levels as well as long-term immunogenicity were not routinely assessed, and it is possible that these effects may become more apparent with increasing follow-up time. In 3 trials, infliximab trough concentrations at the end of follow-up were lower and the proportion of patients with antibodies to infliximab was higher in patients who underwent immunomodulator withdrawal compared with those who continued combination therapy. This may be an infliximab specific effect, however, as similar effects were not appreciated with adalimumab in DIAMOND2.¹³¹ Studies with longer time horizons may better elucidate the impact of long-term

Table 15. Grading of Recommendations Assessment, Development and Evaluation Evidence Profile Comparing Withdrawal of Immunomodulators or Withdrawal of Biologics Compared With Continuing Combination Therapy for Risk of Relapse In Patients With Moderate-to-Severely Active Crohn's Disease in Steroid-Free Remission on Combination Therapy

Outcomes	Study event rates, n/N (%)		Relative effect, RR (95% CI)	Absolute effect	No. of participants (studies)	Quality of the evidence (GRADE) ^a
	Risk with placebo	Risk with IMM withdrawal				
Withdrawal of IMMs (while continuing biologic therapy) compared with continuing combination therapy in adult outpatients with moderate-to-severe CD in steroid-free remission						
Risk of relapse at 12 mo (CRITICAL)	30/202 (14.9)	34/202 (16.8)	1.15 (0.75–1.76)	22 more per 1000 (from 37 fewer to 113 more)	404 (5 RCTs)	⊕⊕○○ ^{b,c} LOW

Outcomes	Study event rates, n/N (%)		Relative effect, RR (95% CI)	Absolute effect	No. of participants (studies)	Quality of the evidence (GRADE) ^a
	Risk with continuing combination therapy	Risk with biologic withdrawal				
Withdrawal of biologics (while continuing IMM monotherapy) compared with continuing combination therapy in adult outpatients with moderate-to-severe CD in steroid-free remission						
Risk of relapse at 12 mo (CRITICAL)	19/169 (11.2)	52/170 (30.6)	2.23 (1.08–4.61)	138 more per 1000 (from 9 more to 406 more)	396 (4 RCTs)	⊕⊕○○ ^{b,c} LOW

IMM, immunomodulator.

^aGRADE Working Group grades of evidence. High quality: We are very confident that the true effect lies close to that of the estimate of the effect. Moderate quality: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low quality: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect. Very low quality: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect.

^bRated down for serious risk of bias (open-level trials with subjective end points).

^cRated down for imprecision because optimal information size not met (<200 events).

Table 16. Grading of Recommendations Assessment, Development and Evaluation Evidence-to-Decision Framework for Use of Biologic Withdrawal or Immunomodulator Withdrawal Over Continuing Combination Therapy in Patients With Moderate-to-Severely Active Crohn's Disease in Steroid-Free Remission for >6 Months

Domain		Criteria		Judgment			
Problem	Is the problem a priority?	No	Probably no	Probably yes	Yes	Varies	Do not know
Desirable effects	How substantial are the desirable anticipated effects?	Trivial (<5%) IMM withdrawal, TNF-antagonist withdrawal	Small	Moderate (10%–30%)	Large (>30%)	Varies	Do not know
Undesirable effects	How substantial are the undesirable anticipated effects?	Trivial IMM withdrawal	Small	Moderate TNF-antagonist withdrawal	Large	Varies	Do not know
Certainty of evidence	What is the overall certainty of the evidence of effects	Very low		Low	Moderate	High	No included studies
Values	Is there important uncertainty about or variability in how much people value the main outcomes?	Important uncertainty or variability		Possibly important uncertainty or variability	Probably no important uncertainty or variability		No important uncertainty or variability
Balance of effects	Does the balance between desirable and undesirable effects favor the intervention or the comparison?	Favors comparison/control	Probably favors comparison TNF-antagonist withdrawal	Does not favor either comparison or intervention	Probably favors intervention IMM Withdrawal	Favors intervention	Varies/do not know
Resource use	Is the incremental cost small relative to the net benefits?	No	Probably No TNF-antagonist withdrawal	Uncertain	Probably yes IMM Withdrawal	Yes	Uncertain
Equity	What would be the impact on health inequities?	Increased	Probably increased TNF-antagonist withdrawal	Probably no impact IMM Withdrawal	Probably reduced	Reduced	Varies/do not know
Acceptability	Is the option acceptable to key stakeholders?	No	Probably No	Probably yes	Yes	Varies	Do not know
Feasibility	Is the option feasible to implement?	No	Probably No	Probably yes	Yes	Varies	Do not know

NOTE. CMD threshold of active interventions was set at 5%. Judgments made by guideline panel are in bold type. IMM, immunomodulator.

therapy withdrawal. Moreover, TNF-antagonist dose adjustment may be needed if drug levels decrease appreciably following immunomodulator withdrawal. Finally, there are no data on immunomodulator withdrawal when using subcutaneous infliximab.

It is also important to individualize the recommendation of immunomodulator withdrawal, incorporating disease-specific factors that may modulate the risk of relapse. Those with more moderate disease, such as those with limited disease extent, absence of concerning features, such as penetrating complications, signs of systemic disease, absence of deep or serpiginous ulceration on endoscopy, without a known history of immunogenicity, and with confirmed biochemical or structural remission for a longer duration may be at lower risk of subsequent relapse after immunomodulator withdrawal. The association between factors such as these and potential risks of relapse were inconsistent across the studies reviewed here, however. Patient preferences regarding the risks of both continued combination therapy and the risks of subsequent relapse with immunomodulator withdrawal should be carefully considered in the shared decision-making process. In patients who value the potential benefit of combination therapy in decreasing risk of relapse, and place lower value on the risks of long-term combination therapy, it would be reasonable to continue combination therapy. Similarly, those at higher risk of immunogenic treatment failure with TNF antagonists, such as those with prior anti-drug antibody development or documented HLA DQA1*05 carriage status may reasonably choose to consider continuing combination therapy. It would also be reasonable to consider therapeutic drug monitoring before and after withdrawal of immunomodulators to ensure that TNF-antagonist levels remain within the therapeutically acceptable range, particularly for infliximab.¹⁴¹ Continued biochemical and/or structural assessment in the post-withdrawal period is appropriate to monitor for evidence of recurrence of disease as well.

TNF-antagonist withdrawal. The deleterious impact of TNF-antagonist withdrawal was noted across the clinical trials reviewed for this study. However, when interpreting these data, it is important to consider the same factors related to heterogeneity across the literature as noted above. As with immunomodulator withdrawal, there was relative consistency across point estimates in the reviewed trials (Supplementary Figure 12). Although SAE rates were similar across studies, there was a significant association between TNF-antagonist withdrawal and subsequent >2-fold increased risk of relapse of disease. There may still be a subgroup of patients who wish to pursue this strategy due to concerns about, for example, adverse events related to long-term treatment, desire to maintain drug-free remission, and financial considerations. Although the risk of flare is likely substantial with this approach, data from the SPARE trial, as well as observational studies suggest that resumption of TNF antagonist at time of flare frequently recaptures clinical remission.¹³⁶ Interpretation of these findings is limited by both sample size

considerations and limited follow-up time, however. In the context of shared decision making, this potential plan of reintroduction of the TNF antagonist may be informative for those contemplating withdrawal of their biologic therapy.

Evidence Gaps and Future Research

As highlighted above, there are several evidence gaps and potential targets for future research. Future pragmatic trials of immunomodulator withdrawal should incorporate longer time horizons, with continued focus on persistence of remission but also uniform incorporation of TNF-antagonist drug level and anti-drug antibody assessment to better understand the long-term implications on drug pharmacokinetics. In addition, future trials are required to assess clinical remission rates for advanced therapy withdrawal, to not only measure persistence of remission, but also rates of medication reinitiation and response rates to reintroduction after withdrawal. Such work should incorporate assessing patient preferences, as well as health-related quality of life over time and evaluate better predictors of who may successfully discontinue therapy in IBD. Similar trials of withdrawal of non-TNF biologics and oral small molecule drugs are also required.

Upfront Use of Advanced Therapies

Question 8: In adult outpatients with moderate-to-severely active CD, is upfront use of advanced therapies superior to step-up therapy for decreasing the risk of disease-related complications?

Recommendation 15: In adult outpatients with moderate-to-severely active CD, the AGA SUGGESTS upfront use of advanced therapy compared with step-up therapy with initial use of corticosteroids and/or immunomodulator monotherapy. (Conditional recommendation, very low certainty of evidence)

Implementation Considerations

- All trials that informed the evidence used combination therapy with TNF antagonists with immunomodulators.
- Some individuals may still derive benefit from the use of corticosteroids for symptomatic management while initiating advanced therapy.

Source of Evidence

Three studies primarily assessing combination therapy with a TNF antagonist and immunomodulator informed these recommendations; 2 of these studies included relative short duration disease and 1 study examined those with longer disease duration. In an open-label study by D'Haens et al¹⁴² of 133 patients with recently diagnosed CD who were naïve to advanced therapies and steroids with <4 years of disease duration,

individuals were randomized to start combination therapy with infliximab with standard induction and episodic dosing for maintenance vs corticosteroids followed by a thiopurine and then TNF antagonist sequentially on therapy failure. REACT-1 was a multicenter open-label cluster RCT consisting of 1982 participants. Individuals were managed with either an algorithm incorporating combination therapy with a TNF antagonist (adalimumab) if not improved after an initial steroid course (CI group), or treated via conventional management (CM group).¹⁴³ Conventional management was not algorithmically defined; treatment selection was at the sole discretion of the treating provider. Adherence rates to the CI protocol were approximately 80%–85%, and adherence to the CM protocol was 10%–15%. There were significantly increased rates of immunomodulator (HR, 1.84; 95% CI, 1.39–2.44), TNF antagonist (HR, 1.72; 95% CI, 1.34–2.22), and combination therapy (HR, 2.29; 95% CI, 1.73–3.04) use in the CI group compared with the CM group over time. Importantly, the pragmatic study design of REACT-1 allowed for enrollment of individuals who were previously exposed to immunomodulators, TNF antagonists, and combination therapy before study entry. This study consisted primarily of those with longstanding disease with a median disease duration in both arms exceeding 12 years (CI 149 months; CM 158 months). Lastly, in the PROFILE study, Noor et al¹⁴⁴ compared combination therapy with infliximab and an immunomodulator to an accelerated step-up algorithm with an initial steroid taper among 386 individuals with CD diagnosed within 6 months of study entry. Rates of sustained steroid- and surgery-free remission were assessed at 48 weeks, as well as rates of disease-related adverse events.¹⁴⁴

Benefits

In the study by D'Haens et al,¹⁴² 133 individuals were enrolled; at 52 weeks of follow-up, 40 of 65 patients (61.5%) in the upfront combined immunosuppression group were in corticosteroid- and surgery-free remission, compared with 23 of 64 patients (42.2%) in the step therapy arm (RR, 0.67; 95% CI, 0.46–0.97). In the multicenter pragmatic REACT-1 study, there was no significant difference in the primary outcome of steroid-free remission at 12 months through 24 months, although there was a significantly lower rate of disease-related adverse events, which were defined as hospitalization-, surgery- or disease-related complications including abscess, fistula, stricture, or serious worsening of disease activity or extra-intestinal manifestations (HR, 0.73; 95% CI, 0.62–0.86).¹⁴³ Lastly, in PROFILE, 386 individuals were enrolled. Similar rates of symptomatic remission were observed at 1 year (RR, 1.12; 95% CI, 0.98–1.30), but there were significantly lower rates of disease-related adverse events with early combination therapy (RR, 0.23; 95% CI, 0.09–0.59).¹⁴⁴ Although clinical remission was selected as the most patient-centric outcome of interest for these guidelines and thus was considered most strongly here, initial combination therapy was

associated with significant benefit in the primary outcomes in PROFILE, which included steroid- and surgery-free remission (79% vs 15%; $P < .001$) and endoscopic healing (67% vs 44%; $P < .001$).

In a meta-analysis of these studies, a total of 2497 individuals were included across 3 clinical trials. With regard to clinical remission at 12 months, the benefit of upfront combined immunosuppression over step therapy was uncertain (66.4% vs 60.6%; RR, 1.18; 95% CI, 0.96–1.46). However, there may be a lower risk of disease-related complications at 12 months with early combination therapy compared with step therapy (17.7% vs 14.8%; RR, 0.62; 95% CI, 0.29–1.34).

Harms

The risk associated with immunomodulators and advanced therapies in CD have been discussed previously. It is important to consider that initial advanced therapy use likely increases the duration of exposure to these agents, and thereby exposure-related risk. However, importantly, these risks should be balanced against the potential consequence of disease-related complications secondary to inadequate treatment of active disease. As noted below, several studies, primarily in pediatric patient populations, have demonstrated an association between delays in initiation of advanced therapies and penetrating complications in CD.^{145–147}

Certainty in Evidence of Effects

The certainty of evidence quality was determined to be very low (for clinical remission) to low (for disease-related complications) (Table 17). Although the overall population included across the 3 included studies was large, risks of bias were introduced through the inclusion of open-label studies, and there was imprecision in the derived estimates. Specifically, the quality of the evidence in relation to clinical remission was rated as very low, due to serious risk of bias due to the inclusion of open-label trials, imprecision, and inconsistency in effect estimates. When considering disease-related complications, the quality of evidence was determined to be low. Although the trials were open-label, we opted not to rate down for risk of bias because the severe disease-related complication end points were objective. The evidence was rated down due to imprecision in estimates. Lastly, there were fairly significant differences in disease duration across the 3 included studies. Individuals enrolled in the study by D'Haens et al¹⁴² were required to have been enrolled within 4 years of diagnosis and the median time from diagnosis to enrollment was 2–2.5 weeks. Similarly, in PROFILE, the mean time from diagnosis to enrollment was 31.2 and 24.1 days in each treatment arm.¹⁴⁴ This is in contrast to REACT-1, where the mean disease duration before enrollment was 149 and 158.1 months in the initial combined immunosuppression and conventional arms, respectively.¹⁴³ The GRADE evidence-to-decision judgments are presented in Table 18.

Table 17. Grading of Recommendations Assessment, Development and Evaluation Evidence Profile Comparing Early Combined Immunosuppression With Step Therapy in Adults With Moderate-to-Severely Active Crohn's Disease

Outcomes	Study event rates, n/N (%)		Relative effect, RR (95% CI)	Absolute effect	No. of participants (studies)	Quality of the evidence (GRADE) ^a
	Risk with placebo	Risk with ECI				
ECI compared with step therapy in adults with active CD						
Clinical remission at 12 mo (CRITICAL)	700/1155 (60.6)	891/1342 (66.4)	1.18 (0.96–1.46)	109 more per 1000 (from 24 fewer to 279 more)	2497 (3 RCTs)	⊕○○○ ^{b,c,d} VERY LOW
Disease complications at 12 mo (CRITICAL)	204/1155 (17.7)	198/1342 (14.8)	0.62 (0.29–1.34)	67 fewer per 1000 (from 125 fewer to 60 more)	2497 (3 RCTs)	⊕⊕○○ ^{b,c,e} LOW

ECI, early combined immunosuppression.

^aGRADE Working Group grades of evidence. High quality: We are very confident that the true effect lies close to that of the estimate of the effect. Moderate quality: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low quality: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect. Very low quality: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect.

^bRated down for serious risk of bias (open-level trials).

^cRated down for imprecision due to wide CIs.

^dRated down for inconsistency in effect size across trials.

^eAlthough the trials were open label, the end point of disease complications was relatively objective, and open-label nature is unlikely to impact this. Hence, we opted not rate down for risk of bias.

Discussion

With the introduction of advanced therapies, the overall trajectory of disease for patients with moderate-to-severely active CD has improved, with decreasing rates of surgical intervention over time.⁴ This is likely secondary to the reduced risk of structural damage, such as penetrating complications like fistula or abscess formation, as well as reduced rates of fibrostenosis that come with adequate control of the robust inflammatory response present in CD. It would therefore stand to reason that reducing overall exposure time to such inflammation through the upfront use of advanced therapies as opposed to a gradual step-up approach would further reduce the risks of these outcomes. However, as highlighted by the data reviewed by the guideline panel here, there are limited prospective clinical trials to assess this effect. Several observational studies have appreciated an association between time to advanced therapy use and reduced rates of surgical intervention. For example, in a retrospective pediatric cohort of 913 patients, individuals receiving a TNF antagonist within 90 days of diagnosis were significantly less likely to experience penetrating complications over time after propensity score adjustment for multiple disease-related risk factors (HR, 0.30; 95% CI, 0.10–0.89).¹⁴⁵ Early TNF antagonist use has also been associated with reduced rates of perianal complications in pediatric CD in 2 separate cohorts.^{146,147} Although observational studies remain at risk of bias, and additional clinical trials would be beneficial to confirm these findings, these data

suggest that patients with moderate-to-severely active CD would benefit from initiation of advanced therapy earlier in the disease course to prevent disease-related complications.

There are several caveats when interpreting both the clinical trial and observational data presented here. First, all trials assessing upfront advanced therapy compared with step-therapy included in the meta-analyses used by the panel focused on the combination of TNF-antagonist and immunomodulators therapy. It remains uncertain whether the early initiation of TNF-antagonist monotherapy or other advanced therapies would provide increased or reduced benefit in reducing downstream complications. Similarly, across studies, there was variability in disease duration and follow-up time was limited. There was heterogeneity in prior treatment exposures, and 1 study allowed for prior immunomodulator and advanced therapy use at cohort entry. It is possible that the benefits of initial use may increase with longer follow-up time, although long-term treatment failures could also mitigate these effects. Benefits of initial advanced therapy use may also be magnified in those who are truly treatment-naïve and who received advanced therapy earliest in their disease course; These effects may be most greatly appreciated in PROFILE, where individuals were treatment-naïve and enrolled shortly after diagnosis. In PROFILE, there was a significant association between upfront advanced therapy use and the primary outcomes of

Table 18. Grading of Recommendations Assessment, Development and Evaluation Evidence-to-Decision Framework for Use of Combination Therapy Over Tumor Necrosis Factor Monotherapy for the Management of Patients With Moderate-to-Severely Active Crohn's Disease

Domain	Criteria	Judgment					
Problem	Is the problem a priority?	No	Probably no	Probably yes	Yes	Varies	Do not know
Desirable effects	How substantial are the desirable anticipated effects?	Trivial to small (<10%)		Moderate (10%–30%)	Large (>30%)	Varies	Do not know
Undesirable effects	How substantial are the undesirable anticipated effects?	Trivial to small		Moderate	Large	Varies	Do not know
Certainty of evidence	What is the overall certainty of the evidence of effects	Very low (clinical remission)		Low (disease complications)	Moderate	High	No included studies
Values	Is there important uncertainty about or variability in how much people value the main outcomes?	Important uncertainty or variability		Possibly important uncertainty or variability	Probably no important uncertainty or variability	No important uncertainty or variability	
Balance of effects	Does the balance between desirable and undesirable effects favor the intervention or the comparison?	Favors comparison/control	Probably favors comparison	Does not favor either comparison or intervention	Probably favors intervention		Favors intervention Varies/do not know
Resource use	Is the incremental cost small relative to the net benefits?	No	Probably no	Uncertain	Probably yes	Yes	Uncertain
Equity	What would be the impact on health inequities?	Increased	Probably increased	Probably no impact	Probably reduced	Reduced	Varies/do not know
Acceptability	Is the option acceptable to key stakeholders?	No	Probably no	Probably yes	Yes	Varies	Do not know
Feasibility	Is the option feasible to implement?	No	Probably no	Probably yes	Yes	Varies	Do not know

NOTE. CMD threshold of active interventions was set at 5%. Judgments made by guideline panel are in bold type.

endoscopic remission and steroid- and surgery-free remission.¹⁴⁴ However, the long-term implications of this approach remain incompletely understood: upfront use of advanced therapies inherently increases exposure time to the medications of interest and the potential risks of immunosuppression. Improved predictive markers to identify individuals who might progress more rapidly may serve to further refine the identification of who might most benefit from earlier advanced therapy initiation. Lastly, although the recommendation of the guideline panel supports the upfront use of initial advanced therapy over a step-up treatment algorithm, this does not exclude the potential benefit of an initial course of corticosteroids while initiating advanced therapy for some individuals. Multiple factors can increase time to advanced therapy initiation, and these acute delays have been shown to both reduce quality of life and increase the risk of health care utilization.¹⁴⁸ Corticosteroids likely still have a role in this initiation process in those most significantly affected by active disease without infection-related complications that would preclude their use.

Evidence Gaps and Future Research

Several evidence gaps should be informed by future research. First, the appropriate window for upfront advanced therapy initiation remains unclear—existing trials analyzed here exhibited significant variability in disease duration, with the strongest potential benefits in terms of endoscopic healing and avoidance of surgery being appreciated in PROFILE, which incorporated the most stringent inclusion criterion related to time from diagnosis. Secondly, there are limited data regarding the impact of upfront initiation of non-TNF-antagonist advanced therapies. Future studies could consider assessing the comparative effectiveness of early use of different classes of advanced therapies with uniform enrollment and initiation time windows to address these concerns. It is also possible that there are subsets of patients that may uniquely benefit from early combination therapy in contrast to others, and unfortunately there are not yet predictive markers that have been shown to clearly identify this population. An example of such an approach would be PROFILE, where individuals were stratified by higher or lower risk for future disease related complications using a panel of 17 genetic markers. Stratified analyses by high- and low-risk designation demonstrated no significant difference in response to early combination therapy.¹⁴⁴ As future predictive markers are assessed and validated in observational research, they should be similarly incorporated into clinical trials, such as those described above. Lastly, it is possible that the benefit derived from early combined immunosuppression increases or erodes over time. When examining longer duration follow-up in the trial by D'Haens et al¹⁴² at 104 weeks, no differences were apparent between top-down therapy vs step therapy. Longer follow-up times in prospectively monitored cohorts would allow for closer monitoring of disease progression over time and ascertain whether there is regression in benefit.

Treating to Endoscopic Remission vs Clinical Remission

Question 9: In adult outpatients with moderate-to-severely active CD, is treat-to-target of endoscopic remission (resolution of inflammation on endoscopy) superior to treat-to-target of symptomatic remission, for maintenance of remission and decreasing risk of disease-related complications?

Recommendation 16: In adult outpatients with moderate-to-severely active CD, the AGA makes **NO RECOMMENDATION** in favor, or against, treating to a target of endoscopic remission, compared with treating to a target of symptomatic remission. (No recommendation, knowledge gap)

Source of Evidence

Two RCTs were reviewed to assess whether treating to a target of endoscopic remission (systematic assessment for symptoms and endoscopic inflammation, followed by treatment escalation in those with evidence of inflammation, regardless of presence or absence of symptoms) vs treating to a target of symptomatic remission (systematic assessment for symptoms, followed by treatment escalation in those with ongoing symptoms) yielded higher rates of long-term remission and reduced CD-related complications. STARDUST was an open-label RCT comparing a treat-to-target strategy incorporating endoscopic monitoring to standard care in individuals with moderate-to-severely active CD who were initiating ustekinumab.¹⁴⁹ Individuals in the treat-to-target monitoring arm received repeated endoscopic assessments at study entry and week 16 in order to guide decision making. Those with <25% improvement in the *Simplified Endoscopic Activity Score for Crohn's Disease* were assigned to receive every 8-week maintenance dosing, whereas those who met that target received every 12-week dosing. Although every 12-week dosing of ustekinumab is approved in other countries, this is not consistent with the FDA-approved dosing interval of every 8 weeks in the United States. In the standard-care arm, maintenance dosing was dictated by symptom response. Both arms received routine biochemical monitoring as well. The primary outcome was endoscopic response at 48 weeks; endoscopic remission rates and clinical remission and response rates were assessed as secondary outcomes.

The second trial assessed was REACT-2, a clustered RCT comparing a strategy of systematic colonoscopic (and symptom) assessment at baseline and then at 6, 12, and 24 months vs systematic evaluation of symptoms, with step-wise algorithmic treatment escalation if the respective target (endoscopic remission vs symptomatic remission) was not met.¹⁵⁰ The measured outcomes at 24 months included clinical and endoscopic remission.

Benefits

Two clinical trials were included in meta-analyses to inform the recommendations for these guidelines. In STARDUST, there was no significant difference between

incorporating endoscopic assessment in a treat-to-target algorithm vs usual care in rates of clinical (62% vs 70%) or endoscopic remission (38% vs 30%) at 48 weeks. Adverse event rates were similar during follow-up in both arms as well (12% vs 13%).¹⁴⁹ In REACT-2, outcomes of clinical and endoscopic remission were assessed at 24 months. There was a higher rate of clinical remission in the endoscopic monitoring arm than in the usual care arm (54.6% vs 45.1%), but no difference in endoscopic outcomes (33.9% vs 31.9%).¹⁵⁰ Based on these 2 studies, there were no significant differences in the outcome of clinical remission (63.2% vs 57.3%; RR, 1.04; 95% CI, 0.78–1.39).

Harms

Risks associated with both immunomodulators, advanced therapy, and combination therapy have been discussed previously. When considering endoscopic remission, one could consider the risks and financial implications of endoscopic evaluation, as well as the disutility of the procedure itself. Although such risks are tangible, they are also relatively self-limited when placed in the context of improving the overall course of disease in CD. Although the studies included here specifically focused on adjusting the dosing of the index therapy for each trial, one could consider this approach in the context of switching to an alternative therapy if dose escalation has already occurred. In these instances, there would be a potential risk of future flares in those who had achieved a clinical

response when striving for endoscopic remission as well. Lastly, although available data to date suggest that dose escalation to reach clinical end points is not associated with significantly modified adverse event risks for biologic advanced therapies, this may not be the case for small molecule therapies, such as JAKs, and does not account for the potential implications from a cost standpoint related to treatment modifications. Prior clinical trials incorporating escalating therapy with adalimumab in response to clinical and biochemical monitoring demonstrated increased adalimumab utilization, but also noted overall decreased costs, likely due to improved disease control and decreased subsequent health care utilization.^{151,152} This may be reflective of system-related effects, but may not fully capture the out-of-pocket experiences of an individual patient.

Certainty in Evidence of Effects

The overall certainty of evidence was determined to be very low due to multiple factors, including serious risk of bias due to the inclusion of studies with an open-label design, imprecision, and inconsistency across effect estimates (Table 19). Importantly, dosing of ustekinumab was also not consistent with the currently approved dosing in the United States; in STARDUST, maintenance dosing was every 12 weeks.¹⁴⁹ In addition, it is important to note heterogeneity across patient populations. Although individuals enrolling in STARDUST had active disease, defined via clinical and endoscopic disease activity, there

Table 19. Grading of Recommendations Assessment, Development and Evaluation Evidence Profile Comparing Treating to a Target of Endoscopic Remission in Comparison With a Target of Clinical Remission in Adults With Moderate-to-Severely Active Crohn's Disease

Outcomes	Study event rates, n/N (%)		Relative effect, RR (95% CI)	Absolute effect	No. of participants (studies)	Quality of the evidence (GRADE) ^a
	Risk with placebo	Risk with TTT endoscopic remission				
TTT of endoscopic remission compared with TTT of clinical remission in adults with CD						
Clinical remission at 12 mo (CRITICAL)	453/790 (57.3)	469/742 (63.2)	1.04 (0.78–1.39)	23 more per 1000 (from 126 fewer to 224 more)	1532 (2 RCTs)	⊕○○○ ^{b,c,d} VERY LOW

TTT, treat-to-target.

^aGRADE Working Group grades of evidence. High quality: We are very confident that the true effect lies close to that of the estimate of the effect. Moderate quality: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low quality: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect. Very low quality: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect.

^bRated down for serious risk of bias (open-level trials).

^cRated down for imprecision due to wide CIs.

^dRated down for inconsistency in effect size across trials ($I^2 = 90\%$).

was a large proportion of individuals enrolling in REACT-2 that did not have active disease.

Discussion

Recent position statements from an international consortium of experts have advised that longitudinal targets for the management of IBD should include not only clinical remission but also endoscopic resolution of inflammation.³¹ Several studies have demonstrated that patients who achieve endoscopic remission (vs those with ongoing endoscopic activity) have favorable long-term outcomes. For example, in a post-hoc analysis of data from a clinical trial incorporating routine biochemical monitoring to inform adalimumab dose modification, Ungaro et al¹⁵³ appreciated a significant reduction in the rate of serious disease-related complications over time in those achieving endoscopic remission compared with those who did not (adjusted HR, 0.19; 95% CI, 0.07–0.31). This is in contrast to a long-term follow-up study of TAILORIX, which examined long-term follow-up of 95 individuals, 45 of whom had achieved deep remission, who had enrolled in a clinical trial assessing the efficacy of regular therapeutic drug monitoring when using infliximab.¹⁵⁴ In this small cohort, there was no difference in rates of CD-related treatment modification, major abdominal or perianal surgery, or CD-related hospitalization over follow-up.

There are limited RCTs assessing whether there is actual benefit in systematically treating toward endoscopic remission target vs symptomatic remission targets (ie, testing whether the target has been achieved, followed by algorithmic treatment adjustment, including escalating index therapy, adding an immunomodulator, followed by switching to an alternative advanced therapy and surgery). There was significant heterogeneity among the 2 reviewed studies, both in terms of the advanced therapy used, algorithms for therapy modification, and the cadence and frequency of endoscopic monitoring that challenge interpretation. Based on the significant uncertainty of evidence with regard to improving maintenance of remission or reducing the risks of adverse events, the guideline panel could not make a recommendation in relation to selecting endoscopic targets over clinical targets.

It is worth emphasizing that in both of the included trials, the majority of individuals in the endoscopic healing arms were not able to meet the goal of endoscopic healing despite an algorithmic approach. For example, in STAR-DUST, only 11% of individuals achieved endoscopic remission.¹⁴⁹ This may be indicative of a particularly challenging patient population to treat. However, it is important to note that it remains conventional practice to assess for clinical, biochemical, and endoscopic response and/or remission after starting and advanced therapy. There are specific patient populations, such as those who have recently undergone intestinal resection,¹⁵⁵ in which endoscopic evaluation may be particularly valuable in clinical decision making. The current question does not address the question related to a tight control strategy focusing on treating to a target of biochemical remission in

order to improve rates of achieving clinical and endoscopic remission. The benefit of a monitoring strategy incorporating biochemical monitoring over clinical monitoring alone was demonstrated in the CALM trial,¹⁵² and has been addressed in previous AGA guidelines on the role of biomarkers in patients with CD.¹²

Evidence Gaps and Future Research

There are several significant evidence gaps regarding this treatment approach that could be addressed by future clinical trials. Providers and investigators have begun to pursue structural outcomes given the well-described discordance between clinical symptoms, captured throughout scoring systems, such as the CDAI, and used to define clinical remission, and biochemical or structural outcomes, such as endoscopic healing. However, there is also variation in endoscopic scoring, and histologic assessment may be more sensitive to persistent residual inflammation. There are compelling observational studies that suggest histologic resolution of inflammation may be more closely correlated with downstream clinical outcomes than endoscopic healing alone in ileal CD, for example.¹⁵⁶ Histologic healing may be an even more challenging target to achieve, however. Furthermore, CD-related inflammation can extend deeply through the bowel wall, and there is growing recognition via technologies such as enterography or intestinal ultrasound that correcting or reducing inflammation within the epithelium of the colon or small bowel may not completely correlate with resolution of the inflammation process transmurally. Such transmural healing may also be more strongly correlated with clinical outcomes than endoscopic healing alone.^{157,158} Further studies of both approved and novel advanced therapies should continue to explore these potential outcomes and their association with clinical outcomes in CD. Future efforts should also continue to refine and standardize outcome definitions for histologic and transmural healing as well.

The appropriate timing and frequency of endoscopic evaluation, as well as its relation to clinical outcomes, including medication persistence, maintenance of remission, and reduction of CD-related adverse events is unclear at this time, and there is significant heterogeneity regarding time to achieving endoscopic healing or other structural outcomes. Understanding not only these temporal associations between treatment duration and structural assessment, but also the factors that might predict an expected earlier or later response is critical; such predictive models would allow clinicians to select the appropriate assessment window and modify current therapies more accurately.

Lastly, ongoing and future clinical trials should assess what treatment modifications should be considered to attain these structural goals. Although dose modification of an existing therapy in an individual with clinical remission is unlikely to result in clinical worsening, transitioning to an entirely different therapeutic mechanism to achieve mucosal, histologic, or transmural healing, particularly in asymptomatic patients, is a much different proposition

given varying safety profiles of medications as well as the risk of loss of response. Similarly, with growing recognition of the potential treatment strategy of targeting multiple inflammatory pathways via combination advanced therapy, one could consider employing these approaches to achieve these structural outcomes. However, although early preliminary data are reassuring,^{124,125} it remains possible that these strategies also carry with them increased risks of serious adverse events related to therapy. Future clinical trials should assess these variable approaches in the context of long-term disease-related complication rates to better understand the incremental value of achieving endoscopic, histologic, or transmural healing. It is imperative that such work be coupled with research exploring patient preferences for these outcomes and that, given the personal and societal financial implications of these therapies, these findings are combined with cost-effectiveness analyses to simulate the best societal approaches as well.

Placing the AGA's Recommendations for the Pharmacologic Management of Moderate-to-Severely Active CD in Context

Providers managing moderate-to-severely active CD have several recently released societal guidelines that they can employ to support clinical decision making. Some of these guidelines addressed different PICO questions (eg, diagnosis, monitoring, mild disease, postoperative management) than those covered by the AGA guidelines. The American College of Gastroenterology recently published their comprehensive guidelines related to the management of CD.¹⁵⁹ For moderate-to-severely active CD, the American College of Gastroenterology's recommendations broadly align with those presented here. Their guideline committee recommended the use of TNF antagonists, combination therapy with TNF antagonists, ustekinumab, risankizumab, mirikizumab, and guselkumab, consistent with the recommendations provided here. They recommended the use of upadacitinib in those who had previously been exposed to a TNF antagonist. They made no recommendation in relation to combination therapy with non-TNF antagonists, and recommendations related to methotrexate and the thiopurines were relatively aligned with these guidelines as well. They did not provide recommendations related to positioning therapies based on prior exposure, but did conditionally recommend the use of risankizumab over ustekinumab in those with prior TNF-antagonist exposure. They recommended that mucosal healing is an important goal for the treatment of CD. They conditionally recommended upfront use of advanced therapy in comparison with step-up therapy, and did not comment on withdrawal of thiopurines or TNF antagonists in those using combination therapy in remission.

The British Society of Gastroenterology recently published a comprehensive guideline for the management of inflammatory bowel disease in 2025.⁸³ With regard to the medical management of moderate-to-severely active CD,

the guideline panel recommended the use of TNF antagonist and combination therapy with thiopurines. They also recommended the use of vedolizumab, ustekinumab, and risankizumab. Upadacitinib use was recommended irrespective of prior TNF-antagonist failure, which may reflect differences in labeling based in Europe in comparison with the United States. In contrast to the recommendations presented here, the British Society of Gastroenterology guideline panel recommended against the use of immunomodulator monotherapy, for both the induction and maintenance of remission. No formal recommendation was provided for mirikizumab. They did not provide recommendations related to prior exposures or positioning. They recommended against TNF-antagonist withdrawal in those receiving combination therapy in remission after 1 year of therapy. Although they did not provide a recommendation regarding thiopurine withdrawal, they did specify that this is generally not associated with an increased risk of flare in those who continue TNF-antagonist therapy and who were in stable remission for more than 6 months. They did not specifically comment on upfront advanced therapies in comparison with step-up therapy, nor did they comment on mucosal healing as an ideal target of therapy.

The European Crohn's and Colitis Organisation also published updated guidelines in October 2024.¹⁶⁰ As with both the American College of Gastroenterology and British Society of Gastroenterology guidelines, recommendations in relation to TNF antagonists, vedolizumab, ustekinumab, and risankizumab were largely similar to those presented in these guidelines. They also recommended against thiopurines for induction therapy, and supported their use for maintenance therapy in CD, and recommended parenteral methotrexate for both induction and maintenance. They recommended combination therapy for infliximab with a thiopurine, and consideration of thiopurine withdrawal after 6–12 months in those who achieve long-term remission. They did not recommend combination therapy with adalimumab, and instead recommended adalimumab monotherapy, however. They uniquely commented specifically on adalimumab and ustekinumab being equally effective in those who are biologic therapy-naïve. As with the British Society of Gastroenterology guidelines, they recommended upadacitinib use regardless of prior advanced therapy exposure. They made no recommendation related to mirikizumab, potentially due to the timing of this guideline development. In contrast to the AGA guidelines, they specifically noted there is insufficient evidence to guide positioning therapies in moderate-to-severely active CD at this time. They recommended upfront use of advanced therapy and treat-to-target monitoring, but noted that there is still debate about what the appropriate target may be.

In summary, recently published guidelines for the management of moderate-to-severely active CD are largely similar across societies. Key differences relate to medication positioning, particularly in relation to upadacitinib, as well as with thiopurine use. As expected, there is heterogeneity in recommendations in relation to queries for which this guideline panel felt there was insufficient evidence. Despite

these minor differences, alignment across the majority of recommendations demonstrates that these are supported by a sound literature base and similar application of recommendation methods. In contrast to other societal guidelines, the AGA guideline panel felt it was appropriate to use current state of the art synthesis methods to attempt to provide clinicians with guidance in relation to positioning these therapies in treatment-naïve and treatment-exposed individuals. Importantly, these recommendations, as well as recommendations related to medications in comparison with placebo, combination therapy with TNF antagonists and non-TNF antagonists, and treatment withdrawal, are currently in a living format. These living guideline recommendations will be updated quarterly, allowing for rapid evolution as new data become available.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Gastroenterology* at www.gastrojournal.org, and at doi:10.1053/j.gastro.2025.09.038.

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Conflicts of interest

Each nominee to this guideline panel underwent a vetting process that required them to report all commercially funded and other relevant

activities within the previous 24 months. All reported financial or intellectual conflicts were reviewed by the Chair of the AGA Institute Clinical Guidelines Committee (CGC) and adjudicated against rules and criteria in the CGC conflict of interest (COI) Policy. Only nominees whose COI status complied with this policy were appointed as guideline panel members. COIs are presented in [Supplementary Table 1](#) for all panel members: Guideline chairs had no conflicts of interest, and <50% of guideline panel members had COIs.

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Supplemental information

AGA Living Clinical Practice Guideline on the Pharmacologic Management of Moderate-to-Severe Crohn's Disease

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ONLINE SUPPLEMENT

**AGA LIVING CLINICAL PRACTICE GUIDELINE ON THE PHARMACOLOGIC MANAGEMENT OF
MODERATE-TO-SEVERE CROHN'S DISEASE**

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Supplementary Tables

eTable 1: Conflicts of interest for the guideline panel members

PANEL MEMBER	PANEL POSITION	INSTITUTION	GRANTS/RESEARCH	MEDICOLEGAL CONSULTING	OTHER CONSULTING	DSMB OR SIMILAR STUDY COMMITTEE	TEACHING/SPEAKING ACTIVITIES	STOCK, OPTIONS, OR OTHER OWNERSHIP
Frank I. Scott, MD,	Panel Chair	University of Colorado	N/A	N/A	N/A	N/A	N/A	N/A
Siddharth Singh, MD, MS	Lead GRADE Methodologist & Co-Chair	University of California San Diego	Pfizer Investigator-initiated Study- PI, \$165,563.78 – ongoing	N/A	N/A	N/A	N/A	N/A
John Haydek, MD, MSCS	Junior Methodologist	University of North Carolina Chapel Hill	N/A	N/A	N/A	N/A	N/A	N/A
Manasi Agrawal, MD	Content Expert	Ioahn School of Medicine at Mount	N/A	N/A	N/A	N/A	N/A	N/A
Ashwin Ananthakrishnan, MD, MPH	Content Expert	Massachusetts General Hospital Research Institute	N/A	N/A	N/A	N/A	N/A	N/A
Benjamin H. Click, MD	Content Expert	University of Colorado	N/A	N/A	AbbVie – Consulting; \$1,625, Oct 2023 ALPCO – Honorarium, \$2,000, May 2024	N/A	N/A	N/A
Gaurav Syal, MD, MS	Content Expert	University of California San Diego	N/A	N/A	N/A	N/A	N/A	N/A
Michael Kappelman, MD	Content Expert	University of North Carolina Chapel Hill	N/A	N/A	Takeda, \$15,000 Abbvie, \$10,000 Pfizer, \$10,000 Lilly, \$15,000	VIVID trial mirikizumab (Member DSMB) SHINE-2: Peds mirikizumab studies (Chair)	N/A	Immediate family member has equity in for-profit health care company
James Lewis, MD	Content Expert	University of Pennsylvania	AbbVie, \$100,000 Takeda, \$400,000 Nestle Health Science, \$470,000 Lilly, \$5,000	Goodell, DeVries, Leech & Dann, LLP representing generic manufacturers of ranitidine, \$38,357	Pfizer, \$3,863, 2022 Merck, Sharpe and Dome, \$2,375, 2022 Gilead / Galapagos (Icon), \$3,163, 2022 Janssen (Apex), \$1,650, 2022 Arena Pharmaceuticals, \$1,742, 2022 Amgen, \$5,157, 2022 Gilead, \$3,623, 2022 Sanofi (Genzyme), \$6,774, 2022 Pfizer, \$10,160, 2023 Merck, Sharpe and Dome, \$1,208, 2023 Gilead / Galapagos (Icon), \$1,145, 2023 Janssen (Apex), \$7,050.00, 2023 Celgene, \$1,400.00, 2023 Amgen, \$5,145.83, 2023 Sanofi (Genzyme), \$3,207.00, 2023 Pfizer, \$3,240, 2024 Janssen, \$11,050, 2024 Amgen, \$5,1170, 2024	ImproveCareNow QI meeting, \$2,000 from Nestle	Dark Canyon Laboratories, stock	
Yuhong Yuan, MD	Trainee Methodologist	London Health Sciences Centre, London, Ontario,	N/A	N/A	N/A	N/A	N/A	

eTable 2: Search strategy for key focus areas addressed in the guidelines

Databases: MEDLINE ALL (Ovid, 1946 to August 14, 2024), Embase (Ovid, 1996 to August 14, 2024); Cochrane CENTRAL (Ovid, Inception to July 2024).

1. Crohn Disease/ use medall,cctr
2. exp Crohn disease/ use oomezd
3. Crohn*.ti,ab,kw,kf,af.
4. ((Inflammatory bowel disease* or IBD) not (ulcerative colitis or UC)).ti.
5. or/1-4 [CD]
6. exp Tumor Necrosis Factor Inhibitors/ use medall,cctr
7. exp tumor necrosis factor antibody/ use oomezd
8. exp Receptors, Tumor Necrosis Factor/ use medall,cctr
9. exp Tumor Necrosis Factors/ use medall,cctr
10. tumor necrosis factor/ use oomezd
11. tumor necrosis factor receptor/ use oomezd
12. ((tumor necrosis or tumour necrosis or TNF or antitumor necrosis or antitumour necrosis or antiTNF* or Cachectin) adj2 (factor* or inhibit* or antibod* or antagonist* or block* or biologic*)).ti,ab,kw,kf.
13. exp Antibodies, Monoclonal/ use medall,cctr
14. exp monoclonal antibody/ use oomezd,medall,cctr
15. (monoclonal adj2 antibod*).ti,ab,kw,kf.
16. (Infliximab or "ABP 710" or ABP710 or Avakine or Avsola or cA2 or Flixabi or "GP 1111" or GP1111 or IFX or Inflectra or INFLIX or Ixifi or "PF 06438179" or "PF 6438179" or PF06438179 or PF6438179 or Remicade or Remsima or Renflexis or Revellex or "TA 650" or TA650 or Zessly or "Mab cA2" or "170277-31-3").ti,ab,kw,kf,rn.
17. (Adalimumab or Humira or Amjevita or Cyltezo or "D2E7 Antibody" or Trudexa or "abp 501" or abp501 or "abt d2e7" or abtd2e7 or adaly or amgevita or "avt 02" or avt02 or "bat 1406" or bat1406 or "bax 2923" or "bax 923" or bax2923 or bax923 or "bi 695501" or bi695501 or "chs 1420" or chs1420 or "ct p17" or ctp17 or "da 3113" or da3113 or "dmb 3113" or dmb3113 or exemptia or "fkb 327" or fkb327 or fyzoclad or "gp 2017" or gp2017 or hadlima or halimatoz or hefiya or "hlx 03" or hlx03 or hulio or hyrimoz or "ibi 303" or ibi303 or idacio or imraldi or kromeya or "lu 200134" or lu200134 or "m 923" or m923 or mabura or "msb 11022" or msb11022 or "ons 3010" or ons3010 or "pf 06410293" or "pf 6410293" or pf06410293 or pf6410293 or raheara or "sb 5" or sb5 or solymbic or "zrc 3197" or zrc3197 or "1446410-95-2" or "331731-18-1").ti,ab,kw,kf,rn.
18. (Certolizumab or "cdp 870" or cdp870 or cimzia or "pha 738144" or pha738144 or "428863-50-7").ti,ab,kw,kf,rn.
19. (Vedolizumab or anti alpha 4* or antialpha 4* or Entyvio or kynteles or "idp 02" or idp02 or "MLN 02" or mln02 or "MLN-0002" or MLN0002 or "pb 016" or pb016 or ro7246311 or "ro 7246311").ti,ab,kw,kf,rn.
20. (Natalizumab or "an 10022" or "an 100226" or an10022 or "an100226" or antegran or antegren or "bg 0002" or bg0002 or "dst 356a1" or dst356a1 or "pb 006" or pb006 or "pbp 2002" or pbp2002 or tyruko or tysabri).ti,ab,kw,kf,rn.
21. (Ustekinumab or "abp 654" or abp654 or "avt 04" or avt04 or "bat 2206" or bat2206 or "bfi 751" or bfi751 or "bmab 1200" or bmab1200 or "bow 090" or bow090 or "cnto 1275" or cnto1275 or "ct p43" or ctp43 or "dmb 3115" or dmb3115 or "eb 1004" or eb1004 or "fyb 202" or fyb202 or "ons 3040" or ons3040 or "pb 007" or pb007 or "ro 723 3920" or "ro 7233920" or ro7233920 or "sb 17" or sb17 or stelara or stellara).ti,ab,kw,kf,rn.

22. Janus Kinase Inhibitors/ use medall,cctr
23. exp Janus kinase inhibitor/ use oomezd
24. ((JAK or Janus Kinase or Janus tyrosine kinase) adj2 (antagonist* or inhibit* or block*)).ti,ab,kw,kf.
25. (upadacitinib or "abt 494" or abt494 or rinvoq).ti,ab,kw,kf,rn.
26. ((IL-23 or interleukin or cytokine) adj2 (antagonist* or inhibit* or block*)).ti,ab,kw,kf.
27. exp cytokine receptor antagonist/ use oomezd
28. (risankizumab or "abbv 066" or abbv066 or "bi 655066 bi655066" or skyrizi).ti,ab,kw,kf,rn.
29. (mirikizumab or "ly 3074828" or ly3074828 or omvoh).ti,ab,kw,kf,rn.
30. (guselkumab or "cnto 1959" or cnto1959 or tremfya).ti,ab,kw,kf,rn.
31. (mercaptopurine or 6MP or "6-MP" or allmercap or bw 57 323h or bw 57323h or bw57323h or empurine or e7wed276i5 or hanixol or ismipur or leukerin or leupurin* or loulla or medipurin or mercaleukin or mercaptopurine* or mercapurene or mern or mycaptine or nsc 755 or nsc755 or pkk6muz20g or puri nethol or purimmun or purine 6 thiol or purine thiol or purimethol or puri-nethol or purinethiol or purinethol or purixan or thiohypoxanthine or thiopurine* or u 4748 or u4748 or xaluprine).ti,ab,kw,kf,rn.
32. exp Mercaptopurine/ use medall,cctr
33. 6 mercaptopurine derivative/ or mercaptopurine/ use oomezd
34. (azathioprine or arathioprin* or aza-q or azafalk or azafor or azahexal or azamedac or azamun* or azanin or azapin or azapress or azaprime or azarex or azasan or azathiodura or azathiopine or azathioprim or azathioprin* or azathropsin or azatioprina or azatox or azatrim or azopi or azoran or azothioprin* or azothioprine or am94r510ms or bw 57 322 or bw 57322 or bw57-322 or bw57322 or colinsan or immuran or immurel or immuthera or imunen or imuprin or Imuran* or imurek or imurel or imuren or jayempi or mrk240iy2l or nsc 39084 or nsc39084 or oraprime or ozathuia or Thioazepine or thioprine or transimune or zytrim).ti,ab,kw,kf,rn.
35. Methotrexate/ use medall,cctr
36. methotrexate derivative/ or methotrexate/ use oomezd
37. (Methotrexate or abitrexate or adx 2191 or adx2191 or amethopterin or amethopterin* or amethopterin* or antifolan or biotrexate or brimexate or canceren or cl 14377 or cl14377 or emt 25299 or emt25299 or emtexate or emthexat* or emtexate or enthexate or farmitrexat* or farmotrex or folex or ifamet or imeth or jylamvo or lantarel or ledertrexate or lumexon or maxtrex or metatrexan or metex or methoblastin or methohexate or methotrate or methotrexate* or methylaminopterin* or metical or metoject or metothrexat* or metotrexat* or metotrexin or metrex or metrotex or mexate or mpi 2505 or mpi 5004 or mpi2505 or mpi5004 or MTX or neotrexate or nordimet or novatrex or nsc 740 or nsc740 or otrexup or r 9985 or r9985 or rasuvo or reditrex or reumatrex or texate or texorate or tremetex or trexall or trexeron or wr 19039 or wr19039 or xaken xatmep or zexate or zlatal).ti,ab,kw,kf,rn.
38. (biologics or biologicals or ((biological or biologic) and (agent* or product* or drug*))).ti.
39. small moledul*.ti.
40. (immunomodulator* or ((immunomodulating or immunotherapeutic) and (agent* or drug* or product*))).ti.
41. or/6-40 [interventions of interest]
42. 5 and 41 [CD and interventions]
43. limit 42 to yr="2020 -Current" [CD and drugs after 2020]
44. exp randomized controlled trial/ or controlled clinical trial.pt. or clinical trials as topic/ or (random* or placebo).ab. or trial.ti.
45. (exp animals/ not humans/) use medall
46. 44 not 45 use medall [MEDLINE RCT filter]

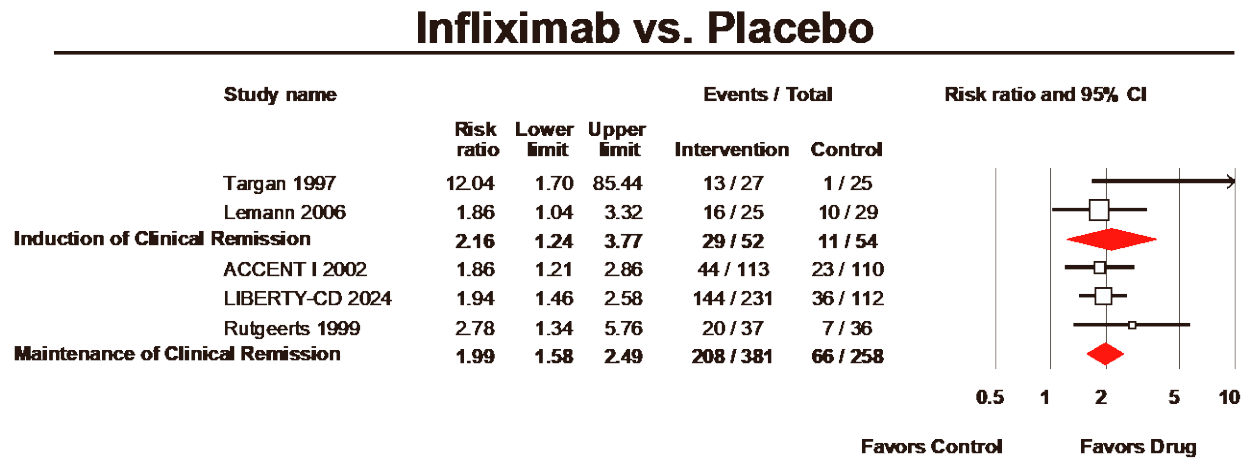
47. 43 and 46 use medall [RCTs in MEDLINE]
48. Randomized controlled trial/ or Controlled clinical study/ or randomization/ or intermethod comparison/ or double blind procedure/ or human experiment/ or (random\$ or placebo or (open adj label) or ((double or single or doubly or singly) adj (blind or blinded or blindly)) or parallel group\$1 or crossover or cross over or ((assign\$ or match or matched or allocation) adj5 (alternate or group\$1 or intervention\$1 or patient\$1 or subject\$1 or participant\$1)) or assigned or allocated or (controlled adj7 (study or design or trial)) or volunteer or volunteers).ti,ab. or (compare or compared or comparison or trial).ti. or ((evaluated or evaluate or evaluating or assessed or assess) and (compare or compared or comparing or comparison)).ab.
49. (random\$ adj sampl\$ adj7 ("cross section\$" or questionnaire\$1 or survey\$ or database\$1)).ti,ab. not (comparative study/ or controlled study/ or randomi?ed controlled.ti,ab. or randomly assigned.ti,ab.)
50. Cross-sectional study/ not (exp randomized controlled trial/ or controlled clinical study/ or controlled study/ or randomi?ed controlled.ti,ab. or control group\$1.ti,ab.)
51. (((case adj control\$) and random\$) not randomi?ed controlled).ti,ab.
52. Systematic review.ti,ab. not (trial or study).ti.
53. (nonrandom\$ not random\$).ti,ab.
54. "random field\$".ti,ab.
55. (random cluster adj3 sampl\$).ti,ab.
56. (review.ab. and review.pt.) not trial.ti.
57. "we searched".ab. and (review.ti. or review.pt.)
58. "update review".ab.
59. (databases adj4 searched).ab.
60. (rat or rats or mouse or mice or swine or porcine or murine or sheep or lambs or pigs or piglets or rabbit or rabbits or cat or cats or dog or dogs or cattle or bovine or monkey or monkeys or trout or marmoset\$1).ti. and animal experiment/
61. Animal experiment/ not (human experiment/ or human/)
62. or/49-61
63. 48 not 62 use oomezd [Embase RCTs filler]
64. 43 and 63 use oomezd [Embase RCTs]
65. limit 47 to dt=20200701-20240814 use medall [Limit not valid in CCTR,Embase; records were retained]
66. limit 47 to ed=20200701-20240814 use medall [Limit not valid in CCTR,Embase; records were retained]
67. limit 64 to dd=20200701-20240814 use oomezd [Limit not valid in CCTR,Ovid MEDLINE(R); records were retained]
68. limit 64 to dc=20200701-20240814 use oomezd [Limit not valid in CCTR; records were retained]
69. 47 and (2021* or 2022* or 2023* or 2024*).ep.
70. 43 use cctr
71. or/65-70 [CD interventions from July 2020 three databases]
72. limit 71 to english language
73. (child/ or Pediatrics/ or Adolescent/ or Infant/ or adolescence/ or newborn/) not (adult/ or aged/)
74. 72 not 73
75. (baby or babies or child or children or pediatric* or paediatric* or peadiatric* or infant* or infancy or neonat* or newborn* or new born* or adolescen* or preschool or pre-school or toddler*).ti. not (adult* or elder* or senior* or men or women).tw.
76. 74 not 75
77. remove duplicates from 75

eTable 3: Interpretation of the Certainty in Evidence of Effects Using the Grading of Recommendations Assessment, Development and Evaluation Framework

Certainty	Description
High	We are very confident that the true effect lies close to that of the estimate of the effect.
Moderate	We are moderately confident in the effect estimate. The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low	Our confidence in the effect estimate is limited. The true effect may be substantially different from the estimate of the effect.
Very Low	We have very little confidence in the effect estimate. The true effect is likely to be substantially different from the estimate of effect.

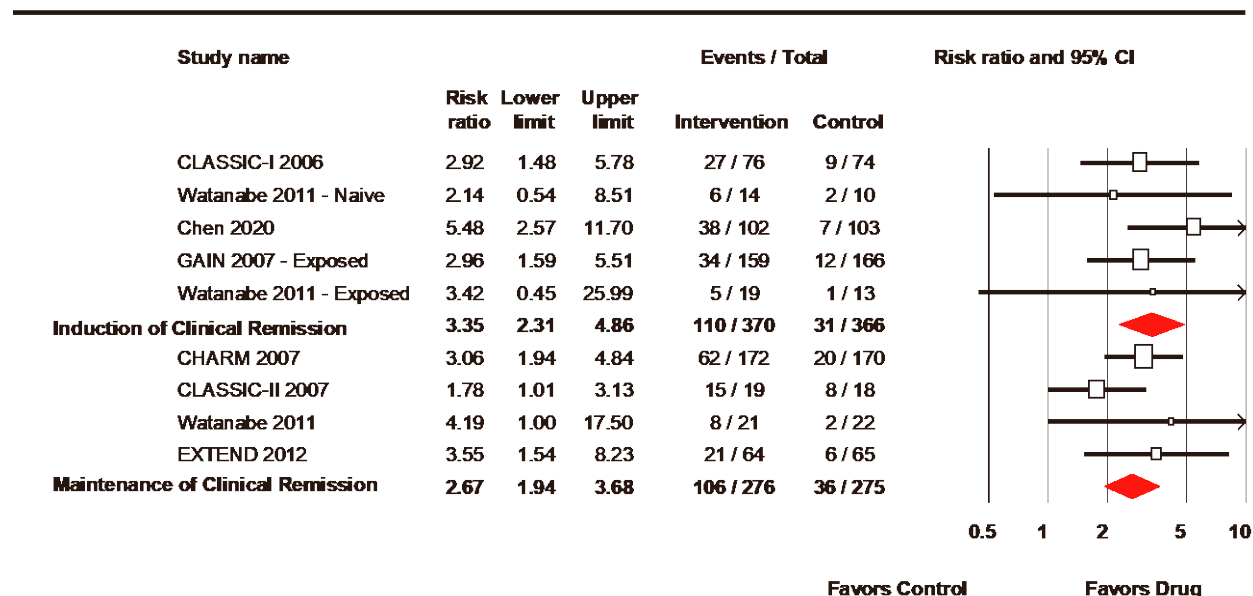
Supplementary Figures

eFigure 1: Forest plot summarizing efficacy of INFLIXIMAB for induction and maintenance of remission in patients with moderately to severely active CD



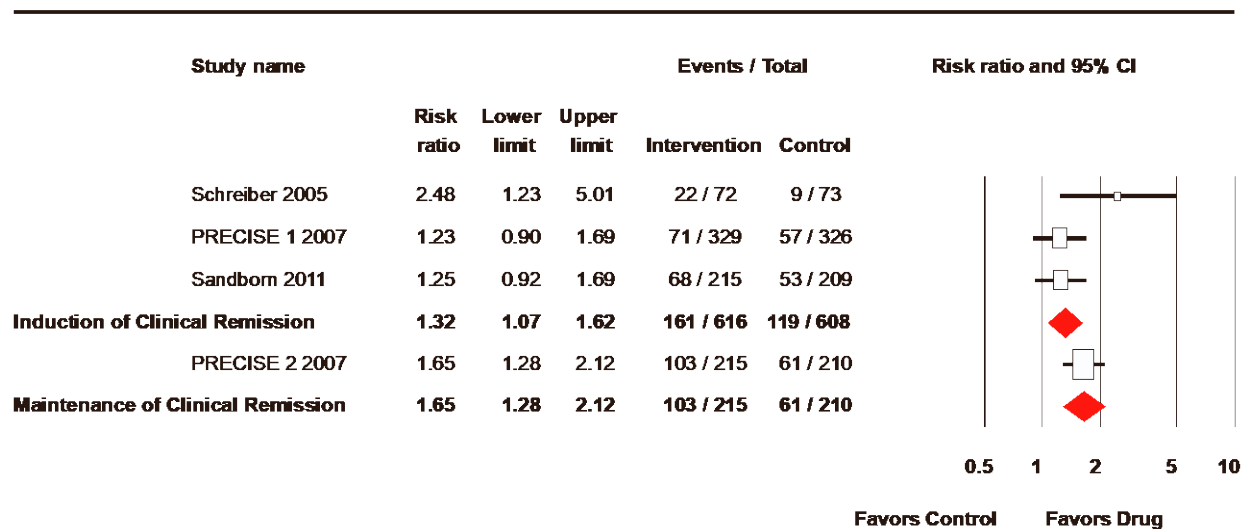
eFigure 2: Forest plot summarizing efficacy of ADALIMUMAB for induction and maintenance of remission in patients with moderately to severely active CD

Adalimumab vs. Placebo



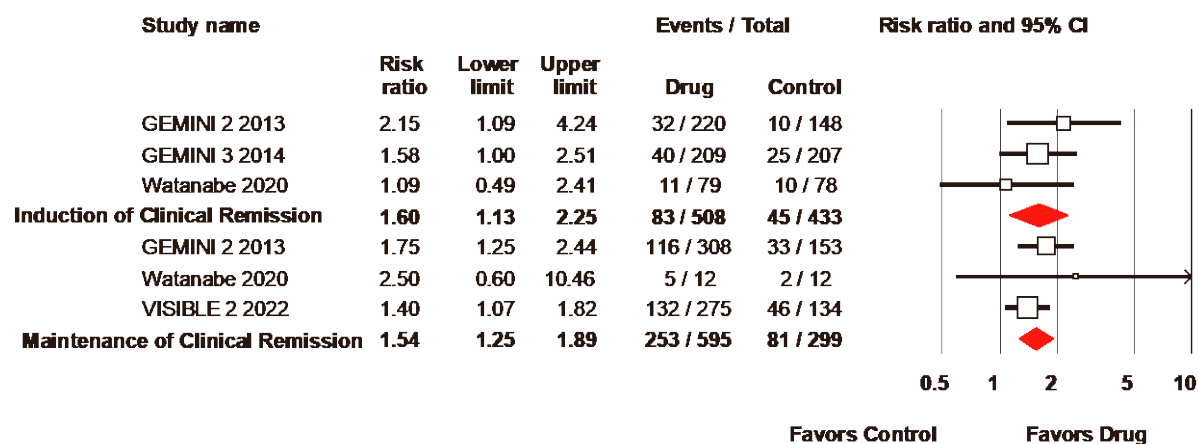
eFigure 3: Forest plot summarizing efficacy of CERTOLIZUMAB PEGOL for induction and maintenance of remission in patients with moderately to severely active CD

Certolizumab pegol vs. Placebo



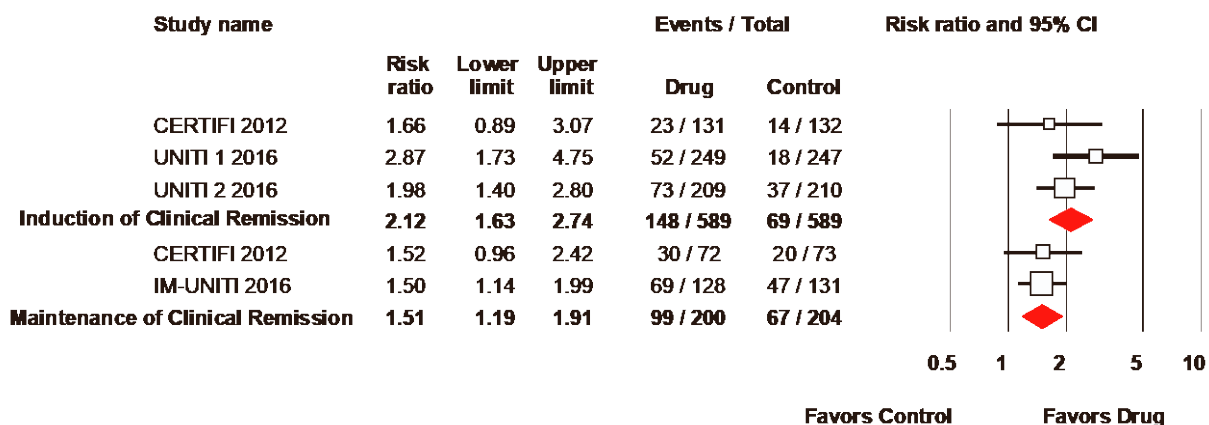
eFigure 4: Forest plot summarizing efficacy of VEDOLIZUMAB for induction and maintenance of remission in patients with moderately to severely active CD

Vedolizumab vs. Placebo



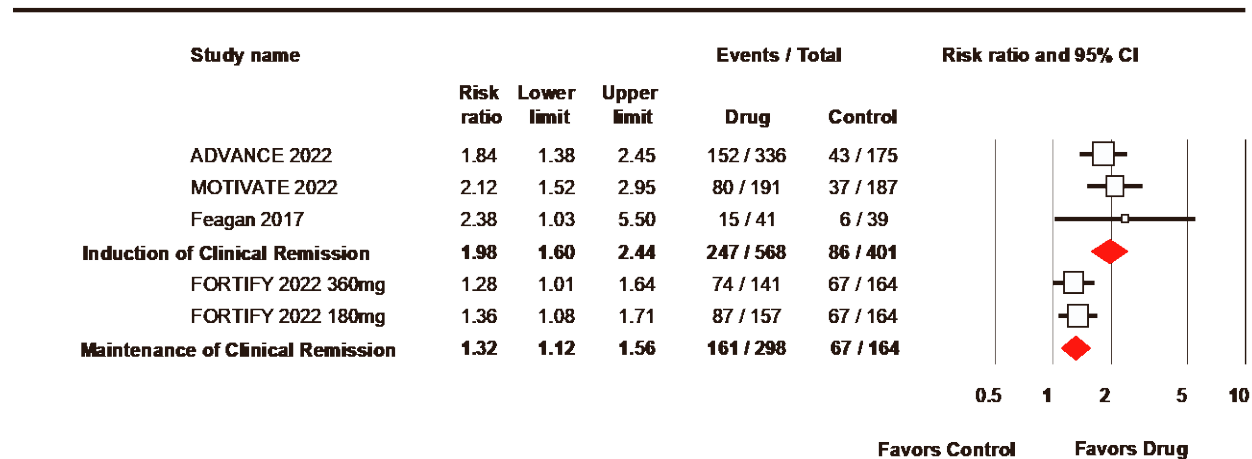
eFigure 5: Forest plot summarizing efficacy of USTEKINUMAB for induction and maintenance of remission in patients with moderately to severely active CD

Ustekinumab vs. Placebo

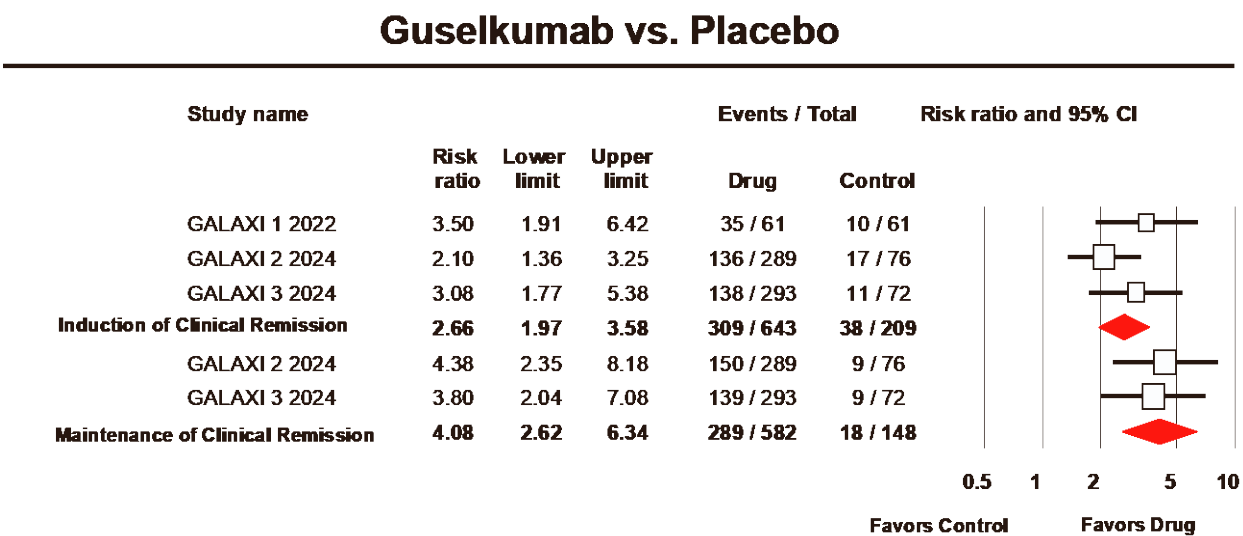


eFigure 6: Forest plot summarizing efficacy of RISANKIZUMAB for induction and maintenance of remission in patients with moderately to severely active CD

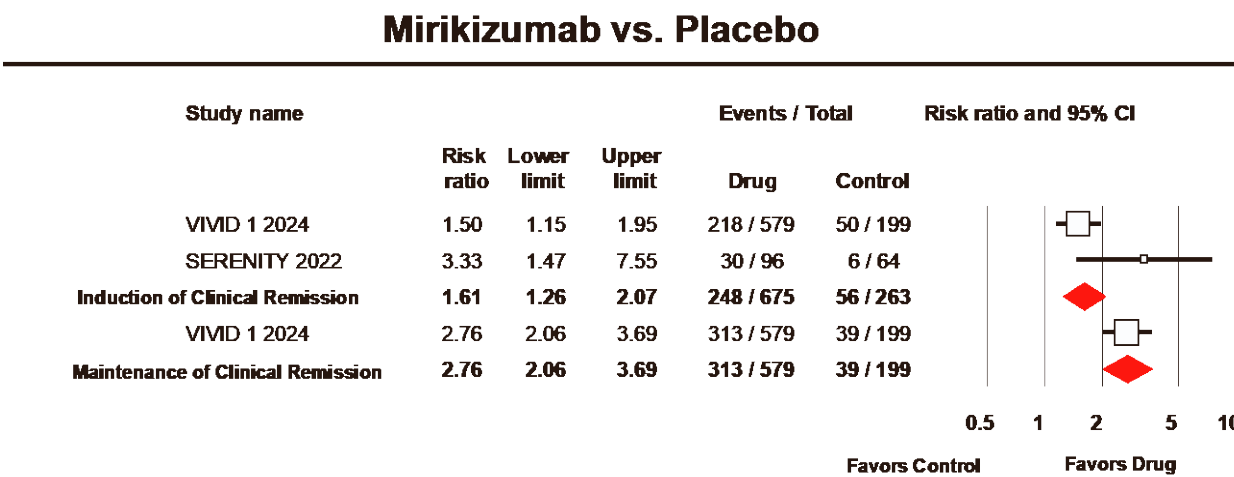
Risankizumab vs. Placebo



eFigure 7: Forest plot summarizing efficacy of GUSELKUMAB for induction and maintenance of remission in patients with moderately to severely active CD



eFigure 8: Forest plot summarizing efficacy of MIRIKIZUMAB for induction and maintenance of remission in patients with moderately to severely active CD



eFigure 9: Forest plot summarizing efficacy of UPADACITINIB for induction and maintenance of remission in patients with moderately to severely active CD

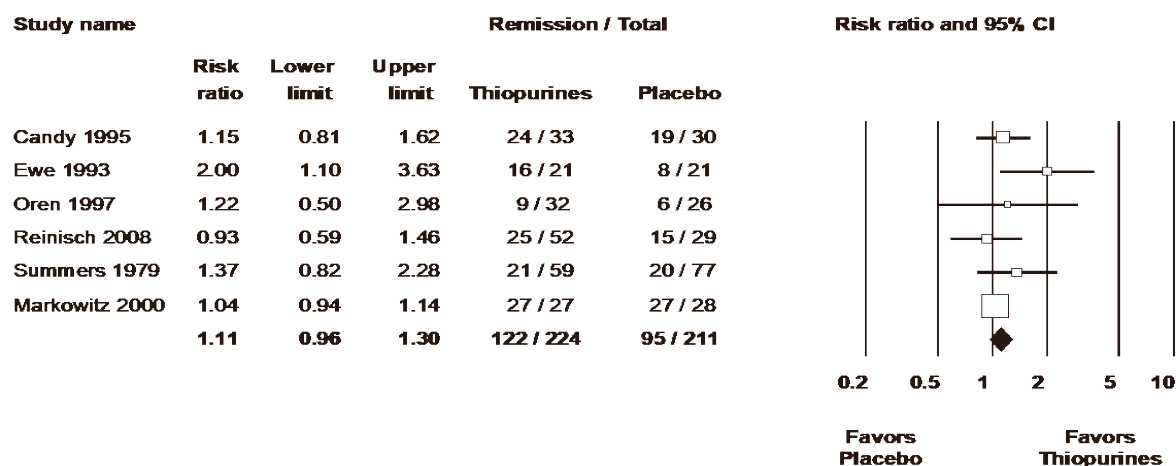
Upadacitinib vs. Placebo



eFigure 10: Forest plot summarizing efficacy of THIOPURINES for (A) induction and (B) maintenance of remission in patients with moderately to severely active CD

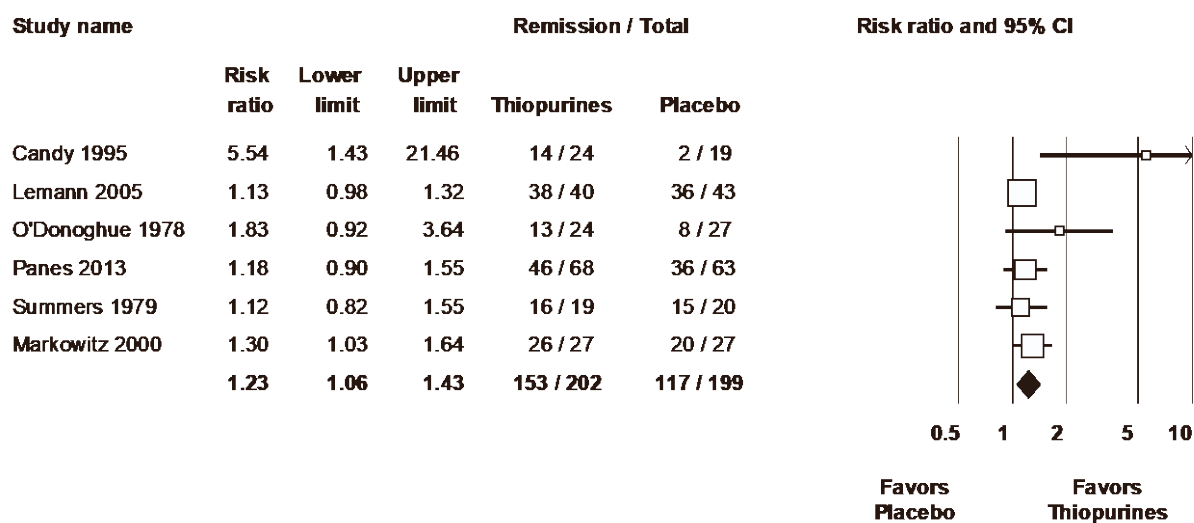
A

Thiopurines for Induction of Remission in Crohn's disease



B

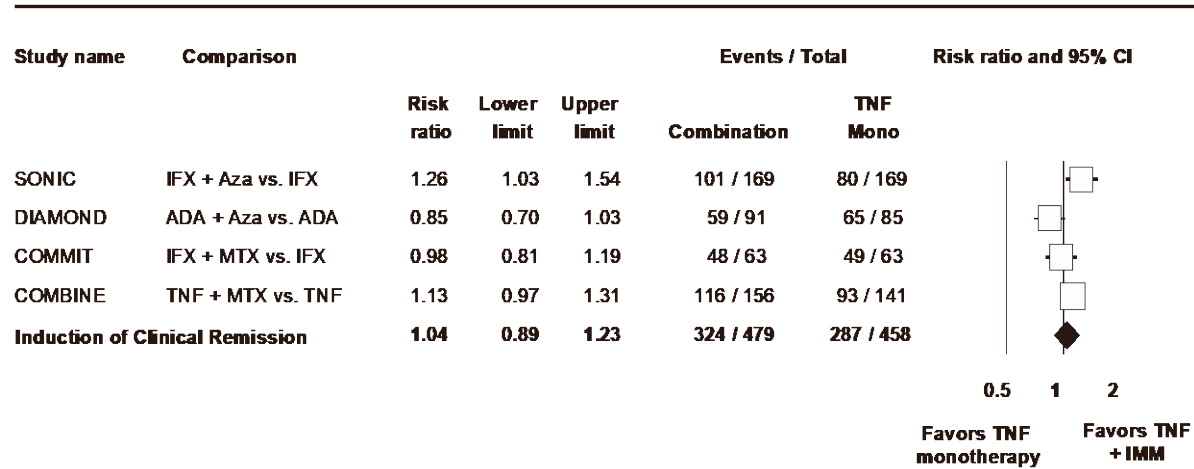
Thiopurines for Maintenance of Remission in Crohn's disease



eFigure 11: Forest plot summarizing efficacy of combined TNF antagonist and immunomodulator versus TNF antagonist monotherapy for (A) induction and (B) maintenance of remission

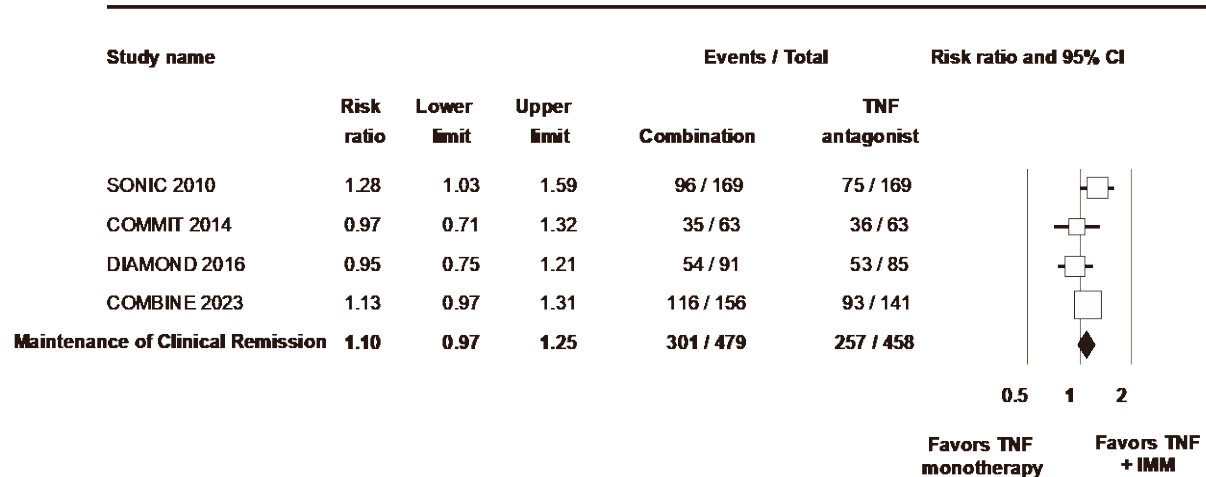
A

Induction of Clinical Remission (W10-16)



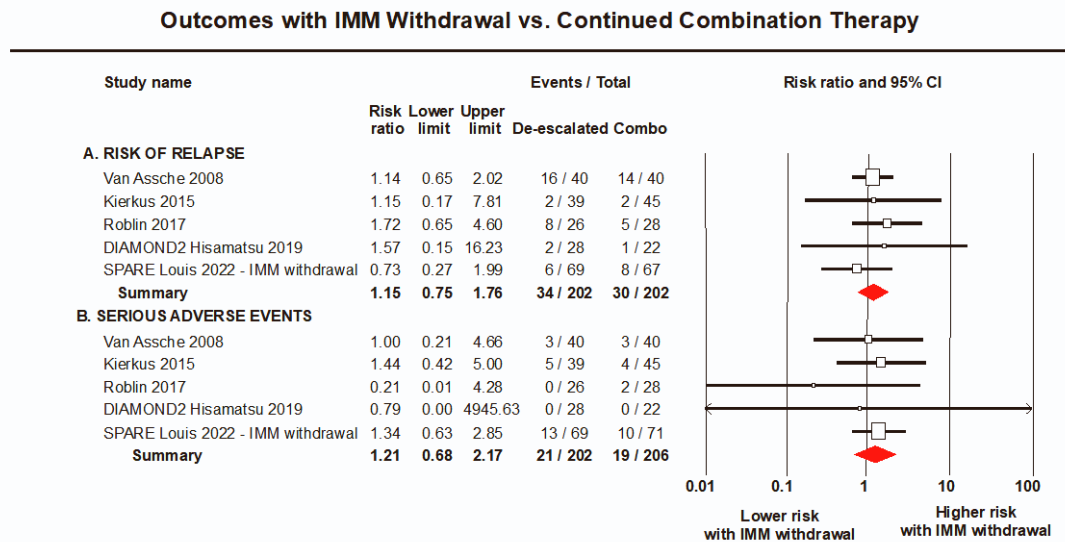
B

Maintenance of Clinical Remission (W26-52)

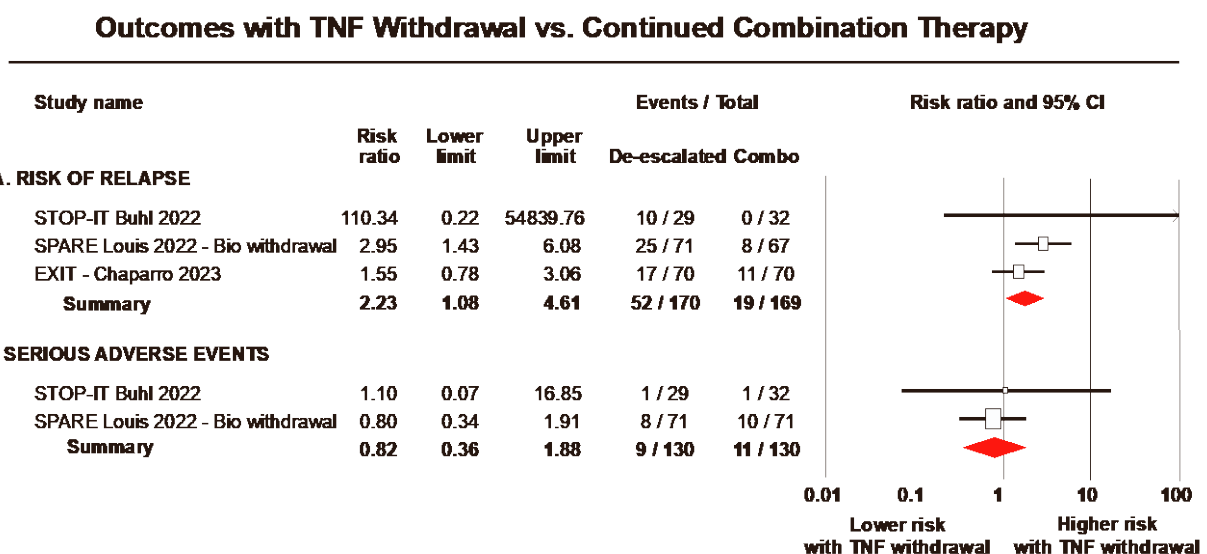


eFigure 12: Forest plot summarizing (A) efficacy and safety of stopping immunomodulator (and continuing biologic monotherapy) vs. continuing combination therapy AND (B) efficacy and safety of stopping TNF antagonist (and continuing immunomodulator monotherapy) vs continuing combination therapy in patients with moderately to severely active Crohn's disease in steroid-free remission on combination therapy

A

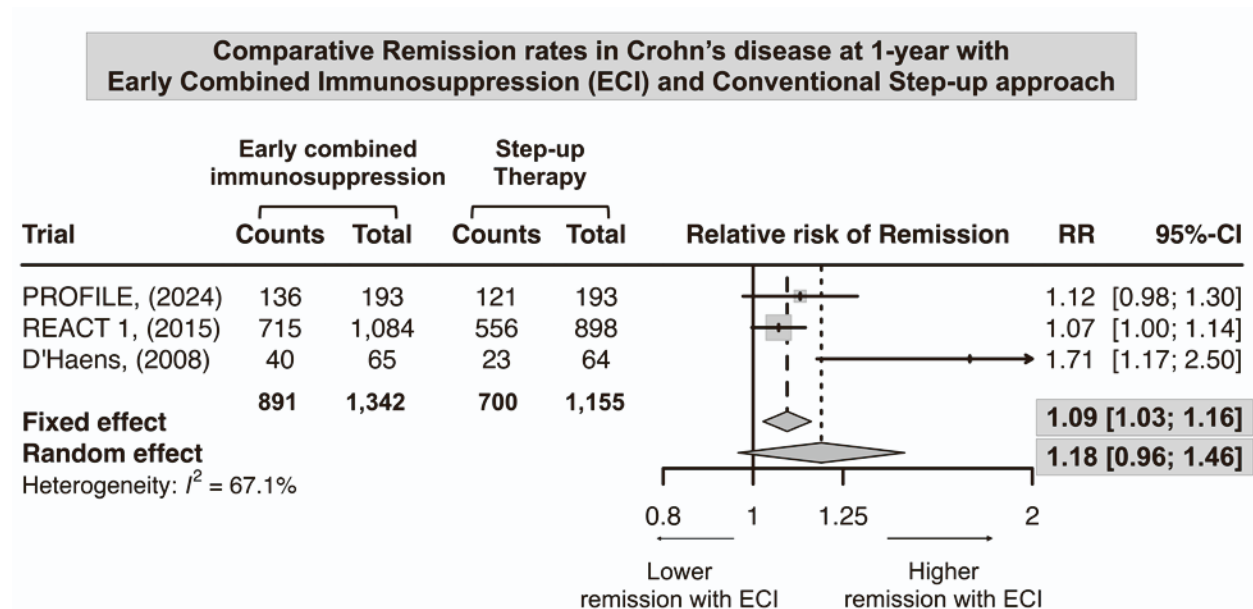


B



eFigure 13: Forest plot summarizing (A) comparative remission rates and (B) comparative disease complication rates at 1 year for early combined immunosuppression compared to conventional step up therapy efficacy in patients with moderately to severely active Crohn's disease in steroid-free remission on combination therapy

A



B

